

Number › Fractions › Simplifying Fractions (Foundation)

Topic 1 of 313 • id: fractions_simplify_F

Q1. Simplify $\frac{3}{18}$.

Q1

Find the highest common factor of 3 and 18: 3

Divide numerator and denominator by 3: $\frac{3 \div 3}{18 \div 3} = \frac{1}{6}$

Answer: $\frac{1}{6}$

Number › Fractions › Equivalent Fractions (Foundation)

Topic 2 of 313 • id: fractions_equivalent_F

Q1. $\frac{7}{10} = \frac{?}{20}$. Find the missing number.

Q1

Find the scale factor between the denominators: $20 \div 10 = 2$

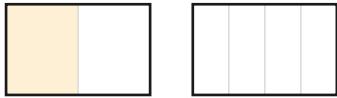
Multiply the numerator by the same scale factor: $7 \times 2 = 14$

Answer: 14

Number › Fractions › Equivalent Fractions (Diagrams) (Foundation)

Topic 3 of 313 • id: fractions_equivalent_diagram_F

Q1. How many parts of Shape B must be shaded to show a fraction equivalent to Shape A?



Q1

Shape A shows $\frac{1}{2}$ shaded; Shape B has 4 equal parts.

Find the scale factor between the denominators: $4 \div 2 = 2$

Multiply the shaded count by the same scale factor: $1 \times 2 = 2$



Answer: 2

Number › Fractions › Ordering Fractions (Foundation)

Topic 4 of 313 • id: fractions_ordering_F

Q1. Write these fractions in descending order: $\frac{3}{4}$, $\frac{1}{4}$, $\frac{5}{9}$, $\frac{1}{2}$

Q1

Convert every fraction to the same denominator, the LCM of 4, 4, 9, 2: 36

Rewrite each fraction over 36: $\frac{3}{4} = \frac{27}{36}$, $\frac{1}{4} = \frac{9}{36}$, $\frac{5}{9} = \frac{20}{36}$, $\frac{1}{2} = \frac{18}{36}$

Order the numerators in descending order and convert back: $\frac{3}{4}$, $\frac{5}{9}$, $\frac{1}{2}$, $\frac{1}{4}$

Answer: $\frac{3}{4}$, $\frac{5}{9}$, $\frac{1}{2}$, $\frac{1}{4}$

Number › Fractions › Adding & Subtracting Fractions (Foundation)

Topic 5 of 313 • id: fractions_add_subtract_F

Q1. Work out $\frac{3}{7} - \frac{1}{4}$. Give your answer as a fraction in its simplest form.

Q1

Find a common denominator: $\text{LCM}(7, 4) = 28$

Convert: $\frac{3}{7} = \frac{12}{28}$, $\frac{1}{4} = \frac{7}{28}$

Subtract: $\frac{12}{28} - \frac{7}{28} = \frac{5}{28}$

Simplify: $\frac{5}{28}$

Answer: $\frac{5}{28}$

Number › Fractions › Multiplying Fractions (Foundation)

Topic 6 of 313 • id: fractions_multiply_F

Q1. Calculate $\frac{2}{8} \times \frac{2}{7}$. Give your answer as a fraction in its simplest form.

Q1

Multiply numerators: $2 \times 2 = 4$

Multiply denominators: $8 \times 7 = 56$

Simplify $\frac{4}{56} = \frac{1}{14}$

Answer: $\frac{1}{14}$

Number › Fractions › Dividing Fractions (Higher)

Topic 7 of 313 • id: fractions_divide_H

Q1. Work out $\frac{5}{2} \div \frac{8}{10}$. Give your answer as a fraction in its simplest form.

Q1

Keep-change-flip: $\frac{5}{2} \div \frac{8}{10} = \frac{5}{2} \times \frac{10}{8}$

Multiply: $\frac{(5 \times 10)}{(2 \times 8)} = \frac{50}{16}$

Simplify: $\frac{25}{8}$

Answer: $\frac{25}{8}$

Number › Fractions › Dividing Fractions (Foundation)

Topic 8 of 313 • id: fractions_divide_F

Q1. Evaluate $\frac{5}{2} \div \frac{3}{8}$. Give your answer as a fraction in its simplest form.

Q1

Keep-change-flip: $\frac{5}{2} \div \frac{3}{8} = \frac{5}{2} \times \frac{8}{3}$

Multiply: $\frac{(5 \times 8)}{(2 \times 3)} = \frac{40}{6}$

Simplify: $\frac{20}{3}$

Answer: $\frac{20}{3}$

Number › Fractions › Mixed Number Arithmetic (Higher)

Topic 9 of 313 • id: fractions_mixed_number_arithmetic_H

Q1. Calculate $5\frac{3}{6} + 1\frac{1}{3}$. Give your answer as a mixed number.

Q1

Convert to improper fractions: $5\frac{3}{6} = \frac{11}{2}$, $1\frac{1}{3} = \frac{4}{3}$

Add the improper fractions: $\frac{11}{2} + \frac{4}{3} = \frac{41}{6}$

Convert back to a mixed number: $6\frac{5}{6}$

Answer: $6\frac{5}{6}$

Number › Fractions › Improper Fractions and Mixed Numbers (Foundation)

Topic 10 of 313 • id: fractions_improper_mixed_F

Q1. Convert $\frac{9}{5}$ to a mixed number.

Q1

Divide the numerator by the denominator: $9 \div 5 = 1$ remainder 4

Write as a mixed number: $1 \frac{4}{5}$

Answer: $1 \frac{4}{5}$

Number › Fractions › Fractions of an Amount (Foundation)

Topic 11 of 313 • id: fractions_of_amount_F

Q1. Calculate $\frac{6}{9}$ of 108.

Q1

Divide by the denominator: $108 \div 9 = 12$

Multiply by the numerator: $12 \times 6 = 72$

Answer: 72

Number › Decimals › Rounding to Decimal Places (Foundation)

Topic 12 of 313 • id: decimals_round_dp_F

Q1. Round 46.341 to 1 decimal place.

Q1

Look at the digit after the 1 decimal place we want: 4, so round down.

46.341 rounded to 1 decimal place = 46.3

Answer: 46.3

Number › Decimals › Rounding to Significant Figures (Foundation)

Topic 13 of 313 • id: decimals_round_sf_F

Q1. Round 546.327 to 2 significant figures.

Q1

Identify the first 2 significant figures of 546.327

546.327 rounded to 2 significant figures = 550

Answer: 550

Number › Decimals › Ordering Decimals, Fractions & Percentages (Foundation)

Topic 14 of 313 • id: decimals_ordering_F

Q1. Write these values in descending order: 59%, 0.56, 0.46, $\frac{49}{100}$

Q1

Convert each value to a decimal: 59% = 0.59, 0.56 = 0.56, 0.46 = 0.46, $\frac{49}{100} = 0.49$

Order in descending order: 59%, 0.56, $\frac{49}{100}$, 0.46

Answer: 59%, 0.56, $\frac{49}{100}$, 0.46

Number › Decimals › Adding and Subtracting Decimals (Foundation)

Topic 15 of 313 • id: decimals_add_subtract_F

Q1. Evaluate $28.72 + 92.5$

Q1

Line up the decimal points, padding with a zero so both numbers have the same number of decimal places.

$$28.72 + 92.5 = 121.22$$

Answer: 121.22

Number › Decimals › Multiplying Decimals (Foundation)

Topic 16 of 313 • id: decimals_multiply_F

Q1. Work out 15.18×9.17

Q1

Ignore the decimal points and multiply the whole numbers: $1518 \times 917 = 1392006$

Count the total decimal places in the question: $2 + 2 = 4$

Place the decimal point 4 places from the right: 139.2006

Answer: 139.2006

Number › Decimals › Dividing Decimals (Foundation)

Topic 17 of 313 • id: decimals_divide_F

Q1. Work out $0.56 \div 0.08$

Q1

Multiply both numbers by 100 so the divisor becomes a whole number: $0.56 \times 100 = 56$, $0.08 \times 100 = 8$

$$56 \div 8 = 7$$

Answer: 7

Number › Decimals › Recurring Decimals to Fractions (One Digit) (Foundation)

Topic 18 of 313 • id: recurring_decimal_single_digit_F

Q1. Write $0.\dot{6}$ as a fraction in its simplest form.

Q1

Let $x = 0.\dot{6}$, so $10x = 6.666\dots$ (only one digit repeats, so multiply by 10 to shift it one place)

Subtract: $10x - x = 6 - 0$, so $9x = 6$

$$x = \frac{6}{9} = \frac{2}{3}$$

Answer: $\frac{2}{3}$

Number › Decimals › Recurring Decimals to Fractions (Two Digits) (Higher)

Topic 19 of 313 • id: recurring_decimal_two_digit_H

Q1. Write $0.\dot{1}\dot{8}$ as a fraction in its simplest form.

Q1

Let $x = 0.\dot{1}\dot{8}$, so $100x = 18.1818\dots$ (two digits repeat, so multiply by 100 to shift a full block)

Subtract: $100x - x = 18 - 0$, so $99x = 18$

$$x = \frac{18}{99} = \frac{2}{11}$$

Answer: $\frac{2}{11}$

Number › Decimals › Recurring Decimals to Fractions (Higher)

Topic 20 of 313 • id: decimals_recurring_to_fraction_H

Q1. Write $0.\dot{8}5714\dot{2}$ as a fraction in its simplest form.

Q1

Let $x = 0.\dot{8}5714\dot{2}$

Multiply x by a power of 10 so the recurring parts line up, then subtract to eliminate the recurring part.

This gives $x = \frac{857142}{999999}$

Simplify: $\frac{6}{7}$

Answer: $\frac{6}{7}$

Number › Order of Operations (BIDMAS) › BIDMAS (2 or 3 operations) (Foundation)

Topic 21 of 313 • id: bidmas_two_three_F

Q1. Work out $8 \times 8 - 60$.

Q1

Multiplication: $8 \times 8 = 64$

Subtraction: $64 - 60 = 4$

Answer: 4

Number › Order of Operations (BIDMAS) › BIDMAS (4+ operations) (Foundation)

Topic 22 of 313 • id: bidmas_F

Q1. Work out $2^2 + 3 \times 6 - 7 \div 7$.

Q1

Indices: $2^2 = 4$

Multiplication: $3 \times 6 = 18$

Division: $7 \div 7 = 1$

Addition: $4 + 18 = 22$

Subtraction: $22 - 1 = 21$

Answer: 21

Number › Standard Form › Converting to Standard Form (Foundation)

Topic 23 of 313 • id: standard_form_to_F

Q1. Write 50000000 in standard form.

Q1

Move the decimal point until there is one non-zero digit before it: 5

Count how many places the decimal point moved: 7

$$50000000 = 5 \times 10^7$$

Answer: 5×10^7

Number › Standard Form › Converting Small Numbers to Standard Form (Foundation)

Topic 24 of 313 • id: standard_form_to_small_F

Q1. Write 0.00000005 in standard form.

Q1

Move the decimal point until there is one non-zero digit before it: 5

Count how many places the decimal point moved: 8 (the number is less than 1, so the power is negative)

$$0.00000005 = 5 \times 10^{-8}$$

Answer: 5×10^{-8}

Number › Standard Form › Converting from Standard Form (Large Numbers) (Foundation)

Topic 25 of 313 • id: standard_form_from_large_F

Q1. Write 5×10^4 as an ordinary number.

Q1

Multiply 5 by 10^4 , which means moving the decimal point 4 place(s) right.

$$5 \times 10^4 = 50000$$

Answer: 50000

Number › Standard Form › Converting from Standard Form (Small Numbers) (Foundation)

Topic 26 of 313 • id: standard_form_from_small_F

Q1. Write 3.81×10^{-3} as an ordinary number.

Q1

Multiply 3.81 by 10^{-3} , which means moving the decimal point 3 place(s) left.

$$3.81 \times 10^{-3} = 0.00381$$

Answer: 0.00381

Number › Standard Form › Multiplying & Dividing in Standard Form (Higher)

Topic 27 of 313 • id: standard_form_multiply_divide_H

Q1. Work out $(5 \times 10^{-4}) \div (4 \times 10^{-3})$. Give your answer in standard form.

Q1

Divide the mantissas: $5 \div 4 = 1.25$

Subtract the powers of 10: $-4 - -3 = -1$

Answer: 1.25×10^{-1}

Number › Standard Form › Multiplying & Dividing in Standard Form (Foundation)

Topic 28 of 313 • id: standard_form_multiply_divide_F

Q1. Work out $(2 \times 10^1) \times (2 \times 10^2)$. Give your answer in standard form.

Q1

Multiply the mantissas: $2 \times 2 = 4$

Add the powers of 10: $1 + 2 = 3$

Answer: 4×10^3

Number › Standard Form › Adding & Subtracting in Standard Form (Higher)

Topic 29 of 313 • id: standard_form_add_subtract_H

Q1. Work out $(9 \times 10^6) + (3 \times 10^5)$. Give your answer in standard form.

Q1

Write both numbers with the same power of 10: $9 \times 10^6 = 90 \times 10^5$, $3 \times 10^5 = 3 \times 10^5$

Add the coefficients: $90 + 3 = 93$

Write in standard form: $93 \times 10^5 = 9.3 \times 10^6$

Answer: 9.3×10^6

Number › Standard Form › Standard Form with a Calculator (Foundation)

Topic 30 of 313 • id: standard_form_calculator_F

Q1. Use a calculator to work out $(9.3 \times 10^7) \div (5.9 \times 10^3)$. Give your answer in standard form, rounded to 3 significant figures if necessary.

Q1

Divide the mantissas on your calculator: $9.3 \div 5.9 = 1.576271$

Subtract the powers of 10: $7 - 3 = 4$

Round the mantissa to 3 significant figures: 1.58×10^4

Answer: 1.58×10^4

Number › Estimation & Bounds › Estimating Calculations (Foundation)

Topic 31 of 313 • id: estimation_rounding_F

Q1. Work out an estimate for $(65.3 \times 5.4) \div 8.5$ by rounding each number to 1 significant figure.

Q1

Round each value to 1 significant figure: $65.3 \approx 70$, $5.4 \approx 5$, $8.5 \approx 9$

$$(70 \times 5) \div 9 = \frac{350}{9}$$

Answer: $\frac{350}{9}$

Number › Estimation & Bounds › Error Intervals (Foundation)

Topic 32 of 313 • id: error_interval_F

Q1. A number, x , is 152 rounded to the nearest whole number. Write down the error interval for x .

Q1

152 is rounded to the nearest whole number, so x can be up to half of that precision above or below 152.

Lower bound: $152 - 0.5 = 151.5$

Upper bound: $152 + 0.5 = 152.5$

Answer: $151.5 \leq x < 152.5$

Number › Estimation & Bounds › Error Intervals (significant figures) (Higher)

Topic 33 of 313 • id: error_interval_H

Q1. A number, x , is 2.4 rounded to 2 significant figures. Write down the error interval for x .

Q1

The last significant figure of 2.4 has a place value of 0.1, so x can be up to half of that above or below 2.4

Lower bound: $2.4 - 0.05 = 2.35$

Upper bound: $2.4 + 0.05 = 2.45$

Answer: $2.35 \leq x < 2.45$

Number › Estimation & Bounds › Bounds of Calculations (Higher)

Topic 34 of 313 • id: bounds_calculation_H

Q1. A rectangle has length 9 cm and width 23 cm, both measured to the nearest cm. Calculate the lower bound for the area of the rectangle.

Q1

Length: 9 cm (to the nearest cm) gives $8.5 \leq \text{length} < 9.5$

Width: 23 cm (to the nearest cm) gives $22.5 \leq \text{width} < 23.5$

For the lower bound of the area, use the lower bound of each measurement: $8.5 \times 22.5 = 191.25 \text{ cm}^2$

Answer: 191.25 cm²

Number › Negative Numbers › Adding and Subtracting Negative Numbers (Foundation)

Topic 35 of 313 • id: negative_add_subtract_F

Q1. Work out $17 + (-6)$.

Q1

$$17 + (-6) = 11$$

Answer: 11

Number › Negative Numbers › Multiplying and Dividing Negative Numbers (Foundation)

Topic 36 of 313 • id: negative_multiply_divide_F

Q1. Work out -10×11 .

Q1

$$-10 \times 11 = -110$$

Answer: -110

Number › Negative Numbers › Ordering Negative Numbers (Foundation)

Topic 37 of 313 • id: negative_ordering_F

Q1. Write these numbers in ascending order: -3, -8, -9, 0, 9

Q1

Place the numbers on a number line: -3, -8, -9, 0, 9.

Reading them in order from smallest to largest gives: -9, -8, -3, 0, 9.

Answer: -9, -8, -3, 0, 9

Number › Multiplying & Dividing by Powers of 10 › Multiplying & Dividing by Powers of 10 (Foundation)

Topic 38 of 313 • id: number_powers_of_ten_F

Q1. Evaluate $70.17 \div 100$.

Q1

Dividing by 100 moves the decimal point 2 places to the left.

$$70.17 \div 100 = 0.7017$$

Answer: 0.7017

Number › Factors, Multiples & Primes › Prime Numbers (Foundation)

Topic 39 of 313 • id: prime_numbers_F

From this list, write down all the prime numbers:

Q1. 33, 62, 54, 26, 14, 86

Q1. 33, 62, 54, 26, 14, 86

Check each number for factors other than 1 and itself: 33, 62, 54, 26, 14, 86

Prime numbers: None

Answer: None

Number › Factors, Multiples & Primes › Multiples (Foundation)

Topic 40 of 313 • id: multiples_F

Q1. Is 68 a multiple of 8?

Q1

$68 \div 8 = 8$ remainder 4

No, 68 is not a multiple of 8.

Answer: No

Number › Factors, Multiples & Primes › Factors (Foundation)

Topic 41 of 313 • id: factors_F

Q1. How many factors does 14 have?

Q1

List every factor of 14 by checking each whole number from 1 to 14: 1, 2, 7, 14.

Count them: 4 factors.

Answer: 4

Number › Factors, Multiples & Primes › Prime Factor Decomposition (Foundation)

Topic 42 of 313 • id: prime_factors_F

Q1. Write 184 as a product of its prime factors.

Q1

Divide 184 repeatedly by the smallest prime that fits: $2 \times 2 \times 2 \times 23$

$$184 = 2 \times 2 \times 2 \times 23$$

Answer: $2 \times 2 \times 2 \times 23$

Number › Factors, Multiples & Primes › Prime Factor Decomposition (Index Form) (Higher)

Topic 43 of 313 • id: prime_factors_H

Q1. Write 200 as a product of its prime factors, giving your answer in index form.

Q1

Divide 200 repeatedly by the smallest prime that fits: $2 \times 2 \times 2 \times 5 \times 5$

Write repeated factors using index notation: $200 = 2^3 \times 5^2$

Answer: $2^3 \times 5^2$

Number › Factors, Multiples & Primes › LCM by Listing (Foundation)

Topic 44 of 313 • id: lcm_by_listing_F

Find the lowest common multiple (LCM) of

Q1. 2 and 8.

Q1. 2 and 8.

Multiples of 2: 2, 4, 6, 8, 10, ...

Multiples of 8: 8, 16, 24, 32, 40, ...

Lowest common multiple = 8

Answer: 8

Number › Factors, Multiples & Primes › HCF by Listing (Foundation)

Topic 45 of 313 • id: hcf_by_listing_F

Find the highest common factor (HCF) of

Q1. 52 and 27.

Q1. 52 and 27.

Factors of 52: 1, 2, 4, 13, 26, 52

Factors of 27: 1, 3, 9, 27

Highest common factor = 1

Answer: 1

Number › Factors, Multiples & Primes › HCF & LCM using Prime Factors (Higher)

Topic 46 of 313 • id: hcf_lcm_by_prime_factors_H

Using prime factorisation, find the

Q1. HCF of 134 and 55.

Q1. HCF of 134 and 55.

$$134 = 2 \times 67$$

$$55 = 5 \times 11$$

HCF: take the lowest power of each prime that appears in both numbers

$$\text{HCF} = 1$$

Answer: 1

Number › Powers, Roots & Indices › Powers & Indices (Foundation)

Topic 47 of 313 • id: powers_F

Q1. Simplify $(4^3)^3$, giving your answer as a single power of 4.

Q1

Multiply the powers: $3 \times 3 = 9$

$$(4^3)^3 = 4^9$$

Answer: 4^9

Number › Powers, Roots & Indices › Negative Indices (Foundation)

Topic 48 of 313 • id: negative_indices_F

Q1. Work out 9^{-3} . Give your answer as a fraction.

Q1

A negative index means take the reciprocal: $9^{-3} = \frac{1}{9^3}$

$$9^3 = 729$$

$$9^{-3} = \frac{1}{729}$$

Answer: $\frac{1}{729}$

Number › Powers, Roots & Indices › Negative & Fractional Indices (Higher)

Topic 49 of 313 • id: powers_H

Q1. Work out 6^{-2} . Give your answer as a fraction.

Q1

$$6^{-2} = \frac{1}{6^2} = \frac{1}{36}$$

Answer: $\frac{1}{36}$

Number › Powers, Roots & Indices › Simplifying Indices (Challenging) (Higher)

Topic 50 of 313 • id: simplifying_indices_challenging_H

Q1. Simplify $(3^1)^2 \div 3^3$, giving your answer as a fraction where necessary.

Q1

Multiply the powers in the bracket: $1 \times 2 = 2$

$$(3^1)^2 = 3^2$$

Subtract the powers to divide: $2 - (3) = -1$

$$3^2 \div 3^3 = 3^{-1} = \frac{1}{3}$$

Answer: $\frac{1}{3}$

Number › Powers, Roots & Indices › Solving Index Equations (Higher)

Topic 51 of 313 • id: indices_common_base_equations_H

Q1. Solve $9^x = 729$.

Q1

Write both sides as powers of 3: $9 = 3^2$ and $729 = 3^6$

$9^x = 729$ becomes $(3^2)^x = 3^6$, i.e. $3^{(2x)} = 3^6$

Since the bases match, the powers must be equal: $2x = 6$

$$x = \frac{6}{2} = 3$$

Answer: 3

Number › Powers, Roots & Indices › Square & Cube Roots (Foundation)

Topic 52 of 313 • id: roots_F

Q1. Calculate the cube root of 729.

Q1

$9^3 = 729$, so the cube root of 729 = 9

Answer: 9

Number › Powers, Roots & Indices › Simplifying Surds (Higher)

Topic 53 of 313 • id: roots_H

Q1. Simplify $\sqrt{125}$, giving your answer in the form $a\sqrt{b}$.

Q1

Find the largest square factor of 125: $5^2 \times 5 = 125$

$$\sqrt{125} = \sqrt{(5^2 \times 5)} = 5\sqrt{5}$$

Answer: $5\sqrt{5}$

Number › Powers, Roots & Indices › Multiplying and Dividing Surds (Higher)

Topic 54 of 313 • id: surds_multiply_divide_H

Q1. Evaluate $\sqrt{20} \div \sqrt{5}$, giving your answer as an integer.

Q1

$$\sqrt{20} \div \sqrt{5} = \sqrt{\left(\frac{20}{5}\right)} = \sqrt{4}$$

$$\sqrt{4} = 2$$

Answer: 2

Number › Powers, Roots & Indices › Surds in Algebraic Expressions (Higher)

Topic 55 of 313 • id: algebraic_surds_H

Q1. Expand and simplify $(7 - \sqrt{2})(2 + \sqrt{2})$, giving your answer in the form $p + q\sqrt{2}$.

Q1

Expand the brackets (FOIL): $(7 - \sqrt{2})(2 + \sqrt{2}) = 14 + 7\sqrt{2} - 2\sqrt{2} - 2$

$\sqrt{2} \times \sqrt{2} = 2$, so the last term is a whole number: -2

Combine the whole-number terms: $14 - 2 = 12$

Combine the $\sqrt{2}$ terms: $+7\sqrt{2} - 2\sqrt{2} = +5\sqrt{2}$

$(7 - \sqrt{2})(2 + \sqrt{2}) = 12 + 5\sqrt{2}$

Answer: $12 + 5\sqrt{2}$

Number › Powers, Roots & Indices › Rationalising the Denominator (Higher)

Topic 56 of 313 • id: rationalise_denominator_H

Q1. Rationalise the denominator of $\frac{18}{\sqrt{3}}$, giving your answer in its simplest form.

Q1

Multiply the top and bottom by $\sqrt{3}$: $\frac{18}{\sqrt{3}} = \frac{18\sqrt{3}}{3}$

Simplify $\frac{18}{3}$ by dividing by 3: $6\sqrt{3}$

Answer: $6\sqrt{3}$

Algebra › Expressions, Formulae, Equations & Identities

› Expressions, Formulae, Equations and Identities (Foundation)

Topic 57 of 313 • id: classify_expressions_F

Is the following an expression, equation, formula, or identity?

Q1. $-7x + 7 = -13$

Q1. $-7x + 7 = -13$

Look at the statement: $-7x + 7 = -13$.

There IS an '=' sign here, so it's a claim that two things are equal - but it's only true for one particular value of x : solving it gives $x = \frac{20}{7}$.

Because it is only true for a specific value (or values) of x rather than for every possible x , this is an equation to be solved, not an identity.

Answer: equation

Algebra › Solving Linear Equations › One-Step Equations (Foundation)

Topic 58 of 313 • id: linear_one_step_F

Q1. Solve: $5x = 35$

Q1

$$5x = 35$$

Divide both sides by 5:

$$x = 7$$

Answer: 7

Algebra › Solving Linear Equations › Two-Step Equations (Foundation)

Topic 59 of 313 • id: linear_two_step_F

Q1. Solve: $3x + 15 = 45$

Q1

$$3x + 15 = 45$$

Subtract 15 from both sides:

$$3x = 30$$

Divide both sides by 3:

$$x = 10$$

Answer: 10

Algebra › Solving Linear Equations › Multi-Step Equations (Foundation)

Topic 60 of 313 • id: linear_multi_step_F

Q1. Solve: $3x + 10 + 6x - 9 = 91$

Q1

Collect like terms: $3x + 6x = 9x$, $10 - 9 = 1$

$$9x + 1 = 91$$

Subtract 1 from both sides:

$$9x = 90$$

Divide both sides by 9:

$$x = 10$$

Answer: 10

Algebra › Solving Linear Equations › Unknowns on Both Sides (Foundation)

Topic 61 of 313 • id: linear_both_sides_F

Q1. Solve: $2x + 9 = 5x - 21$

Q1

$$2x + 9 = 5x - 21$$

Subtract $5x$ from both sides:

$$-3x + 9 = -21$$

Subtract 9 from both sides:

$$-3x = -30$$

Divide both sides by -3 :

$$x = 10$$

Answer: 10

Algebra › Solving Linear Equations › Unknowns on Both Sides (Higher)

Topic 62 of 313 • id: linear_both_sides_H

Q1. Solve: $7x - 9 = x - 21$

Q1

$$7x - 9 = x - 21$$

Subtract $1x$ from both sides:

$$6x - 9 = -21$$

Add 9 to both sides:

$$6x = -12$$

Divide both sides by 6:

$$x = -2$$

Answer: -2

Algebra › Solving Linear Equations › Equations with Brackets (Foundation)

Topic 63 of 313 • id: linear_brackets_F

Q1. Solve: $4(3x + 4) = 100$

Q1

Expand the bracket: $4(3x + 4) = 12x + 16$

$$12x + 16 = 100$$

Subtract 16 from both sides:

$$12x = 84$$

Divide both sides by 12:

$$x = 7$$

Answer: 7

Algebra › Solving Linear Equations › Equations with Brackets (Higher)

Topic 64 of 313 • id: linear_brackets_H

Q1. Solve: $3(4x + 6) = 42$

Q1

Expand the bracket: $3(4x + 6) = 12x + 18$

$$12x + 18 = 42$$

Subtract 18 from both sides:

$$12x = 24$$

Divide both sides by 12:

$$x = 2$$

Answer: 2

Algebra › Solving Linear Equations › Equations Involving Fractions (Foundation)

Topic 65 of 313 • id: fractional_equations_F

Solve the following equation.

Q1. $\frac{x}{2} = 9$

Q1. $\frac{x}{2} = 9$

Multiply both sides by 2: $x = 9 \times 2$

$$x = 18$$

Answer: $x = 18$

Algebra › Solving Linear Equations › Equations Involving Fractions (Higher)

Topic 66 of 313 • id: fractional_equations_H

Solve the following equation.

Q1. $\frac{x}{2} + \frac{x}{6} = 5$

Q1. $\frac{x}{2} + \frac{x}{6} = 5$

Multiply every term by the LCM of 2 and 6, which is 6:

$$3x + 1x = 30$$

$$4x = 30$$

$$x = \frac{15}{2}$$

Answer: $x = \frac{15}{2}$

Algebra › Solving Linear Equations › Cross Multiplication (Foundation)

Topic 67 of 313 • id: cross_multiplication_F

Solve the following equation.

Q1. $\frac{7}{x} = \frac{2}{4}$

Q1. $\frac{7}{x} = \frac{2}{4}$

Cross-multiply: $7 \times 4 = 2 \times x$

$$x = 14$$

Answer: $x = 14$

Algebra › Solving Linear Equations › Cross Multiplication (Higher)

Topic 68 of 313 • id: cross_multiplication_H

Solve the following equation.

Q1. $\frac{x-1}{6} = \frac{2x-1}{4}$

Q1. $\frac{x-1}{6} = \frac{2x-1}{4}$

Cross-multiply: $4(x-1) = 6(2x-1)$

$$4x - 4 = 12x - 6$$

Subtract $12x$ from both sides:

$$-8x - 4 = -6$$

Add 4 to both sides:

$$-8x = -2$$

Divide both sides by -8:

$$x = \frac{1}{4}$$

Answer: $x = \frac{1}{4}$

Algebra › Forming and Solving Equations › Forming and Solving Equations (Foundation)

Topic 69 of 313 • id: forming_equations_F

Q1. I think of a number, multiply it by 3 and add 20. The result is 62. Form an equation and solve it to find the number.

Q1

Let the number be x .

$$3x + 20 = 62$$

Subtract 20 from both sides:

$$3x = 42$$

Divide both sides by 3:

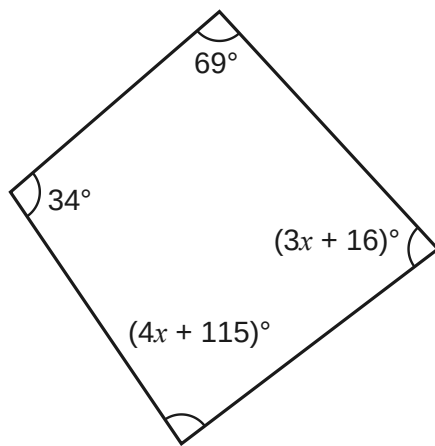
$$x = 14$$

Answer: 14

Algebra › Forming and Solving Equations › Forming and Solving Equations (Higher)

Topic 70 of 313 • id: forming_equations_H

Q1. A quadrilateral has angles 34° , 69° , $(3x + 16)^\circ$, and $(4x + 115)^\circ$. Form an equation and solve it to find x .



Q1

Angles in a quadrilateral sum to 360° .

$$34 + 69 + (3x + 16) + (4x + 115) = 360$$

Collect the x -terms and constants together:

$$7x + 131 = 257$$

Subtract 131 from both sides:

$$7x = 126$$

Divide both sides by 7:

$$x = 18$$

Answer: 18

Algebra › Changing the Subject of a Formula › Changing the Subject of a Formula (Foundation)

Topic 71 of 313 • id: change_subject_F

Q1. Make x the subject of the formula $y = 2x + 7$.

Q1

$$y = 2x + 7$$

Subtract 7 from both sides: $y - 7 = 2x$

Divide both sides by 2: $x = \frac{y - 7}{2}$

Answer: $x = \frac{y - 7}{2}$

Algebra › Changing the Subject of a Formula › Changing the Subject of a Formula (Higher)

Topic 72 of 313 • id: change_subject_H

Q1. Make x the subject of the formula $y = 6x + 8x + 14$.

Q1

$$y = 6x + 8x + 14$$

Subtract 14 from both sides: $y - 14 = 6x + 8x$

Factorise the right-hand side: $y - 14 = (14)x$

Divide both sides by 14: $x = \frac{y - 14}{14}$

Answer: $x = \frac{y - 14}{14}$

Algebra › Changing the Subject of a Formula › Changing the Subject by Factorising (Higher)

Topic 73 of 313 • id: change_subject_factorise_H

Make x the subject of the formula

Q1. $wx + 12x = 14$.

Q1. $wx + 12x = 14$.

$$wx + 12x = 14$$

Factorise the left-hand side: $x(w + 12) = 14$

Divide both sides by $(w + 12)$: $x = \frac{14}{w + 12}$

Answer: $x = \frac{14}{w + 12}$

Algebra › Substitution into Formulae › Substituting into Formulae (Foundation)

Topic 74 of 313 • id: substitution_F

Q1. $A = LW$. Find the value of A when $L = 15$ and $w = 4$.

Q1

$$A = LW$$

$$A = 15 \times 4$$

$$A = 60$$

Answer: $A = 60$

Algebra › Substitution into Formulae › Substituting into Formulae (Higher)

Topic 75 of 313 • id: substitution_H

Q1. $a = \frac{v-u}{t}$. Find the value of a when $u = 47$, $v = 54$ and $t = 7$.

Q1

$$a = \frac{v-u}{t}$$

$$a = \frac{54-47}{7}$$

$$a = \frac{7}{7}$$

$$a = 1$$

Answer: $a = 1$

Algebra › Substitution into Formulae › Rearranging and Substituting into Formulae (Foundation)

Topic 76 of 313 • id: substitution_rearrange_F

Q1. $v = u + at$. Make a the subject of the formula, then find the value of a when $v = 55$, $u = 10$ and $t = 5$.

Q1

$$v = u + at$$

Rearrange to make a the subject: $a = \frac{v - u}{t}$

Substitute $v = 55$, $u = 10$ and $t = 5$: $a = \frac{55 - 10}{5}$

$$a = \frac{45}{5} = 9$$

Answer: $a = 9$

Algebra › Substitution into Formulae › Rearranging and Substituting into Formulae (Higher)

Topic 77 of 313 • id: substitution_rearrange_H

Q1. $a = \frac{v-u}{t}$. Make u the subject of the formula, then find the value of u when $v = 41$, $a = -1$ and $t = 8$.

Q1

$$a = \frac{v-u}{t}$$

Rearrange to make u the subject: $u = v - at$

Substitute $v = 41$, $a = -1$ and $t = 8$: $u = 41 - (-1) \times 8$

$$u = 49$$

Answer: $u = 49$

Algebra › Expanding Brackets › Single Bracket (Foundation)

Topic 78 of 313 • id: expand_single_bracket_F

Q1. Expand: $8(-5x - 3)$

Q1

Multiply 8 by $-5x$: $-40x$

Multiply 8 by -3 : -24

Combine: $-40x - 24$

Answer: $-40x - 24$

Algebra › Expanding Brackets › Double Brackets (Foundation)

Topic 79 of 313 • id: expand_double_brackets_F

Expand and simplify:

Q1. $(4x + 7)(2x + 5)$

Q1. $(4x + 7)(2x + 5)$

First: $4x \times 2x = 8x^2$

Outer: $4x \times 5 = 20x$

Inner: $7 \times 2x = 14x$

Last: $7 \times 5 = 35$

Combine like terms: $8x^2 + 34x + 35$

Answer: $8x^2 + 34x + 35$

Algebra › Expanding Brackets › Double Brackets (No Coefficient) (Foundation)

Topic 80 of 313 • id: expand_double_brackets_no_coefficient_F

Expand and simplify:

Q1. $(x + 7)(x + 9)$

Q1. $(x + 7)(x + 9)$

First: $x \times x = x^2$

Outer: $x \times 9 = 9x$

Inner: $7 \times x = 7x$

Last: $7 \times 9 = 63$

Combine like terms: $x^2 + 16x + 63$

Answer: $x^2 + 16x + 63$

Algebra › Expanding Brackets › Double Brackets (Higher)

Topic 81 of 313 • id: expand_double_brackets_H

Expand and simplify:

Q1. $(x - 3)(4x + 3)$

Q1. $(x - 3)(4x + 3)$

First: $1x \times 4x = 4x^2$

Outer: $1x \times 3 = 3x$

Inner: $-3 \times 4x = -12x$

Last: $-3 \times 3 = -9$

Combine like terms: $4x^2 - 9x - 9$

Answer: $4x^2 - 9x - 9$

Algebra › Expanding Brackets › Triple Brackets (Higher)

Topic 82 of 313 • id: expand_triple_brackets_H

Expand and simplify:

Q1. $(3x - 2)(4x - 3)(4x + 4)$

Q1. $(3x - 2)(4x - 3)(4x + 4)$

First expand two of the brackets: $(3x - 2)(4x - 3) = 12x^2 - 17x + 6$

Multiply the result by $(4x + 4)$ and collect like terms

$$= 48x^3 - 20x^2 - 44x + 24$$

Answer: $48x^3 - 20x^2 - 44x + 24$

Algebra › Expanding Brackets › Triple Brackets (no coefficient) (Higher)

Topic 83 of 313 • id: expand_triple_brackets_no_coefficient_H

Expand and simplify:

Q1. $(x + 1)(x - 3)(x - 1)$

Q1. $(x + 1)(x - 3)(x - 1)$

First expand two of the brackets: $(x + 1)(x - 3) = x^2 - 2x - 3$

Multiply the result by $(x - 1)$ and collect like terms

$$= x^3 - 3x^2 - x + 3$$

Answer: $x^3 - 3x^2 - x + 3$

Algebra › Factorising › Common Factor (Foundation)

Topic 84 of 313 • id: factorise_common_factor_F

Q1. Factorise: $-12x - 18$

Q1

Find the highest common factor of -12 and -18: 6

Factorise: $6(-2x - 3)$

Answer: $6(-2x - 3)$

Algebra › Factorising › Quadratics (Foundation)

Topic 85 of 313 • id: factorise_quadratics_F

Q1. Factorise: $x^2 + 4x + 3$

Q1

Find two numbers that multiply to 3 and add to 4: 3 and 1

Write as two brackets: $(x + 3)(x + 1)$

Answer: $(x + 3)(x + 1)$

Algebra › Factorising › Quadratics (Higher)

Topic 86 of 313 • id: factorise_quadratics_H

Q1. Factorise: $x^2 + 10x + 16$

Q1

Find two numbers that multiply to 16 and add to 10: 2 and 8

Write as two brackets: $(x + 2)(x + 8)$

Answer: $(x + 2)(x + 8)$

Algebra › Solving Quadratic Equations › Solving by Factorising (Foundation)

Topic 87 of 313 • id: solve_quadratic_factorising_F

Q1. Solve $x^2 + 10x + 16 = 0$ by factorising.

Q1

Find two numbers that multiply to 16 and add to 10: 8 and 2

Factorise: $(x + 8)(x + 2) = 0$

Either $x + 8 = 0$ or $x + 2 = 0$

$x = -8$ or $x = -2$

Answer: $x = -8$ or $x = -2$

Algebra › Solving Quadratic Equations › Solving by Factorising (Higher)

Topic 88 of 313 • id: solve_quadratic_factorising_H

Q1. Solve $x^2 - 9x + 20 = 0$ by factorising.

Q1

Find two numbers that multiply to 20 and add to -9: -4 and -5

Factorise: $(x - 4)(x - 5) = 0$

Either $x - 4 = 0$ or $x - 5 = 0$

$x = 4$ or $x = 5$

Answer: $x = 4$ or $x = 5$

Algebra › Solving Quadratic Equations › Solving by Completing the Square (Higher)

Topic 89 of 313 • id: solve_quadratic_completing_square_H

Q1. Solve $x^2 + 18x + 70 = 0$ by completing the square, giving your answers in exact surd form.

Q1

Half the coefficient of x : $18 \div 2 = 9$

Complete the square: $x^2 + 18x + 70 = (x + 9)^2 - 11$

Set $(x + 9)^2 - 11 = 0$, so $(x + 9)^2 = 11$

$(x + 9) = \pm\sqrt{11}$

$x = -9 + \sqrt{11}$ or $x = -9 - \sqrt{11}$

Answer: $x = -9 + \sqrt{11}$ or $x = -9 - \sqrt{11}$

Algebra › Solving Quadratic Equations › The Quadratic Formula (Higher)

Topic 90 of 313 • id: quadratic_formula_H

Q1. Solve $3x^2 + 13x + 13 = 0$, giving your answer in the form $x = (-b \pm k\sqrt{m})/2a$, fully simplified.

Q1

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$$

$$a = 3, b = 13, c = 13$$

$$x = \frac{-13 \pm \sqrt{13^2 - 4 \times 3 \times 13}}{6}$$

$$x = \frac{-13 \pm \sqrt{13}}{6}$$

$$\sqrt{13} = \sqrt{1^2 \times 13} = \sqrt{13}$$

$$x = \frac{-13 \pm \sqrt{13}}{6}$$

$$\text{Answer: } x = \frac{-13 \pm \sqrt{13}}{6}$$

Algebra › Algebraic Indices › Laws of Indices with Algebra (Foundation)

Topic 91 of 313 • id: algebraic_indices_F

Q1. Simplify $4x^6 \times 6x^6$

Q1

Multiplying powers of the same base: add the exponents, and multiply the coefficients.

$$4x^6 \times 6x^6 = (4 \times 6) x^{(6+6)}$$

$$= 24x^{12}$$

Answer: $24x^{12}$

Algebra › Algebraic Indices › Laws of Indices with Algebra (Negative and Fractional) (Higher)

Topic 92 of 313 • id: algebraic_indices_H

Q1. Simplify $(4x^5)^3$

Q1

Raising a power to another power: multiply the exponents (this still works when the inner exponent is negative or zero), and raise the coefficient to that power too.

$$(4x^5)^3 = 4^3 x^{(5 \times 3)}$$

$$= 64x^{15}$$

Answer: $64x^{15}$

Algebra › Completing the Square › Completing the Square (Higher)

Topic 93 of 313 • id: completing_the_square_H

Q1. Write $x^2 + 16x - 1$ in the form $(x + p)^2 + q$.

Q1

Half the coefficient of x : $16 \div 2 = 8$

$$(x + 8)^2 = x^2 + 16x + 64$$

$$x^2 + 16x - 1 = (x + 8)^2 + (-1 - 64) = (x + 8)^2 - 65$$

Answer: $(x + 8)^2 - 65$

Algebra › Turning Point of a Graph › Turning Point of a Graph (Higher)

Topic 94 of 313 • id: turning_point_of_graph_H

Q1. Find the coordinates of the turning point of the curve $y = x^2 - 8x + 8$.

Q1

Complete the square: $x^2 - 8x + 8 = (x - 4)^2 - 8$

The turning point of $y = (x + p)^2 + q$ is $(-p, q)$.

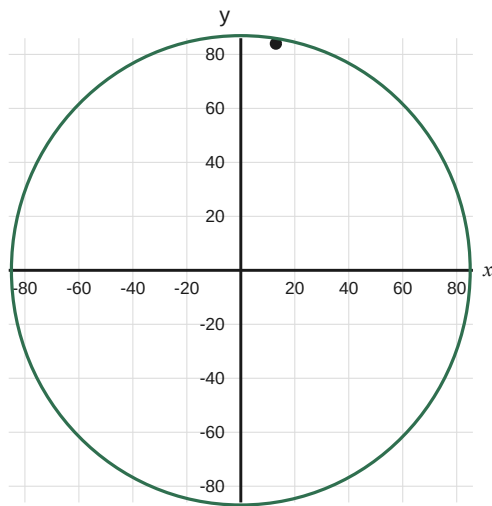
Turning point = $(4, -8)$

Answer: $(4, -8)$

Algebra › Equation of a Circle › Equation of a Circle (Higher)

Topic 95 of 313 • id: circle_equation_H

Q1. The point (13, 84) lies on the circle $x^2 + y^2 = 7225$. Find the equation of the tangent to the circle at this point.



Q1

The point (13, 84) lies on the circle $x^2 + y^2 = 7225$, since $13^2 + 84^2 = 7225$.

The radius to (13, 84) has gradient $\frac{84}{13}$, so the tangent (perpendicular to the radius) has gradient $-\frac{13}{84}$.

The tangent line through (13, 84) with that gradient simplifies to $13x + 84y = 7225$.

Answer: $13x + 84y = 7225$

Algebra › Functions › Function Notation (Foundation)

Topic 96 of 313 • id: functions_evaluate_F

Q1. $f(x) = 4x - 2$. Find the value of x for which $f(x) = 10$.

Q1

$$4x - 2 = 10$$

Add 2 to both sides:

$$4x = 12$$

Divide both sides by 4:

$$x = 3$$

Answer: 3

Algebra › Functions › Composite and Inverse Functions (Higher)

Topic 97 of 313 • id: functions_composite_inverse_H

Q1. $f(x) = -2x + 6$ and $g(x) = -5x + 1$. Find $fg(x)$.

Q1

$$fg(x) = f(-5x + 1)$$

$$= -2(-5x + 1) + 6$$

$$= 10x + 4$$

Answer: $10x + 4$

Algebra › Functions › Evaluating Inverse Functions (Higher)

Topic 98 of 313 • id: functions_inverse_evaluate_H

Q1. $f(x) = 4x + 1$. Find $f^{-1}(0)$.

Q1

$$f^{-1}(x) = \frac{x-1}{4}$$

$$f^{-1}(0) = \frac{0-1}{4}$$

$$f^{-1}(0) = -\frac{1}{4}$$

Answer: $-\frac{1}{4}$

Algebra › Algebraic Fractions › Adding and Subtracting Algebraic Fractions (Higher)

Topic 99 of 313 • id: algebraic_fractions_add_subtract_H

Q1. Simplify $\frac{9}{x-4} - \frac{2}{x-1}$, giving your answer as a single fraction.

Q1

Write both fractions over the common denominator $(x-4)(x-1)$:

$$\frac{9}{x-4} - \frac{2}{x-1} = \frac{9(x-1) - 2(x-4)}{(x-4)(x-1)}$$

Expand the numerator: $9(x-1) - 2(x-4) = 7x - 1$

$$= \frac{7x-1}{(x-4)(x-1)}$$

Answer: $\frac{7x-1}{(x-4)(x-1)}$

Algebra › Algebraic Fractions › Multiplying and Dividing Algebraic Fractions (Higher)

Topic 100 of 313 • id: algebraic_fractions_multiply_divide_H

Q1. Simplify $\frac{x^2 - 16}{x - 2} \times \frac{x - 2}{x - 4}$, giving your answer in its simplest form.

Q1

Factorise the difference of two squares: $x^2 - 16 = (x - 4)(x + 4)$

$$\frac{(x - 4)(x + 4)}{x - 2} \times \frac{x - 2}{x - 4}$$

Cancel the common factors $x - 2$ and $x - 4$:

$$= x + 4$$

Answer: $x + 4$

Algebra › Algebraic Fractions › Fractional Equations (Unknown in the Denominator) (Higher)

Topic 101 of 313 • id: fractional_equations_advanced_H

Solve the following equation, giving your answers to 2 decimal places where necessary.

Q1. $\frac{8}{x-1} = \frac{x+2}{5}$

Q1. $\frac{8}{x-1} = \frac{x+2}{5}$

Multiply both sides by the denominator(s) to clear the fractions.

This rearranges to $x^2 + x - 42 = 0$.

Solving the quadratic: $x = -7$ or $x = 6$.

Answer: $x = -7$ or $x = 6$

Algebra › Simultaneous Equations › Common Coefficient (Foundation)

Topic 102 of 313 • id: simultaneous_common_coefficient_F

Solve the simultaneous equations:

Q1. $2x + 4y = -18$ and $4x + 4y = -8$

Q1. $2x + 4y = -18$ and $4x + 4y = -8$

$$2x + 4y = -18 \quad (1)$$

$$4x + 4y = -8 \quad (2)$$

Subtract the equations to eliminate y : $-2x = -10$, so $x = 5$

Substitute $x = 5$ into (1): $4y = -28$, so $y = -7$

Answer: $x = 5$, $y = -7$

Algebra › Simultaneous Equations › Different Coefficients (Higher)

Topic 103 of 313 • id: simultaneous_different_coefficient_H

Solve the simultaneous equations:

Q1. $5x + 2y = 51$ and $4x + 4y = 48$

Q1. $5x + 2y = 51$ and $4x + 4y = 48$

$$5x + 2y = 51 \quad (1)$$

$$4x + 4y = 48 \quad (2)$$

$$(1) \times 4: 20x + 8y = 204 \quad (3)$$

$$(2) \times 2: 8x + 8y = 96 \quad (4)$$

$$(3) - (4): 12x = 108, \text{ so } x = 9$$

Substitute $x = 9$ into (1): $2y = 6$, so $y = 3$

Answer: $x = 9$, $y = 3$

Algebra › Simultaneous Equations › Forming and Solving (Foundation)

Topic 104 of 313 • id: simultaneous_forming_and_solving_F

Q1. 5 tickets and 5 programmes cost £6.10 in total. 5 tickets and 2 programmes cost £4.21 in total. Form and solve a pair of simultaneous equations to find the cost of one ticket and one programme.

Q1

Let x = cost of one ticket (pence), y = cost of one programme (pence).

$$5x + 5y = 610 \quad (1)$$

$$5x + 2y = 421 \quad (2)$$

Solving simultaneously: $x = 59$, $y = 63$

Answer: ticket = £0.59, programme = £0.63

Algebra › Simultaneous Equations › Quadratic Simultaneous Equations (Higher)

Topic 105 of 313 • id: simultaneous_quadratic_H

Solve the simultaneous equations:

Q1. $y = x^2 + 2x + 7$ and $y = -3x + 1$

Q1. $y = x^2 + 2x + 7$ and $y = -3x + 1$

Substitute $y = -3x + 1$ into $y = x^2 + 2x + 7$: $-3x + 1 = x^2 + 2x + 7$

Rearrange: $x^2 + 5x + 6 = 0$

Factorise: $(x + 3)(x + 2) = 0$

$x = -3$ or $x = -2$

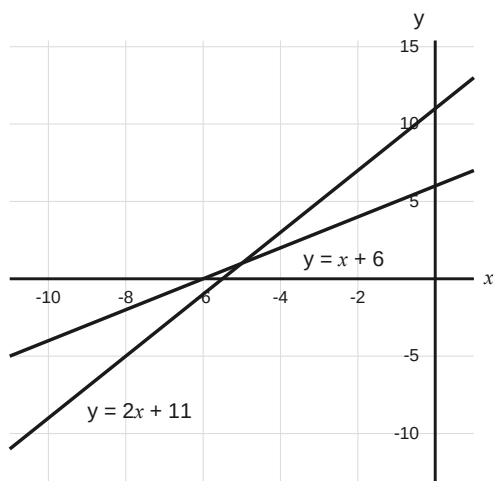
When $x = -3$, $y = 10$. When $x = -2$, $y = 7$.

Answer: (-3, 10) and (-2, 7)

Algebra › Simultaneous Equations › Solving Graphically (Foundation)

Topic 106 of 313 • id: simultaneous_graphically_F

Q1. The graphs of $y = x + 6$ and $y = 2x + 11$ are shown below. Use the graphs to write down the solution to the simultaneous equations.



Q1

The solution to a pair of simultaneous equations is the point where their graphs intersect.

The lines cross at $x = -5$, $y = 1$.

Answer: $x = -5$, $y = 1$

Algebra › Inequalities › Solving Linear Inequalities (Foundation)

Topic 107 of 313 • id: solving_inequalities_F

Solve the inequality:

Q1. $2x + 9 > 5$

Q1. $2x + 9 > 5$

$$2x + 9 > 5$$

Subtract 9 from both sides:

$$2x > -4$$

Divide both sides by 2:

$$x > -2$$

Answer: $x > -2$

Algebra › Inequalities › Solving Linear Inequalities (Higher)

Topic 108 of 313 • id: solving_inequalities_H

Solve the inequality:

Q1. $-6(x + 9) < 6$

Q1. $-6(x + 9) < 6$

Expand the bracket: $-6(x + 9) < 6$

$$-6x - 54 < 6$$

Add 54 to both sides:

$$-6x < 60$$

Divide both sides by -6 (dividing by a negative number reverses the inequality sign):

$$x > -10$$

Answer: $x > -10$

Algebra › Inequalities › Satisfying an Inequality (Foundation)

Topic 109 of 313 • id: satisfying_inequalities_F

Q1. Solve the inequality $-7 \leq 2x - 7 < 11$, then list the integer values of x that satisfy it.

Q1

$$-7 \leq 2x - 7 < 11$$

Add 7 to all three parts:

$$0 \leq 2x < 18$$

Divide all three parts by 2:

$$0 \leq x < 9$$

The integers satisfying this are: 0, 1, 2, 3, 4, 5, 6, 7, 8

Answer: 0, 1, 2, 3, 4, 5, 6, 7, 8

Algebra › Inequalities › Satisfying an Inequality (Higher)

Topic 110 of 313 • id: satisfying_inequalities_H

Q1. x satisfies both $2x - 8 \geq 3$ and $2x - 7 \leq 7$. List the integer values of x that satisfy both.

Q1

Solve the first inequality:

$$2x - 8 \geq 3$$

Add 8 to both sides:

$$2x \geq 11$$

Divide both sides by 2:

$$x \geq \frac{11}{2}$$

Solve the second inequality:

$$2x - 7 \leq 7$$

Add 7 to both sides:

$$2x \leq 14$$

Divide both sides by 2:

$$x \leq 7$$

Combine both results into a single range: $\frac{11}{2} \leq x \leq 7$

The integers satisfying both inequalities are: 6, 7

Answer: 6, 7

Algebra › Inequalities › Solving Quadratic Inequalities (Higher)

Topic 111 of 313 • id: quadratic_inequalities_H

Solve the inequality:

Q1. $x^2 + 6x < 0$

Q1. $x^2 + 6x < 0$

Factorise: $x^2 + 6x = (x)(x + 6)$

The critical values are $x = -6$ and $x = 0$.

The coefficient of x^2 is positive, so the graph is a U-shaped parabola, and < 0 is satisfied between the two roots.

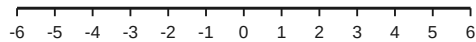
Answer: $-6 < x < 0$

Answer: $-6 < x < 0$

Algebra › Inequalities › Inequalities on a Number Line (Foundation)

Topic 112 of 313 • id: inequalities_number_line_F

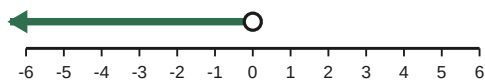
Q1. Draw the solution set of $x < 0$ on a number line.



Q1

Draw an open circle at 0 (the boundary value is not included).

Shade the line to the left of 0, with an arrow showing it continues indefinitely.

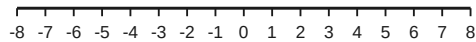


Answer: $x < 0$

Algebra › Inequalities › Compound Inequalities on a Number Line (Higher)

Topic 113 of 313 • id: inequalities_number_line_H

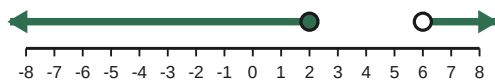
Q1. Draw the inequality/ies of $x \leq 2$ or $x > 6$ on a number line.



Q1

Draw a filled circle at 2 and an open circle at 6.

Shade to the left of 2 and to the right of 6, with arrows showing both continue indefinitely.

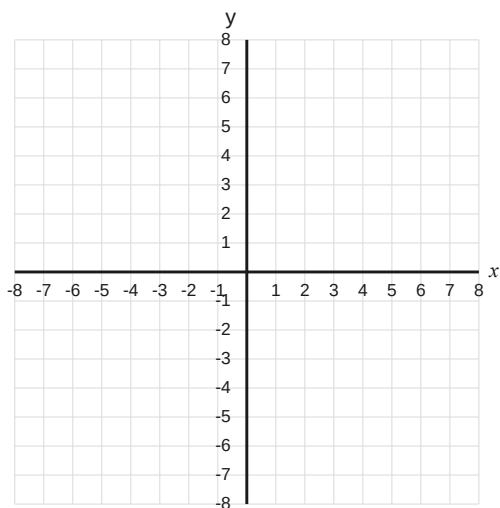


Answer: $x \leq 2$ or $x > 6$

Algebra › Inequalities › Shading Inequality Regions (Higher)

Topic 114 of 313 • id: inequalities_region_H

Q1. On the grid, shade the region that satisfies both $y \geq \frac{x}{2} + 1$ and $y \geq 2x - 2$.

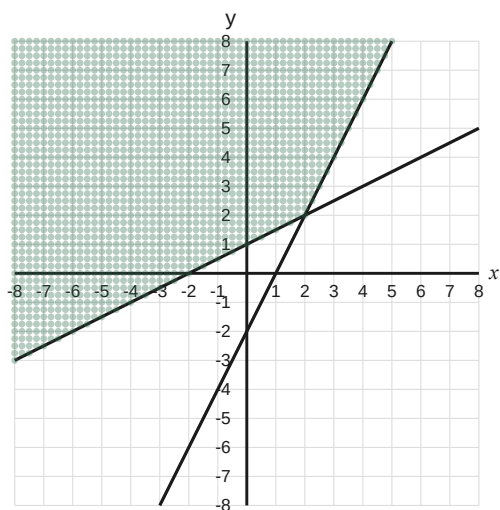


Q1

Draw the line $y = \frac{x}{2} + 1$ (solid - the line itself is included in the region).

Draw the line $y = 2x - 2$ (solid - the line itself is included in the region).

Shade the region where both $y \geq \frac{x}{2} + 1$ and $y \geq 2x - 2$ are true at the same time.



Answer: $y \geq \frac{x}{2} + 1$ and $y \geq 2x - 2$

Algebra › Algebraic Proof › Algebraic Proof (Higher)

Topic 115 of 313 • id: algebraic_proof_H

Q1. Prove algebraically that $(n + 1)^3 - n^3 - 1$ is always a multiple of 6.

Q1

Let n be any integer.

Write the given expression algebraically: $(n + 1)^3 - n^3 - 1$.

Expand and simplify: $(n + 1)^3 - n^3 - 1 = 3n^2 + 3n$.

Factorise/rearrange to expose the structure: $3n^2 + 3n = 3n(n + 1)$, and $n(n + 1)$ is always even, so $3n(n + 1)$ always carries a factor of 2 as well as its factor of 3.

$3n(n + 1)$ is clearly a multiple of 3. But $n(n + 1)$ is also always even (one of n , $n + 1$ is always even), so $3n(n + 1)$ is a multiple of $3 \times 2 = 6$ as well.

Answer: Proven: $(n + 1)^3 - n^3 - 1 = 3n(n + 1)$, always a multiple of 6.

Algebra › Sequences › Next Term (Foundation)

Topic 116 of 313 • id: sequences_next_term_F

Find the next term in the sequence:

Q1. 6, 10, 14, 18, 22, ...

Q1. 6, 10, 14, 18, 22, ...

Add 4 each time.

Next term = $22 + 4 = 26$

Answer: 26

Algebra › Sequences › Term-to-Term Rule (Foundation)

Topic 117 of 313 • id: sequences_term_to_term_rule_F

Q1. Here are the first four terms of a sequence: -1, -3, -9, -27. Describe the term-to-term rule, and find the next term.

Q1

Term-to-term rule: multiply by 3

Next term = -81

Answer: Rule: multiply by 3. Next term = -81

Algebra › Sequences › nth Term of a Linear Sequence (Foundation)

Topic 118 of 313 • id: sequences_nth_term_F

Q1. Here are the first four terms of a sequence: 8, 13, 18, 23.
Find an expression for the n th term.

Q1

Common difference = 5

n th term = $5n + 3$

n th term = $5n + 3$

Answer: $5n + 3$

Algebra › Sequences › nth Term of a Quadratic Sequence (Higher)

Topic 119 of 313 • id: sequences_quadratic_nth_term_H

Q1. Here are the first four terms of a quadratic sequence: -5, -9, -15, -23.
Find an expression for the n th term.

Q1

First differences: -4, -6, -8

Second difference is constant: -2, so the coefficient of n^2 is $-2 \div 2 = -1$

n th term = $-n^2 - n - 3$

Answer: $-n^2 - n - 3$

Algebra › Sequences › Special Sequences (Foundation)

Topic 120 of 313 • id: special_sequences_F

Q1. The first 4 triangular numbers are: 1, 3, 6, 10. Find the 6th term of the sequence.

Q1

These are the triangular numbers: the n th term is given by $\frac{n(n+1)}{2}$.

$$\text{Term } 6 = \frac{6(6+1)}{2} = \frac{42}{2} = 21$$

Answer: 21

Algebra › Sequences › Fibonacci-Type and Geometric Sequences (Higher)

Topic 121 of 313 • id: special_sequences_H

Q1. The first 5 terms of a Fibonacci-type sequence are: 8, 10, 18, 28, 46. Each term (from the 3rd onwards) is found by adding the two terms before it. Find the next two terms of the sequence.

Q1

Rule: each term (from the 3rd onwards) is found by adding the two terms before it.

$$28 + 46 = 74$$

$$46 + 74 = 120$$

Answer: 74, 120

Algebra › Iteration › Iteration (Higher)

Topic 122 of 313 • id: iteration_H

Q1. The iterative formula $x_{n+1} = \frac{17 - x_n^2}{9}$ is used with $x_0 = -1$. Find the values of x_1 , x_2 and x_3 , giving each answer to 3 decimal places.

Q1

Iterative formula: $x_{n+1} = \frac{17 - x_n^2}{9}$

$$x_0 = -1$$

$$x_1 = g(x_0) = \frac{17 - (-1)^2}{9} = 1.778$$

$$x_2 = g(x_1) = \frac{17 - 1.778^2}{9} = 1.538$$

$$x_3 = g(x_2) = \frac{17 - 1.538^2}{9} = 1.626$$

Answer: $x_1 = 1.778$, $x_2 = 1.538$, $x_3 = 1.626$

Algebra › Iteration › Trial and Improvement (Higher)

Topic 123 of 313 • id: trial_and_improvement_H

Q1. $f(x) = x^3 - 2x - 72$. Show that $f(x) = 0$ has a root between $x = 4$ and $x = 5$, then use trial and improvement to find this root correct to 1 decimal place.

Q1

$$f(x) = x^3 - 2x - 72$$

$$f(4) = -16.00 \text{ (negative)}$$

$$f(5) = 43.00 \text{ (positive)}$$

$$f(4.1) = -11.28 \text{ (negative)}$$

$$f(4.2) = -6.31 \text{ (negative)}$$

$$f(4.3) = -1.09 \text{ (negative)}$$

$$f(4.4) = 4.38 \text{ (positive)}$$

The sign changes between $x = 4.3$ and $x = 4.4$, so try the midpoint $x = 4.35$: $f(4.35) = 1.61$

$f(4.35)$ has the same sign as $f(4.4)$, so the root lies between 4.3 and 4.35 - to 1 decimal place, this rounds down to 4.3

Answer: $x = 4.3$

Algebra › Kinematics (SUVAT) › Kinematics (SUVAT) (Higher)

Topic 124 of 313 • id: kinematics_suvat_H

Formulae

$$v = u + at$$

$$s = ut + \left(\frac{1}{2}\right)at^2$$

$$v^2 = u^2 + 2as$$

Q1. An object's velocity increases from 25 m/s (u) to 19 m/s (v) over a time (t) of 6 s at a constant acceleration. Find this acceleration (a).

Q1

This question gives three of the four SUVAT quantities u, v, a and t (and asks for the fourth), so use $v = u + at$, the equation linking exactly those four.

$$v = u + at$$

Rearrange to make a the subject: $a = \frac{v - u}{t}$

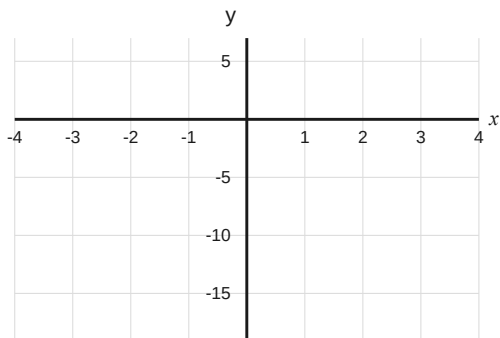
Substitute $u = 25$, $v = 19$, $t = 6$: $a = \frac{19 - 25}{6} = -\frac{6}{6} = -1$

Answer: -1 m/s²

Algebra › Plotting Graphs › Plotting Straight Line Graphs (Foundation)

Topic 125 of 313 • id: plot_straight_line_F

Q1. Complete a table of values for $y = 4x - 6$ for $x = -3$ to 3 , then plot the graph.



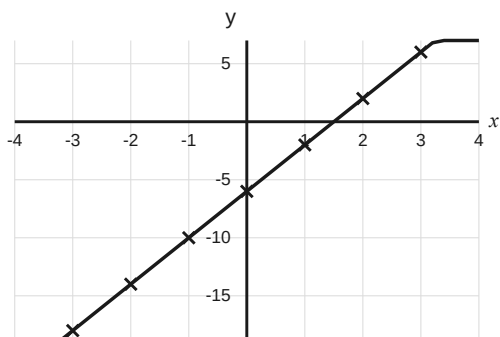
Q1

Table of values for $y = 4x - 6$:

x : -3, -2, -1, 0, 1, 2, 3

y : -18, -14, -10, -6, -2, 2, 6

Plot each (x, y) point and join them with a single ruled straight line.

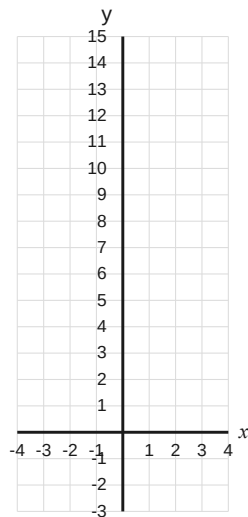


Answer: Straight line: $y = 4x - 6$

Algebra › Plotting Graphs › Plotting Quadratic Graphs (Foundation)

Topic 126 of 313 • id: plot_quadratic_F

Q1. Complete a table of values for $y = x^2 - 2x - 1$ for $x = -3$ to 3 , then plot the graph.



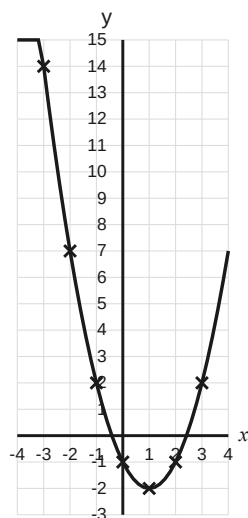
Q1

Table of values for $y = x^2 - 2x - 1$:

x : -3, -2, -1, 0, 1, 2, 3

y : 14, 7, 2, -1, -2, -1, 2

Plot each (x, y) point and join them with a single smooth curve.

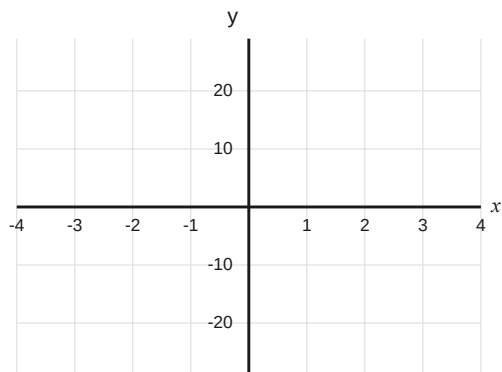


Answer: Quadratic curve: $y = x^2 - 2x - 1$

Algebra › Plotting Graphs › Plotting Cubic Graphs (Higher)

Topic 127 of 313 • id: plot_cubic_H

Q1. Complete a table of values for $y = -x^3$ for $x = -3$ to 3 , then plot the graph.



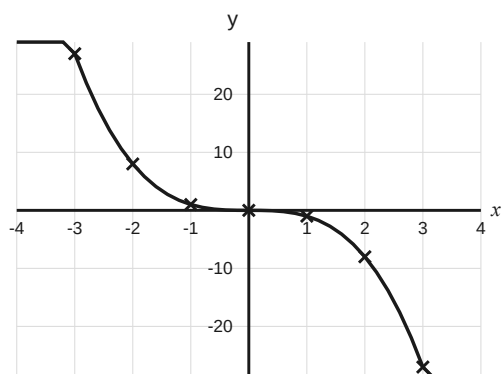
Q1

Table of values for $y = -x^3$:

x : -3, -2, -1, 0, 1, 2, 3

y : 27, 8, 1, 0, -1, -8, -27

Plot each (x, y) point and join them with a single smooth curve.

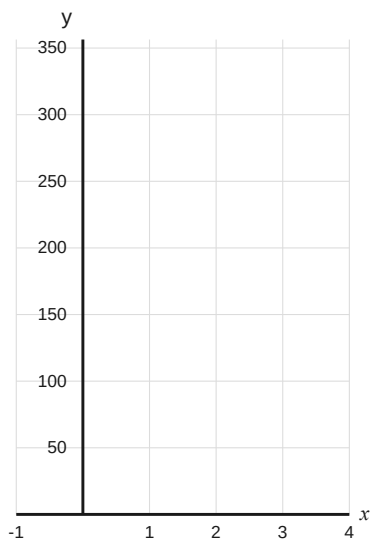


Answer: Cubic curve: $y = -x^3$

Algebra › Plotting Graphs › Plotting Exponential Graphs (Higher)

Topic 128 of 313 • id: plot_exponential_H

Q1. Complete a table of values for $y = 4 \times 3^x$ for $x = 0$ to 4, then plot the graph.



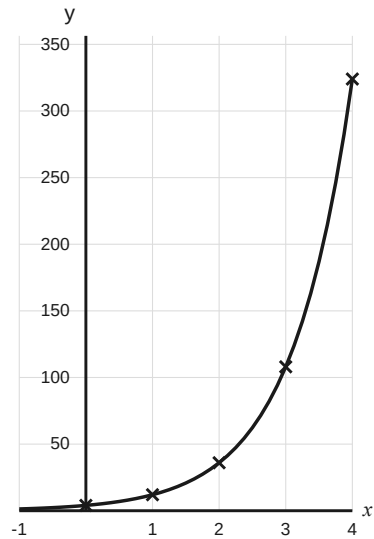
Q1

Table of values for $y = 4 \times 3^x$:

x : 0, 1, 2, 3, 4

y : 4, 12, 36, 108, 324

Plot each (x, y) point and join them with a single smooth curve, getting steeper as x increases and approaching (but never touching) the x -axis as x decreases.

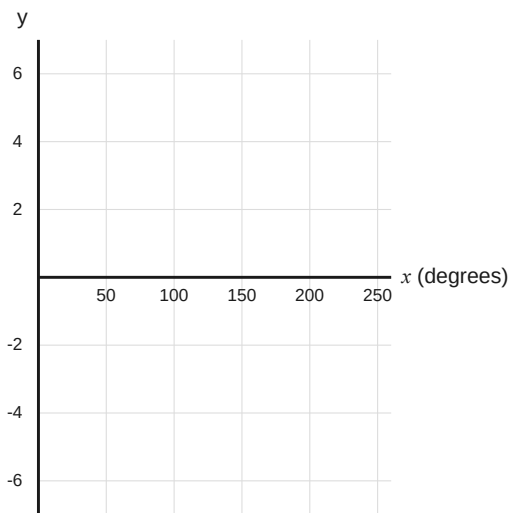


Answer: Exponential curve: $y = 4 \times 3^x$

Algebra › Plotting Graphs › Plotting Trigonometric Graphs (Higher)

Topic 129 of 313 • id: trig_graph_H

Q1. Complete a table of values for $y = -\tan(x^\circ)$ for $x = 100$ to 260 (in steps of 40°), then plot the graph.



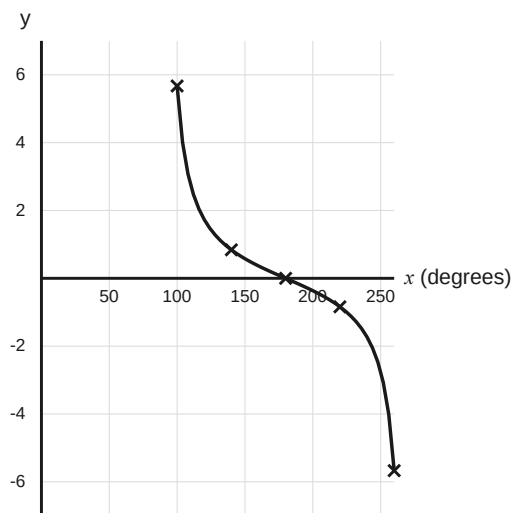
Q1

Table of values for $y = -\tan(x^\circ)$:

x : 100, 140, 180, 220, 260

y : 5.67, 0.84, 0.0, -0.84, -5.67

Plot each (x, y) point and join them with a single smooth curve - $y = -\tan(x^\circ)$ repeats the same shape every 180° , and rises/falls very steeply near the edges of this window (approaching, but never reaching, a vertical asymptote just outside it).

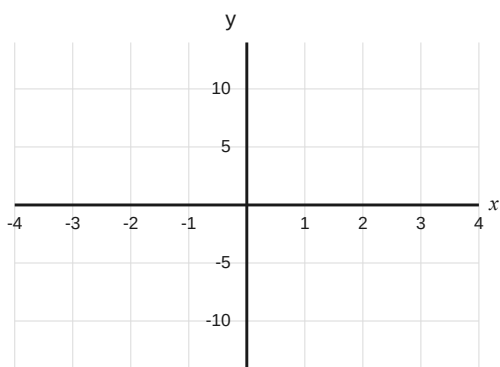


Answer: $y = -\tan(x^\circ)$

Algebra › Plotting Graphs › Plotting Reciprocal Graphs (Higher)

Topic 130 of 313 • id: plot_reciprocal_H

Q1. Complete a table of values for $y = \frac{12}{x}$ for $x = -4$ to 4 , then plot the graph.



Q1

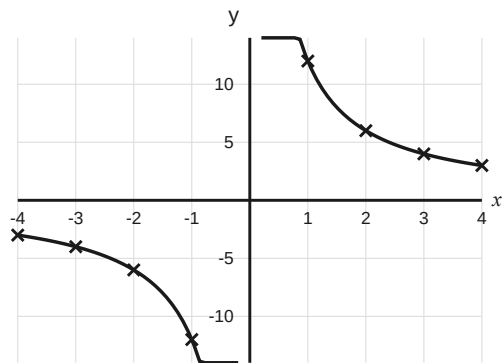
Table of values for $y = \frac{12}{x}$:

x : -4, -3, -2, -1, 1, 2, 3, 4

y : -3, -4, -6, -12, 12, 6, 4, 3

There is no value at $x = 0$: dividing by 0 is undefined, so leave that column out of the table.

Plot each (x, y) point. The graph has two separate curved branches and never touches $x = 0$.

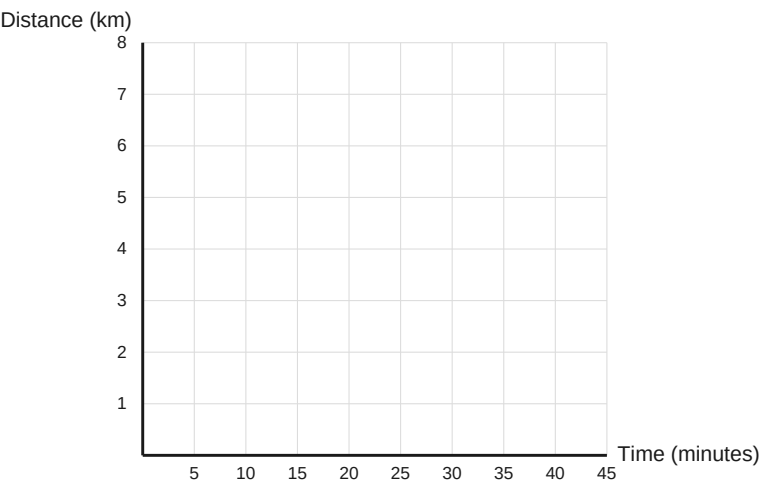


Answer: Reciprocal curve: $y = \frac{12}{x}$

Algebra › Plotting Graphs › Plotting Distance-Time Graphs (Foundation)

Topic 131 of 313 • id: plot_distance_time_F

Q1. A cyclist leaves home at 0 minutes. They cycle at a constant speed, reaching a shop 6 km away after 15 minutes. They stay at the shop until 25 minutes, then cycle straight back home at a constant speed, arriving home at 40 minutes. Draw a distance-time graph to show this journey, and find the total distance travelled.



Q1

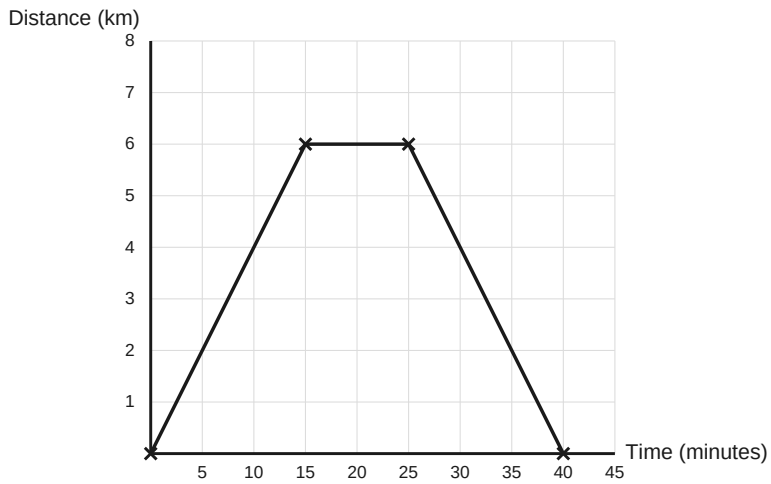
Set out at 0 minutes, 0 km. Travel at a constant speed, arriving at 6 km after 15 minutes.

Stay at 6 km until 25 minutes.

Return at a constant speed to 0 km, arriving home at 40 minutes.

Plot these points and join each stage with a straight line segment.

Total distance travelled = 6 km out + 6 km back = 12 km

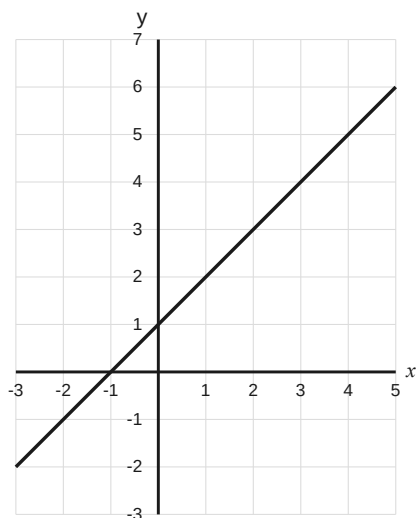


Answer: 12 km

Algebra › Equation of a Line › Equation of a Line from a Graph (Foundation)

Topic 132 of 313 • id: line_equation_from_graph_F

Q1. The graph shows a straight line. Find the equation of the line in the form $y = mx + c$.



Q1

$$\text{Gradient} = \text{rise/run} = \frac{5 - -1}{4 - -2} = 1$$

Using $y = mx + c$ with the point $(-2, -1)$: $-1 = 1 \times (-2) + c$, so $c = 1$

Equation of the line: $y = x + 1$

Answer: $y = x + 1$

Algebra › Equation of a Line › Parallel Lines (Foundation)

Topic 133 of 313 • id: parallel_lines_equation_F

Q1. A line is parallel to $y = 3x + 2$ and passes through the point $(-1, 4)$. Find the equation of the line.

Q1

Parallel lines have the same gradient, so $m = 3$.

Substitute $(-1, 4)$ into $y = 3x + c$: $4 = 3 \times (-1) + c$, so $c = 7$

Equation of the line: $y = 3x + 7$

Answer: $y = 3x + 7$

Algebra › Equation of a Line › Perpendicular Lines (Higher)

Topic 134 of 313 • id: perpendicular_lines_equation_H

Q1. A line is perpendicular to $y = -x + 1$ and passes through the point $(-2, -3)$. Find the equation of the line.

Q1

Perpendicular gradients multiply to give -1 : $m \times -1 = -1$, so $m = 1$

Substitute $(-2, -3)$ into $y = 1x + c$: $-3 = 1 \times (-2) + c$, so $c = -1$

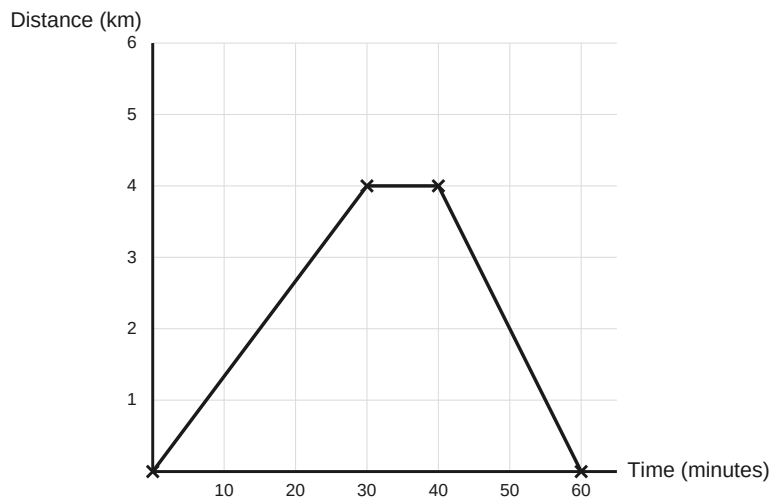
Equation of the line: $y = x - 1$

Answer: $y = x - 1$

Algebra › Real-Life Graphs › Interpreting Distance-Time Graphs (Foundation)

Topic 135 of 313 • id: distance_time_interpret_F

Q1. The distance-time graph shows a delivery van's journey. Find its speed during the outbound leg of the journey, in km/h.



Q1

Outbound: 4 km in 30 minutes = 4 km in $\frac{1}{2}$ hours

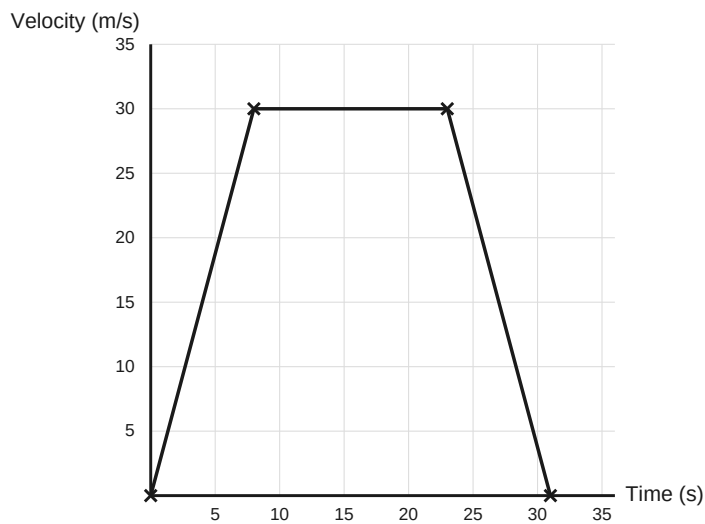
Speed = distance $\div$ time = $4 \div \frac{1}{2} = 8$ km/h

Answer: 8 km/h

Algebra › Real-Life Graphs › Interpreting Velocity-Time Graphs (Higher)

Topic 136 of 313 • id: velocity_time_interpret_H

Q1. The velocity-time graph shows a car's journey. Find the total distance travelled.



Q1

Total distance = total area under the graph (a trapezium)

$$\text{Distance} = \frac{1}{2} \times (31 + 15) \times 30 = 690 \text{ m}$$

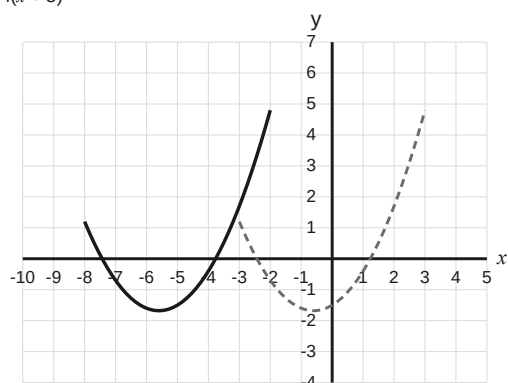
Answer: 690 m

Algebra › Transformations of Graphs › Transformations of Graphs (Higher)

Topic 137 of 313 • id: graph_transformations_H

Q1. The graph of $y = f(x)$ is transformed to give the graph of $y = f(x + 5)$. Describe this transformation fully.

$y = f(x)$
 $y = f(x + 5)$



Q1

$y = f(x + 5)$ corresponds to a translation by vector $(-5, 0)$.

Answer: Translation by vector $(-5, 0)$

Ratio & Proportion › Percentages › Percentage of an Amount (Foundation)

Topic 138 of 313 • id: percentage_of_amount_F

Q1. Find 20% of 200.

Q1

Convert 20% to a fraction: $\frac{20}{100}$

Multiply: $\frac{20}{100} \times 200 = 40$

Answer: 40

Ratio & Proportion › Percentages › Percentage Change (Foundation)

Topic 139 of 313 • id: percentage_change_F

Q1. Increase 240 by 20%.

Q1

Convert 20% to a multiplier: 1.20

Multiply: $240 \times 1.20 = 288$

Answer: 288

Ratio & Proportion › Percentages › Finding a Percentage Change (Foundation)

Topic 140 of 313 • id: find_percentage_change_F

Q1. The price of an item changes from £20 to £23. Find the percentage increase.

Q1

Find the difference between the two values: £23 and £20 differ by £3

Divide by the ORIGINAL value and convert to a percentage: $£3 \div £20 \times 100 = 15\%$

Answer: 15% increase

Ratio & Proportion › Percentages › Percentage Increase/Decrease (Calculator) (Higher)

Topic 141 of 313 • id: percentage_increase_decrease_calculator_H

Q1. Using a calculator, decrease £56.00 by 9.75%. Give your answer to 2 decimal places.

Q1

Convert 9.75% to a multiplier: 0.9025

Multiply: $£56.00 \times 0.9025 = £50.54$ (calculator, 2 d.p.)

Answer: £50.54

Ratio & Proportion › Percentages › Reverse Percentage (Foundation)

Topic 142 of 313 • id: reverse_percentage_F

Q1. After a 20% increase, an item costs £384. Find the original price.

Q1

Convert 20% to a multiplier: 1.20 (original $\times$ multiplier = new value)

Divide the new value by the multiplier: $384 \div 1.20 = 320$

Answer: £320

Ratio & Proportion › Percentages › Reverse Percentage (Higher)

Topic 143 of 313 • id: reverse_percentage_H

Q1. After a 12% decrease, an item costs £274.56. Find the original price.

Q1

Convert 12% to a multiplier: 0.88 (original $\times$ multiplier = new value)

Divide the new value by the multiplier: $274.56 \div 0.88 = 312$

Answer: £312

Ratio & Proportion › Percentages › Compound Percentage Change (Higher)

Topic 144 of 313 • id: compound_percentage_H

Q1. A price of £260 decreases by 60%, then increases by 60%. Find the final price.

Q1

After the first change: $260 \times 0.40 = 104$

After the second change: $104 \times 1.60 = 166.40$

Answer: £166.40

Ratio & Proportion › Percentages › Compound Percentage Change (Foundation)

Topic 145 of 313 • id: compound_percentage_F

Q1. A price of £40 increases by 25%, then increases by 50%. Find the final price.

Q1

After the first change: $40 \times 1.25 = 50$

After the second change: $50 \times 1.50 = 75$

Answer: £75

Ratio & Proportion › Percentages › Simple Interest (Foundation)

Topic 146 of 313 • id: simple_interest_F

Q1. £2700 is invested at a simple interest rate of 5% per year for 1 years. Find the total amount in the account after 1 years.

Q1

Simple interest formula: $I = P \times r \times t \div 100$

$$I = 2700 \times 5 \times 1 \div 100 = £135$$

$$\text{Total in the account} = P + I = £2700 + £135 = £2835$$

Answer: £2835

Ratio & Proportion › Percentages › Mixed Percentages Practice (Higher)

Topic 147 of 313 • id: mixed_percentages_H

Q1. Decrease 380 by 20%.

Q1

Convert 20% to a multiplier: 0.80

Multiply: $380 \times 0.80 = 304$

Answer: 304

Ratio & Proportion › Best Buys › Best Buys (Foundation)

Topic 148 of 313 • id: best_buys_F

Q1. A shop sells orange juice in different sizes: A: £12.50 for 1.25L; B: £25.20 for 1L; C: £22.80 for 1.5L. Which option is the better value for money? Show your working.

Q1

Work out the price per 100ml for each option:

A: 1.25L = 1250ml

A: $£12.50 \div 1250 \times 100 = 100.00\text{p}$ per 100ml

B: 1L = 1000ml

B: $£25.20 \div 1000 \times 100 = 252.00\text{p}$ per 100ml

C: 1.5L = 1500ml

C: $£22.80 \div 1500 \times 100 = 152.00\text{p}$ per 100ml

The lowest price per 100ml is A at 100.00p per 100ml, so A is the best value.

Answer: A (£12.50 for 1.25L)

Ratio & Proportion › Best Buys › Best Buys (Non-Calculator) (Foundation)

Topic 149 of 313 • id: best_buys_noncalculator_F

Q1. A shop sells breakfast cereal in different sizes: A: £1.50 for 100g; B: £2.20 for 200g. Which option is the better value for money? Show your working.

Q1

Work out the price per 100g for each option:

A: $£1.50 \div 100 \times 100 = 150\text{p}$ per 100g

B: $£2.20 \div 200 \times 100 = 110\text{p}$ per 100g

The lowest price per 100g is B at 110p per 100g, so B is the best value.

Answer: B (£2.20 for 200g)

Ratio & Proportion › Ratio › Share a Two-Part Ratio (Foundation)

Topic 150 of 313 • id: ratio_share_two_part_F

Q1. Share 130 in the ratio 4:9.

Q1

Total parts = $4 + 9 = 13$

Value of one part = $130 \div 13 = 10$

Share 1 = $4 \times 10 = 40$

Share 2 = $9 \times 10 = 90$

Answer: 40 : 90

Ratio & Proportion › Ratio › Find a Missing Share (Foundation)

Topic 151 of 313 • id: ratio_find_missing_share_F

Q1. $p : q = 2 : 3$. $p = 18$. What is the value of q ?

Q1

Value of one part = $18 \div 2 = 9$

$q = 3 \times 9 = 27$

Answer: 27

Ratio & Proportion › Ratio › Share a Three-Part Ratio (Higher)

Topic 152 of 313 • id: ratio_share_three_part_H

Q1. Share 72 in the ratio 3:1:4.

Q1

Total parts = $3 + 1 + 4 = 8$

Value of one part = $72 \div 8 = 9$

Share 1 = $3 \times 9 = 27$

Share 2 = $1 \times 9 = 9$

Share 3 = $4 \times 9 = 36$

Answer: 27 : 9 : 36

Ratio & Proportion › Ratio › Share a Three-Part Ratio (Foundation)

Topic 153 of 313 • id: ratio_share_three_part_F

Q1. Share 70 in the ratio 8:5:1.

Q1

Total parts = $8 + 5 + 1 = 14$

Value of one part = $70 \div 14 = 5$

Share 1 = $8 \times 5 = 40$

Share 2 = $5 \times 5 = 25$

Share 3 = $1 \times 5 = 5$

Answer: 40 : 25 : 5

Ratio & Proportion › Ratio › Combine Ratios (Higher)

Topic 154 of 313 • id: ratio_combine_H

Q1. Given that $a:b = 5:9$ and $b:c = 8:2$, find $a:b:c$ in its simplest form.

Q1

Scale $a:b = 5:9$ by 8 so $b = 72$: $a:b = 40:72$

Scale $b:c = 8:2$ by 9 so $b = 72$: $b:c = 72:18$

Combine: $a:b:c = 40:72:18$

Simplify by dividing by 2: $a:b:c = 20:36:9$

Answer: 20 : 36 : 9

Ratio & Proportion › Ratio › Ratio in the Form 1:n (Foundation)

Topic 155 of 313 • id: ratio_1_to_n_F

Q1. Write the ratio 11:4 in the form n:1.

Q1

Divide both parts of the ratio by 4 (the second part) so the second part becomes 1.

$$11 \div 4 \approx 2.75, 4 \div 4 = 1$$

Answer: 2.75 : 1 (to 2 d.p.)

Ratio & Proportion › Ratio › Ratio Given the Difference (Foundation)

Topic 156 of 313 • id: ratio_difference_F

Q1. $p : q = 4 : 7$.

$q - p = 15$.

Find p and q , and their total.

Q1

Difference in parts = $7 - 4 = 3$ parts

Value of one part = $15 \div 3 = 5$

$p = 4 \times 5 = 20$

$q = 7 \times 5 = 35$

$p + q = 20 + 35 = 55$

Answer: $p = 20$, $q = 35$ (total 55)

Ratio & Proportion › Ratio › Ratio Given the Difference (Three Parts) (Higher)

Topic 157 of 313 • id: ratio_difference_H

Q1. $p : q : r = 5 : 3 : 2$.

$p - r = 6$.

Find p , q and r , and their total.

Q1

Difference in parts = $5 - 2 = 3$ parts

Value of one part = $6 \div 3 = 2$

$p = 5 \times 2 = 10$

$q = 3 \times 2 = 6$

$r = 2 \times 2 = 4$

$p + q + r = 10 + 6 + 4 = 20$

Answer: $p = 10$, $q = 6$, $r = 4$ (total 20)

Ratio & Proportion › Ratio › Ratio Problems with Algebra (Higher)

Topic 158 of 313 • id: ratio_to_equation_H

Q1. $(x + 1) : (3x - 6) = 2:3$. Find the value of x .

Q1

Cross-multiply: $3(x + 1) = 2(3x - 6)$

Expand: $3x + 3 = 6x - 12$

Collect x terms: $-3x = -15$

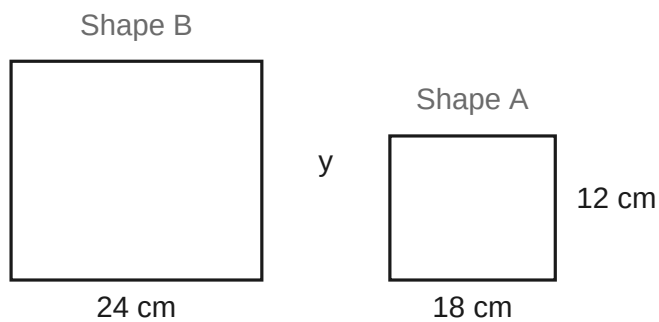
$x = -15 \div -3 = 5$

Answer: $x = 5$

Ratio & Proportion › Ratio › Similar Shapes: Lengths (Foundation)

Topic 159 of 313 • id: ratio_shape_similar_F

Q1. Shape A and Shape B are similar. Find the length of side y.



Q1

$$\text{Scale factor} = 24 \div 18 = \frac{4}{3}$$

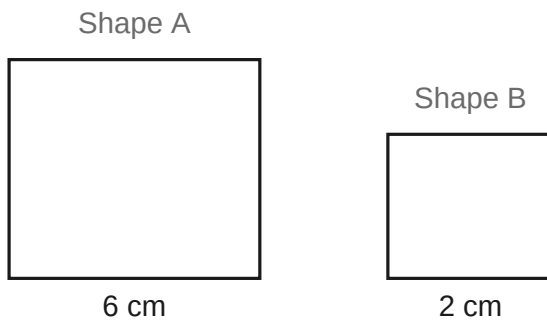
$$\text{Corresponding length} = 12 \times \frac{4}{3} = 16$$

Answer: 16 cm

Ratio & Proportion › Ratio › Similar Shapes: Area and Volume (Higher)

Topic 160 of 313 • id: ratio_shape_similar_H

Q1. Shape A and Shape B are similar. The volume of shape A is 216 cm^3 . Find the volume of shape B.



Q1

$$\text{Linear scale factor} = 2 \div 6 = \frac{1}{3}$$

$$\text{Volume scale factor} = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$$

$$\text{Volume of B} = 216 \times \frac{1}{27} = 8 \text{ cm}^3$$

Answer: 8 cm^3

Ratio & Proportion › Proportion › Direct Proportion (Foundation)

Topic 161 of 313 • id: direct_proportion_F

Q1. If 7 cm on a map represents 133 km in real life, how many km does 10 cm represent?

Q1

Unit value = $133 \text{ km} \div 7 = 19 \text{ km}$

Answer = $19 \text{ km} \times 10 = 190 \text{ km}$

Answer: 190 km

Ratio & Proportion › Proportion › Direct Proportion (Non-Calculator) (Foundation)

Topic 162 of 313 • id: direct_proportion_noncalculator_F

Q1. If 10 cm on a map represents 160 km in real life, how many km does 5 cm represent?

Q1

Unit value = $160 \text{ km} \div 10 = 16 \text{ km}$

Answer = $16 \text{ km} \times 5 = 80 \text{ km}$

Answer: 80 km

Ratio & Proportion › Proportion › Inverse Proportion (Foundation)

Topic 163 of 313 • id: inverse_proportion_F

Q1. It takes 8 machines 8 hours to complete a production run. Working at the same rate, how long would it take 2 machines?

Q1

$$\text{Constant} = 8 \times 8 = 64$$

$$\text{Time for 2 machines} = 64 \div 2 = 32$$

Answer: 32 hours

Ratio & Proportion › Proportion › Inverse Proportion (Non-Calculator) (Foundation)

Topic 164 of 313 • id: inverse_proportion_noncalculator_F

Q1. It takes 6 workers 20 hours to build a wall. Working at the same rate, how long would it take 5 workers?

Q1

$$\text{Constant} = 6 \times 20 = 120$$

$$\text{Time for 5 workers} = 120 \div 5 = 24$$

Answer: 24 hours

Ratio & Proportion › Proportion › Direct Proportion with Powers (Higher)

Topic 165 of 313 • id: algebraic_direct_proportion_H

Q1. y is directly proportional to x^2 . When $x = 6$, $y = 216$. Find the value of y when $x = 4$.

Q1

$$y = kx^2$$

$$216 = k \times 6^2 = k \times 36$$

$$k = 216 \div 36 = 6$$

$$y = 6 \times 4^2 = 6 \times 16 = 96$$

Answer: 96

Ratio & Proportion › Proportion › Inverse Proportion with Powers (Higher)

Topic 166 of 313 • id: algebraic_inverse_proportion_H

Q1. y is inversely proportional to x^2 . When $x = 5$, $y = 4$. Find the value of y when $x = 2$.

Q1

$$y = k \div x^2$$

$$k = y \times x^2 = 4 \times 5^2 = 4 \times 25 = 100$$

$$y = k \div x^2 = 100 \div 2^2 = 100 \div 4 = 25$$

Answer: 25

Ratio & Proportion › Compound Measures › Speed, Distance and Time (Foundation)

Topic 167 of 313 • id: sdt_mixed_F

Q1. A delivery van travels 3432 m at an average speed of 26 m/s. Find how long the journey takes.

Q1

time = distance $\div$ speed

time = $3432 \div 26 = 132$ seconds

Answer: 132 seconds

Ratio & Proportion › Compound Measures › Speed with Unit Conversions (Higher)

Topic 168 of 313 • id: speed_with_conversions_H

Q1. A train travels 2100 m in 75 seconds. Find its average speed in km/h, correct to 1 decimal place.

Q1

speed = distance $\div$ time = $2100 \div 75 = 28$ m/s

Convert to km/h: $28 \times 3.6 = 100.8$ km/h (1 d.p.)

Answer: 100.8 km/h

Ratio & Proportion › Compound Measures › Unit Conversions (Foundation)

Topic 169 of 313 • id: unit_conversions_F

Q1. Convert 431 cm to mm.

Q1

$$1 \text{ cm} = 10 \text{ mm}$$

$$431 \text{ cm} = 431 \times 10 = 4310 \text{ mm}$$

Answer: 4310 mm

Ratio & Proportion › Compound Measures › Area and Volume Unit Conversions (Higher)

Topic 170 of 313 • id: unit_conversions_H

Q1. Convert 42 m^3 to cm^3 .

Q1

Linear conversion factor: $1 \text{ m} = 100 \text{ cm}$, so for volume, $1 \text{ m}^3 = 100^3 = 1000000 \text{ cm}^3$

$42 \text{ m}^3 = 42 \times 1000000 = 42000000 \text{ cm}^3$

Answer: 42000000 cm^3

Ratio & Proportion › Compound Measures › Density (Foundation)

Topic 171 of 313 • id: density_F

Q1. A sculpture has a mass of 4368 kg and a density of 772 kg/m³. Find its volume, correct to 2 decimal places.

Q1

volume = mass ÷ density

volume = $4368 \div 772 = 5.66 \text{ m}^3$ (2 d.p.)

Answer: 5.66 m³

Ratio & Proportion › Compound Measures › Density (Multi-Step) (Higher)

Topic 172 of 313 • id: density_H

Q1. A stone has a mass of 16 kg and a volume of 1017 cm³. Find its density in g/cm³, correct to 2 decimal places.

Q1

Convert mass to grams: 16 kg = 16 × 1000 = 16000 g

density = mass ÷ volume = 16000 ÷ 1017 = 15.73 g/cm³ (2 d.p.)

Answer: 15.73 g/cm³

Ratio & Proportion › Compound Measures › Pressure (Foundation)

Topic 173 of 313 • id: pressure_F

Q1. A concrete block exerts a force of 1284 N over a contact area of 26 m². Find the pressure, correct to 2 decimal places.

Q1

pressure = force ÷ area

pressure = $1284 \div 26 = 49.38 \text{ N/m}^2$ (2 d.p.)

Answer: 49.38 N/m²

Ratio & Proportion › Compound Measures › Pressure (Multi-Step) (Higher)

Topic 174 of 313 • id: pressure_H

Q1. A crate has a rectangular base measuring 29 cm × 4 cm and exerts a force of 3499 N. Find the pressure it exerts, correct to 2 decimal places.

Q1

$$\text{area} = \text{length} \times \text{width} = 29 \times 4 = 116 \text{ cm}^2$$

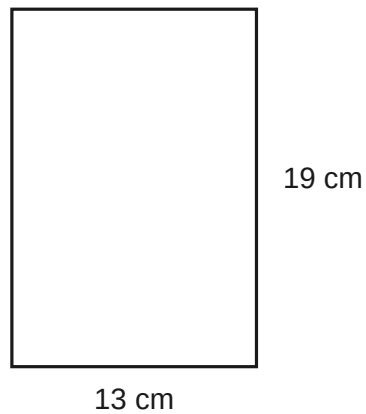
$$\text{pressure} = \text{force} \div \text{area} = 3499 \div 116 = 30.16 \text{ N/cm}^2 \text{ (2 d.p.)}$$

Answer: 30.16 N/cm²

Geometry › Area & Perimeter › Rectangles (Foundation)

Topic 175 of 313 • id: area_rectangle_F

Q1. Find the perimeter of the following rectangle.



Q1

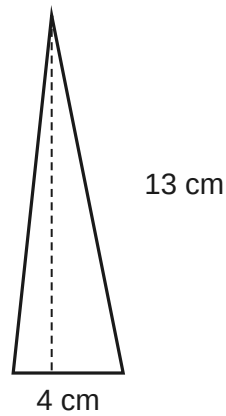
$$\text{Perimeter} = 2 \times (\text{length} + \text{width}) = 2 \times (13 + 19) = 64 \text{ cm}$$

Answer: 64 cm

Geometry › Area & Perimeter › Triangles (Foundation)

Topic 176 of 313 • id: area_triangle_F

Q1. A triangle has base 4 cm and height 13 cm. Find its area.



Q1

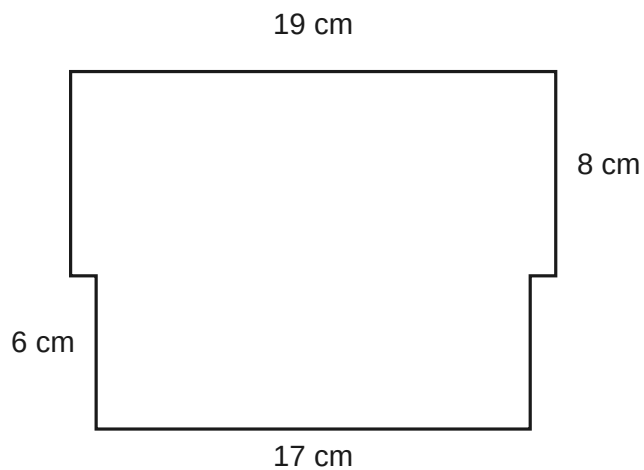
$$\text{Area} = \left(\frac{1}{2}\right) \times \text{base} \times \text{height} = \left(\frac{1}{2}\right) \times 4 \times 13 = 26 \text{ cm}^2$$

Answer: 26 cm²

Geometry › Area & Perimeter › Composite Rectangles (Foundation)

Topic 177 of 313 • id: area_composite_rectangles_F

Q1. Find the area of this compound shape.



Q1

Area of the top bar = $19 \times 8 = 152 \text{ cm}^2$

Area of the stem = $17 \times 6 = 102 \text{ cm}^2$

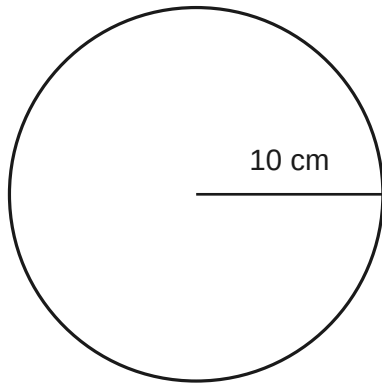
Total area = $152 + 102 = 254 \text{ cm}^2$

Answer: 254 cm^2

Geometry › Area & Perimeter › Circles (Calculator) (Foundation)

Topic 178 of 313 • id: area_circle_F

Q1. Find the circumference of the following circle, correct to 2 decimal places.



Q1

$$\text{Circumference} = 2 \times \pi \times r = 2 \times \pi \times 10$$

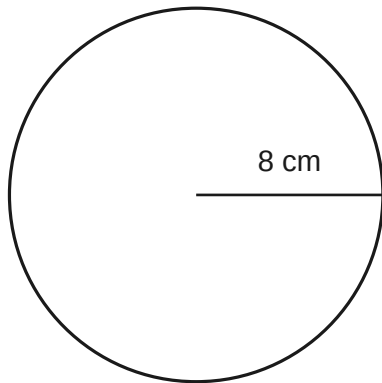
$$= 62.83 \text{ cm (2 d.p., using a calculator value of } \pi \text{)}$$

Answer: 62.83 cm

Geometry › Area & Perimeter › Circles (Higher)

Topic 179 of 313 • id: area_circle_H

Q1. Find the circumference of the following circle, in terms of π .



Q1

$$\text{Circumference} = 2 \times \pi \times r = 2 \times \pi \times 8 = 16\pi \text{ cm}$$

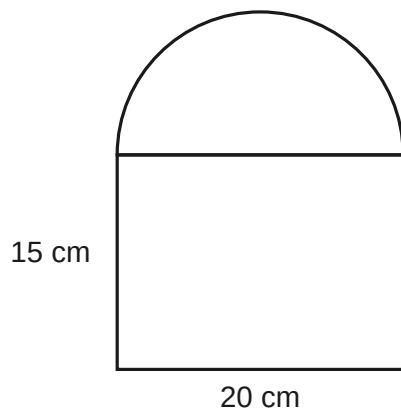
$$\approx 50.3 \text{ cm (3 s.f.)}$$

Answer: 16π cm (≈ 50.3 cm)

Geometry › Area & Perimeter › Semicircle Compound Shapes (Calculator) (Foundation)

Topic 180 of 313 • id: area_semicircle_compound_F

Q1. A shape is made from a rectangle 20 cm by 15 cm with a semicircle of diameter 20 cm attached to one side. Find the total area, correct to 3 s.f.



Q1

$$\text{Area of rectangle} = 20 \times 15 = 300 \text{ cm}^2$$

$$\text{Radius of semicircle} = 20 \div 2 = 10 \text{ cm}$$

$$\text{Area of semicircle} = \left(\frac{1}{2}\right) \times \pi \times 10^2 = 50\pi \text{ cm}^2$$

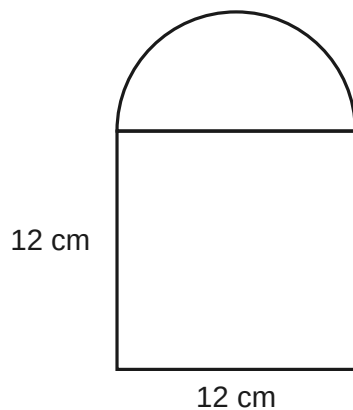
$$\text{Total area} = 300 + 50\pi \approx 457 \text{ cm}^2 \text{ (3 s.f.)}$$

Answer: $\approx 457 \text{ cm}^2$

Geometry › Area & Perimeter › Semicircle Compound Shapes (Higher)

Topic 181 of 313 • id: area_semicircle_compound_H

Q1. A shape is made from a rectangle 12 cm by 12 cm with a semicircle of diameter 12 cm attached to one side. Find the total area, giving your answer in terms of π .



Q1

$$\text{Area of rectangle} = 12 \times 12 = 144 \text{ cm}^2$$

$$\text{Radius of semicircle} = 12 \div 2 = 6 \text{ cm}$$

$$\text{Area of semicircle} = \left(\frac{1}{2}\right) \times \pi \times 6^2 = 18\pi \text{ cm}^2$$

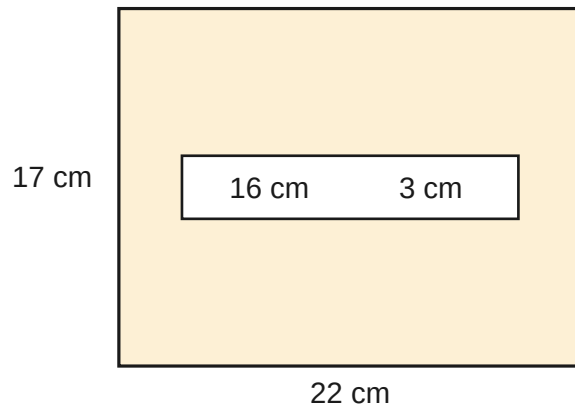
$$\text{Total area} = 144 + 18\pi \text{ cm}^2 \text{ (exact form)}$$

Answer: $(144 + 18\pi) \text{ cm}^2$

Geometry › Area & Perimeter › Subtractive Compound Shapes (Higher)

Topic 182 of 313 • id: area_subtract_compound_H

Q1. Find the shaded area.



Q1

Area of large rectangle = $22 \times 17 = 374 \text{ cm}^2$

Area of rectangular hole = $16 \times 3 = 48 \text{ cm}^2$

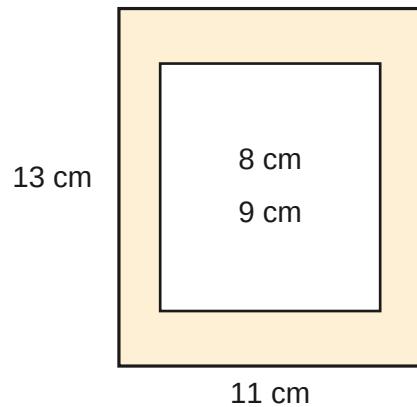
Shaded area = $374 - 48 = 326 \text{ cm}^2$

Answer: 326 cm^2

Geometry › Area & Perimeter › Subtractive Compound Shapes (Foundation)

Topic 183 of 313 • id: area_subtract_compound_F

Q1. Find the shaded area.



Q1

Area of large rectangle = $11 \times 13 = 143 \text{ cm}^2$

Area of rectangular hole = $8 \times 9 = 72 \text{ cm}^2$

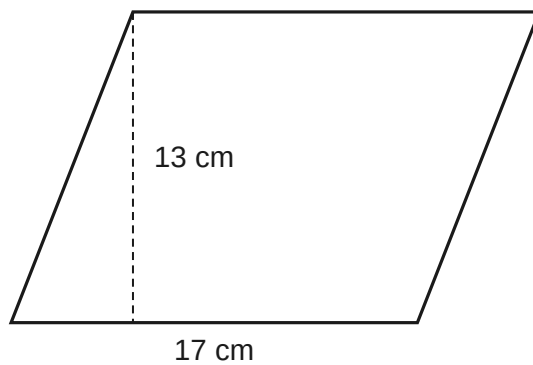
Shaded area = $143 - 72 = 71 \text{ cm}^2$

Answer: 71 cm^2

Geometry › Area & Perimeter › Parallelograms (Foundation)

Topic 184 of 313 • id: area_parallelogram_F

Q1. Find the area of the following parallelogram.



Q1

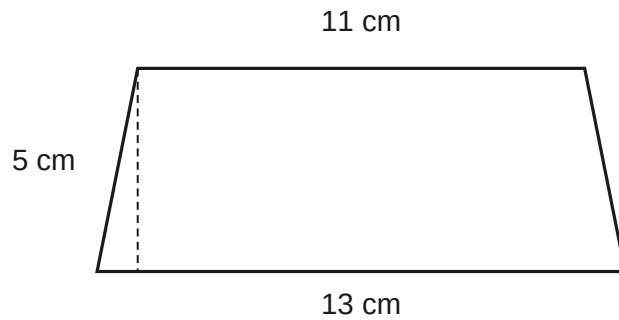
$$\text{Area} = \text{base} \times \text{height} = 17 \times 13 = 221 \text{ cm}^2$$

Answer: 221 cm²

Geometry › Area & Perimeter › Trapeziums (Foundation)

Topic 185 of 313 • id: area_trapezium_F

Q1. Find the area of the following trapezium.



Q1

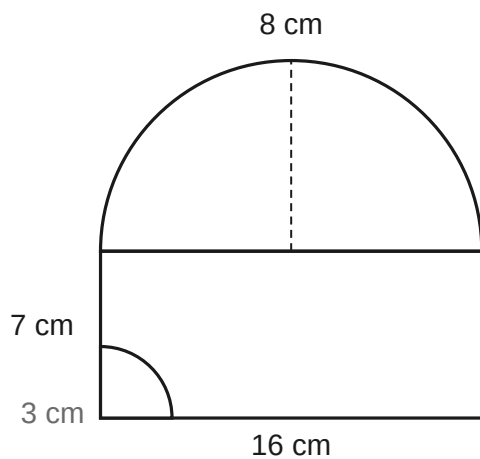
$$\text{Area} = \frac{1}{2} \times (a + b) \times \text{height} = \frac{1}{2} \times (11 + 13) \times 5 = 60 \text{ cm}^2$$

Answer: 60 cm²

Geometry › Area & Perimeter › Mixed Compound Shapes (Higher)

Topic 186 of 313 • id: area_mixed_compound_H

Q1. Find the total area of the following compound shape, correct to 1 decimal place.



Q1

$$\text{Rectangle area} = 16 \times 7 = 112 \text{ cm}^2$$

$$\text{Semicircular top area} = \frac{1}{2} \times \pi \times 8^2 \approx 101 \text{ cm}^2$$

$$\text{Quarter-circle cut area} = (\pi \times 3^2) \div 4 \approx 7.07 \text{ cm}^2$$

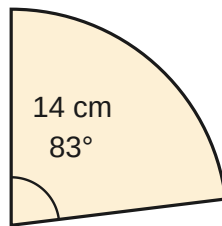
$$\text{Total area} = 112 + 101 - 7.07 \approx 205.5 \text{ cm}^2 \text{ (1 d.p.)}$$

Answer: 205.5 cm²

Geometry › Area & Perimeter › Arc Length (Calculator) (Foundation)

Topic 187 of 313 • id: arc_length_F

Q1. Find the arc length of the following sector, correct to 2 decimal places.



Q1

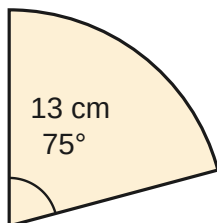
$$\begin{aligned}\text{Arc length} &= (\theta \div 360) \times 2 \times \pi \times r = (83 \div 360) \times 2 \times \pi \times 14 \\ &= 20.28 \text{ cm (2 d.p., using a calculator value of } \pi)\end{aligned}$$

Answer: 20.28 cm

Geometry › Area & Perimeter › Arc Length (Higher)

Topic 188 of 313 • id: arc_length_H

Q1. Find the exact arc length of the following sector, in terms of π .



Q1

$$\text{Arc length} = (\theta \div 360) \times 2 \times \pi \times r = (75 \div 360) \times 2 \times \pi \times 13$$

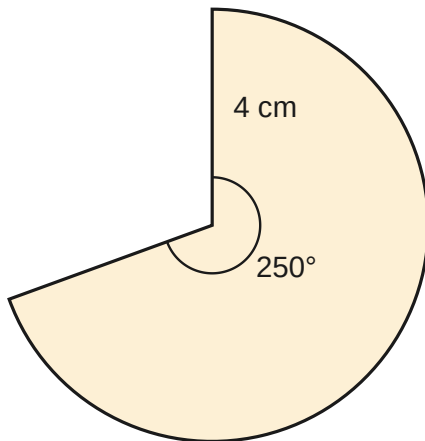
$$= \left(\frac{65}{12}\right)\pi \text{ cm}$$

$$\text{Answer: } \left(\frac{65}{12}\right)\pi \text{ cm}$$

Geometry › Area & Perimeter › Area of a Sector (Calculator) (Foundation)

Topic 189 of 313 • id: area_sector_F

Q1. Find the area of the following sector, correct to 2 decimal places.



Q1

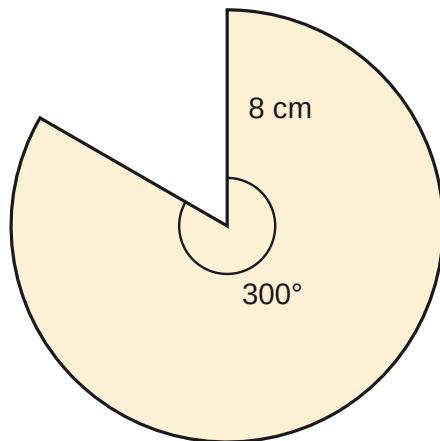
$$\begin{aligned}\text{Area} &= (\theta \div 360) \times \pi \times r^2 = (250 \div 360) \times \pi \times 4^2 \\ &= 34.91 \text{ cm}^2 \text{ (2 d.p., using a calculator value of } \pi)\end{aligned}$$

Answer: 34.91 cm²

Geometry › Area & Perimeter › Area of a Sector (Higher)

Topic 190 of 313 • id: area_sector_H

Q1. Find the exact area of the following sector, in terms of π .



Q1

$$\text{Area} = (\theta \div 360) \times \pi \times r^2 = (300 \div 360) \times \pi \times 8^2$$

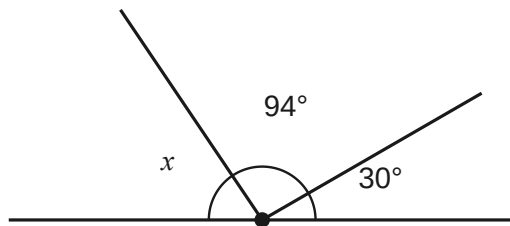
$$= \left(\frac{160}{3}\right)\pi \text{ cm}^2$$

$$\text{Answer: } \left(\frac{160}{3}\right)\pi \text{ cm}^2$$

Geometry › Angles › On a Straight Line (Foundation)

Topic 191 of 313 • id: angles_straight_line_F

Q1. The angles 30° , 94° and x° lie on a straight line. Find x .



Q1

Angles on a straight line sum to 180° .

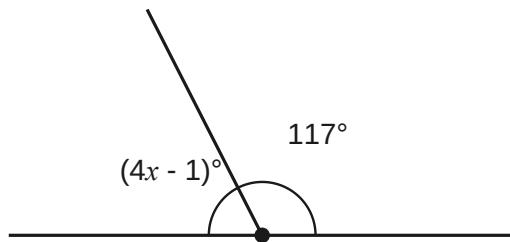
$$x = 180 - (30 + 94) = 180 - 124 = 56$$

Answer: 56

Geometry › Angles › On a Straight Line (Algebraic) (Higher)

Topic 192 of 313 • id: angles_straight_line_H

Q1. The angles 117° and $(4x - 1)^\circ$ lie on a straight line. Find x .



Q1

Angles on a straight line sum to 180° .

$$4x - 1 = 63$$

Add 1 to both sides:

$$4x = 64$$

Divide both sides by 4:

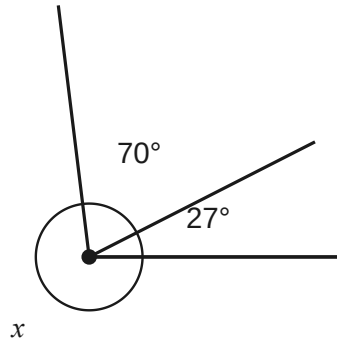
$$x = 16$$

Answer: 16

Geometry › Angles › Around a Point (Foundation)

Topic 193 of 313 • id: angles_around_point_F

Q1. The angles 27° , 70° and x° are angles around a point. Find x .



Q1

Angles around a point sum to 360° .

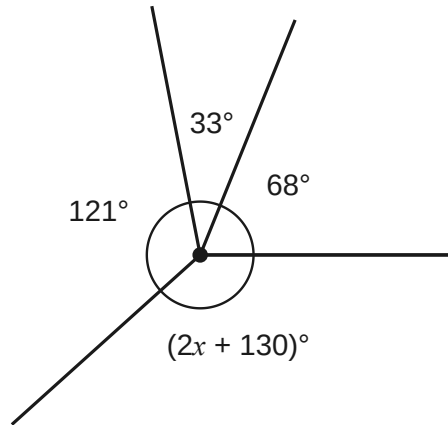
$$x = 360 - (27 + 70) = 360 - 97 = 263$$

Answer: 263

Geometry › Angles › Around a Point (Algebraic) (Higher)

Topic 194 of 313 • id: angles_around_point_H

Q1. The angles 68° , 33° , 121° and $(2x + 130)^\circ$ are angles around a point. Find x .



Q1

Angles around a point sum to 360° .

$$2x + 130 = 138$$

Subtract 130 from both sides:

$$2x = 8$$

Divide both sides by 2:

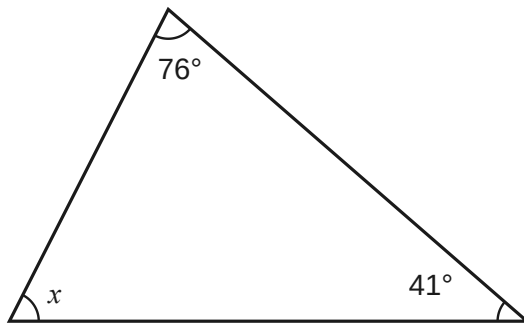
$$x = 4$$

Answer: 4

Geometry › Angles › In a Triangle (Foundation)

Topic 195 of 313 • id: angles_triangle_F

Q1. A triangle has angles 76° , 41° , and x° . Find x .



Q1

Angles in a triangle sum to 180° .

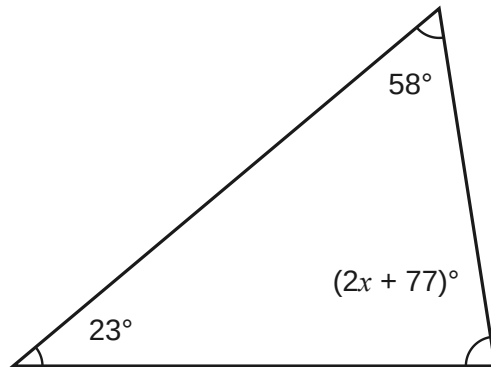
$$x = 180 - (76 + 41) = 180 - 117 = 63$$

Answer: 63

Geometry › Angles › In a Triangle (Algebraic) (Higher)

Topic 196 of 313 • id: angles_triangle_H

Q1. A triangle has angles 58° , 23° , and $(2x + 77)^\circ$. Find x .



Q1

Angles in a triangle sum to 180° .

$$2x + 77 = 99$$

Subtract 77 from both sides:

$$2x = 22$$

Divide both sides by 2:

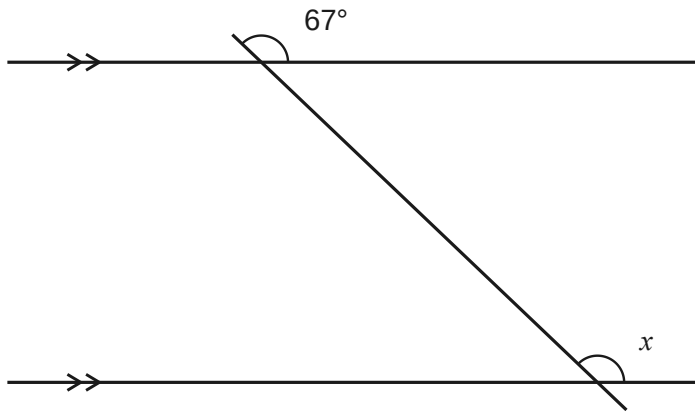
$$x = 11$$

Answer: 11

Geometry › Angles › Parallel Lines (Foundation)

Topic 197 of 313 • id: angles_parallel_lines_F

Q1. A line crosses two parallel lines. The angle 67° and the angle x° are corresponding angles. Find x .



Q1

Corresponding angles are equal.

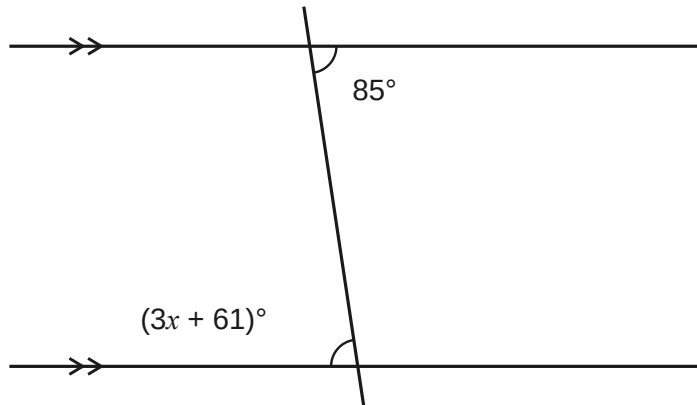
$$x = 67$$

Answer: 67

Geometry › Angles › Parallel Lines (Higher)

Topic 198 of 313 • id: angles_parallel_lines_H

Q1. A line crosses two parallel lines. The angle 85° and the angle $(3x + 61)^\circ$ are alternate angles. Find x .



Q1

Alternate angles are equal.

$$3x + 61 = 85$$

Subtract 61 from both sides:

$$3x = 24$$

Divide both sides by 3:

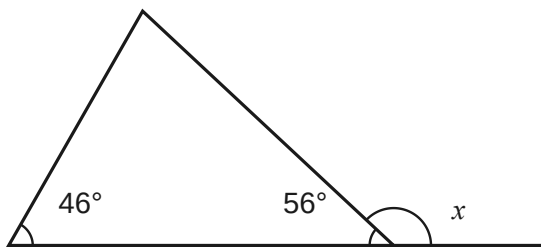
$$x = 8$$

Answer: 8

Geometry › Angles › Exterior Angle (Foundation)

Topic 199 of 313 • id: angles_exterior_F

Q1. An exterior angle of a triangle is x° . The two remote interior angles are 46° and 56° . Find x .



Q1

The exterior angle of a triangle equals the sum of the two remote interior angles.

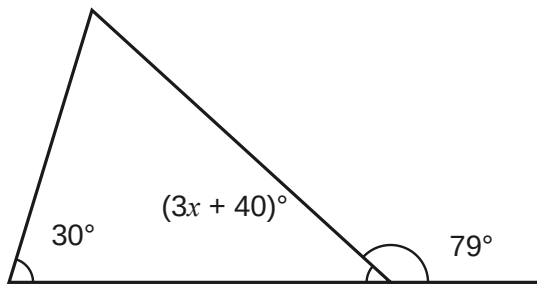
$$x = 46 + 56 = 102$$

Answer: 102

Geometry › Angles › Exterior Angle (Higher)

Topic 200 of 313 • id: angles_exterior_H

Q1. An exterior angle of a triangle is 79° . The two remote interior angles are 30° and $(3x + 40)^\circ$. Find x .



Q1

The exterior angle of a triangle equals the sum of the two remote interior angles.

$$3x + 40 = 49$$

Subtract 40 from both sides:

$$3x = 9$$

Divide both sides by 3:

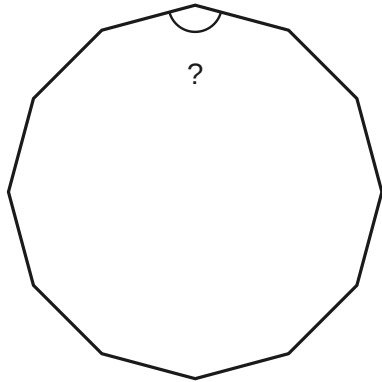
$$x = 3$$

Answer: 3

Geometry › Angles › Polygon Angles (Foundation)

Topic 201 of 313 • id: angles_polygon_interior_F

Q1. A regular polygon has 12 sides. Find the size of one interior angle.



Q1

Sum of interior angles = $(n - 2) \times 180 = (12 - 2) \times 180 = 1800^\circ$

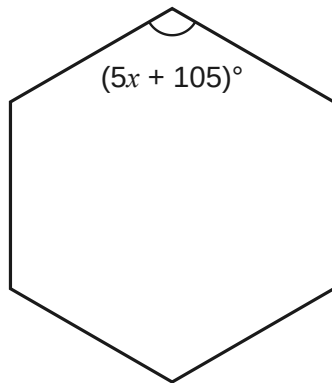
This is a regular polygon, so each interior angle = $1800 \div 12 = 150^\circ$

Answer: 150°

Geometry › Angles › Polygon Interior Angles (Higher)

Topic 202 of 313 • id: angles_polygon_interior_H

Q1. A polygon has 6 sides. 5 of its interior angles are each 114° , and the remaining interior angle is $(5x + 105)^\circ$. Find x .



Q1

Sum of interior angles = $(n - 2) \times 180 = (6 - 2) \times 180 = 720^\circ$

Sum of the 5 equal angles = $5 \times 114 = 570^\circ$

Remaining angle = $720 - 570 = 150^\circ$

$5x + 105 = 150$

Subtract 105 from both sides:

$5x = 45$

Divide both sides by 5:

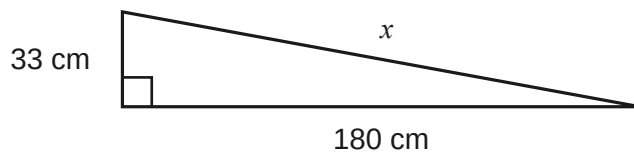
$x = 9$

Answer: 9

Geometry › Pythagoras' Theorem › Hypotenuse (Triple) (Foundation)

Topic 203 of 313 • id: pythagoras_hypotenuse_triple_F

Q1. A right-angled triangle has sides 33 cm and 180 cm. Find the length of x .



Q1

$$c^2 = a^2 + b^2 = 33^2 + 180^2 = 1089 + 32400 = 33489$$

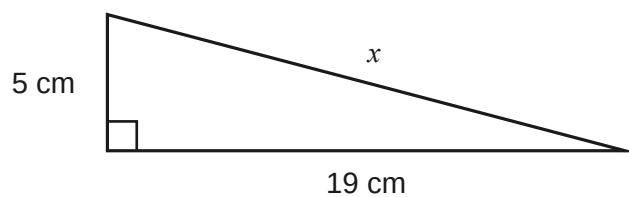
$$c = \sqrt{33489} = 183$$

Answer: 183 cm

Geometry › Pythagoras' Theorem › Hypotenuse (Decimal) (Foundation)

Topic 204 of 313 • id: pythagoras_hypotenuse_decimal_F

Q1. A right-angled triangle has sides 5 cm and 19 cm. Find the length of x , correct to 1 decimal place.



Q1

$$c^2 = a^2 + b^2 = 5^2 + 19^2 = 25 + 361 = 386$$

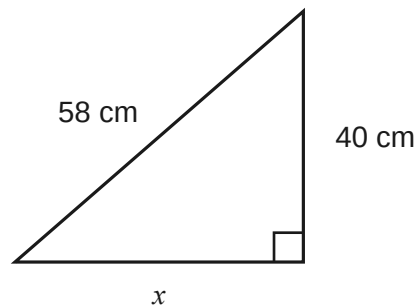
$$c = \sqrt{386} \approx 19.6 \text{ cm (1 d.p.)}$$

Answer: 19.6 cm

Geometry › Pythagoras' Theorem › Shorter Side (Foundation)

Topic 205 of 313 • id: pythagoras_shorter_leg_F

Q1. A right-angled triangle has hypotenuse 58 cm and one side 40 cm. Find the length of x .



Q1

$$b^2 = c^2 - a^2 = 58^2 - 40^2 = 3364 - 1600 = 1764$$

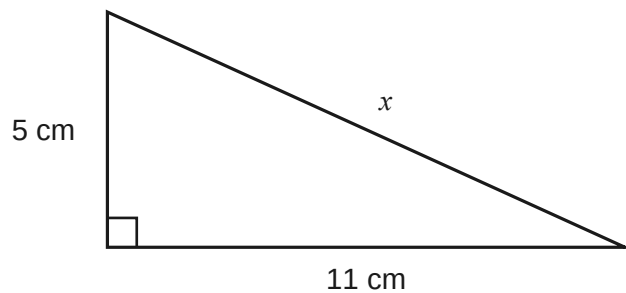
$$b = \sqrt{1764} = 42$$

Answer: 42 cm

Geometry › Pythagoras' Theorem › Exact Surd Hypotenuse (Higher)

Topic 206 of 313 • id: pythagoras_surd_hypotenuse_H

Q1. A right-angled triangle has sides 5 cm and 11 cm. Find the exact length of x , giving your answer as a simplified surd.



Q1

$$c^2 = a^2 + b^2 = 5^2 + 11^2 = 25 + 121 = 146$$

$$c = \sqrt{146} = \sqrt{146} \text{ cm (exact form)}$$

Answer: $\sqrt{146}$ cm

Geometry › Pythagoras' Theorem › Ladder Context (Higher)

Topic 207 of 313 • id: pythagoras_ladder_context_H

Q1. A ladder of length 15 m leans against a vertical wall with its base 8 m from the wall. Find the height the ladder reaches up the wall, giving your answer as a surd in its simplest form.

Q1

$$\text{height}^2 = \text{ladder}^2 - \text{base}^2 = 15^2 - 8^2 = 225 - 64 = 161$$

$$\text{height} = \sqrt{161} = \sqrt{161} \text{ m (exact form)}$$

Answer: $\sqrt{161}$ m

Geometry › Pythagoras' Theorem › Ladder Context (Foundation)

Topic 208 of 313 • id: pythagoras_ladder_context_F

Q1. A ladder of length 111 m leans against a vertical wall with its base 36 m from the wall. Find the height the ladder reaches up the wall.

Q1

$$\text{height}^2 = \text{ladder}^2 - \text{base}^2 = 111^2 - 36^2 = 12321 - 1296 = 11025$$

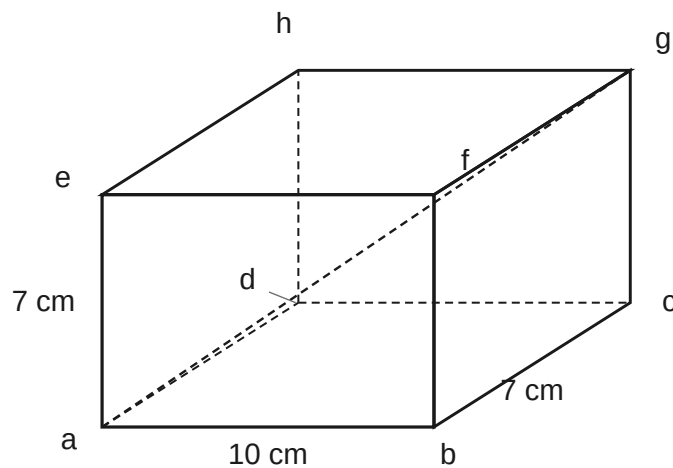
$$\text{height} = \sqrt{11025} = 105 \text{ m}$$

Answer: 105 m

Geometry › Pythagoras' Theorem › 3D Pythagoras (Higher)

Topic 209 of 313 • id: pythagoras_3d_H

Q1. A cuboid has length 7 cm, width 10 cm and height 7 cm. Find the length of the diagonal ag that runs from one corner of the cuboid to the opposite corner, correct to 1 decimal place.



Q1

First find the diagonal across the base: $\text{base diagonal}^2 = l^2 + w^2 = 7^2 + 10^2 = 149$.

Then apply Pythagoras again using the height: $ag^2 = \text{base diagonal}^2 + h^2 = 149 + 7^2 = 198$.

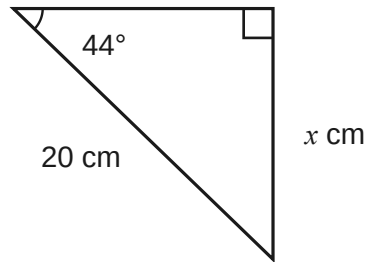
$ag = \sqrt{198} = 14.1 \text{ cm (1 d.p.)}$

Answer: 14.1 cm

Geometry › Trigonometry › Missing Sides (Foundation)

Topic 210 of 313 • id: trig_missing_side_F

Q1. In the triangle shown, find the length of x , correct to 1 decimal place.



Q1

We know the hypotenuse and want the opposite, so use sin.

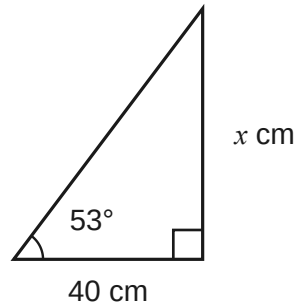
$$x = 20 \times \sin(44^\circ) = 13.9 \text{ (1 d.p.)}$$

Answer: 13.9 cm

Geometry › Trigonometry › Missing Sides (Higher)

Topic 211 of 313 • id: trig_missing_side_H

Q1. In the triangle shown, find the length of x , correct to 3 significant figures.



Q1

We know the adjacent and want the opposite, so use $\tan$.

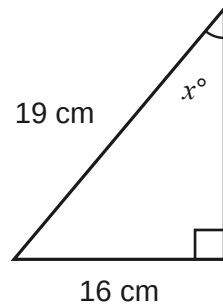
$$x = 40 \times \tan(53^\circ) = 53.1 \text{ (3 s.f.)}$$

Answer: 53.1 cm

Geometry › Trigonometry › Missing Angles (Foundation)

Topic 212 of 313 • id: trig_missing_angle_F

Q1. In the triangle shown, find the angle x , correct to 1 decimal place.



Q1

Use $\sin$, since we know two of the three sides.

$$\sin(x) = 16 \div 19$$

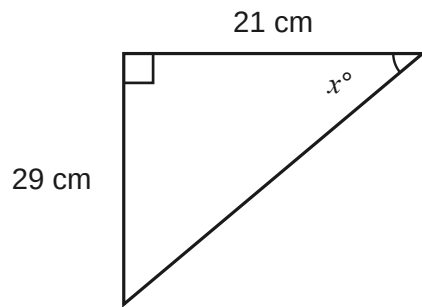
$$x = \sin^{-1}(16 \div 19) = 57.4^\circ \text{ (1 d.p.)}$$

Answer: 57.4°

Geometry › Trigonometry › Missing Angles (Higher)

Topic 213 of 313 • id: trig_missing_angle_H

Q1. In the triangle shown, find the angle x , correct to 1 decimal place.



Q1

Use $\tan$, since we know two of the three sides.

$$\tan(x) = 29 \div 21$$

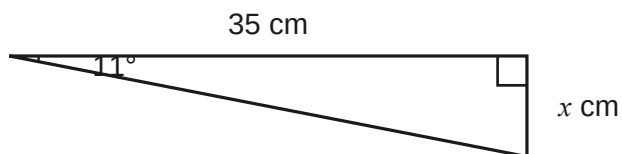
$$x = \tan^{-1}(29 \div 21) = 54.1^\circ \text{ (1 d.p.)}$$

Answer: 54.1°

Geometry › Trigonometry › Mixed Sides and Angles (Higher)

Topic 214 of 313 • id: trig_mixed_H

Q1. In the triangle shown, find the length of x , correct to 1 decimal place.



Q1

We know the adjacent and want the opposite, so use $\tan$.

$$x = 35 \times \tan(11^\circ) = 6.8 \text{ (1 d.p.)}$$

Answer: 6.8 cm

Geometry › Trigonometry › Exact Trig Values (Higher)

Topic 215 of 313 • id: exact_trig_values_H

Write down the exact value of

Q1. $\cos(90^\circ)$.

Q1. $\cos(90^\circ)$.

Recall the exact-value triangles: 30° - 60° - 90° (sides 1, $\sqrt{3}$, 2) and 45° - 45° - 90° (sides 1, 1, $\sqrt{2}$) - no calculator needed for these special angles.

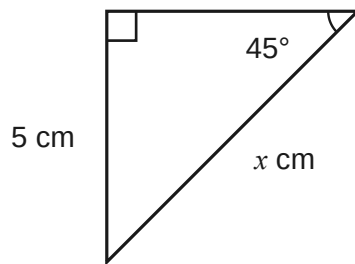
$$\cos(90^\circ) = 0$$

Answer: 0

Geometry › Trigonometry › Exact Trig Values in Triangles (Higher)

Topic 216 of 313 • id: exact_trig_values_triangles_H

Q1. In the triangle shown, find the exact length of x . Give your answer in surd form where appropriate.



Q1

We know the opposite and want the hypotenuse, so use $\sin(45^\circ) = \frac{\sqrt{2}}{2}$ - its exact value, not a decimal approximation.

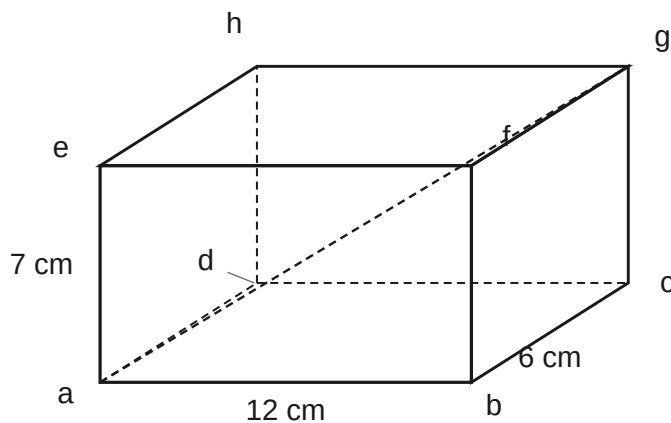
$$x = 5 \div \frac{\sqrt{2}}{2} = 5\sqrt{2} \text{ cm}$$

Answer: $5\sqrt{2}$ cm

Geometry › Trigonometry › 3D Trigonometry (Higher)

Topic 217 of 313 • id: trig_3d_H

Q1. A cuboid has length 6 cm, width 12 cm and height 7 cm. Find the angle between the diagonal ag (from one corner to the opposite corner) and the base of the cuboid, correct to 1 decimal place.



Q1

First find the diagonal across the base: base diagonal = $\sqrt{l^2 + w^2} = \sqrt{6^2 + 12^2} = 13.416$ cm - this is the projection of ag onto the base.

The angle between ag and the base, height and base diagonal form a right-angled triangle:
 $\tan(\text{angle}) = \text{height} \div \text{base diagonal} = 7 \div 13.416$

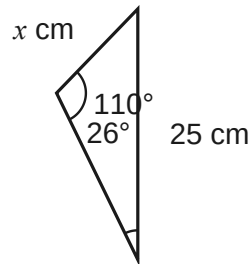
$\text{angle} = \tan^{-1}(7 \div 13.416) = 27.6^\circ$ (1 d.p.)

Answer: 27.6°

Geometry › Sine Rule › Sine Rule (Higher)

Topic 218 of 313 • id: sine_rule_H

Q1. In triangle ABC, angle A = 110° , angle B = 26° , and side a = 25 cm. Find the length of side b, correct to 3 significant figures.



Q1

Use the sine rule: $a / \sin(A) = b / \sin(B)$

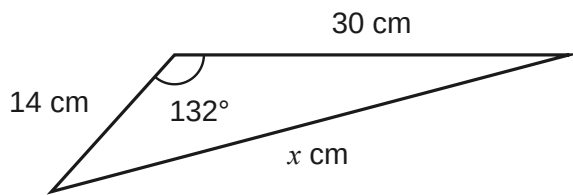
$$b = (25 \times \sin(26^\circ)) \div \sin(110^\circ) = 11.7 \text{ cm (3 s.f.)}$$

Answer: 11.7 cm

Geometry › Cosine Rule › Cosine Rule (Higher)

Topic 219 of 313 • id: cosine_rule_H

Q1. In triangle ABC, side $b = 14$ cm, side $c = 30$ cm, and the angle between them, angle $A = 132^\circ$. Find the length of side a , correct to 1 decimal place.



Q1

Use the cosine rule: $a^2 = b^2 + c^2 - 2bc \cos(A)$

$$a^2 = 14^2 + 30^2 - 2 \times 14 \times 30 \times \cos(132^\circ) = 1658$$

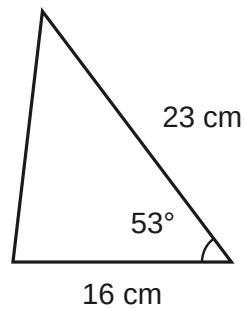
$$a = \sqrt{1658} = 40.7 \text{ cm (1 d.p.)}$$

Answer: 40.7 cm

Geometry › Area of a Triangle › Area of a Triangle (Higher)

Topic 220 of 313 • id: triangle_area_sine_rule_H

Q1. Triangle ABC has side $a = 23$ cm, side $b = 16$ cm, and the angle between them, angle $C = 53^\circ$. Find the area of the triangle, correct to 1 decimal place.



Q1

Use $\text{Area} = \left(\frac{1}{2}\right) \times a \times b \times \sin(C)$

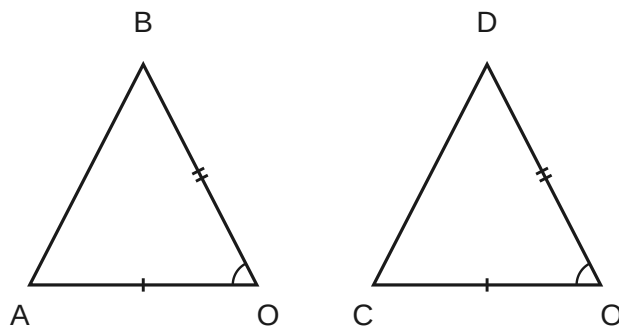
$\text{Area} = 0.5 \times 23 \times 16 \times \sin(53^\circ) = 146.9 \text{ cm}^2$ (1 d.p.)

Answer: 146.9 cm²

Geometry › Congruence Proof › Congruent Triangle Proof (Higher)

Topic 221 of 313 • id: congruent_triangle_proof_H

Q1. Lines AC and BD cross at point O, with $OA = OC$ and $OB = OD$. Prove that the two triangles are congruent.



Q1

$OA = OC$ (given)

$\angle AOB = \angle COD$ (vertically opposite angles)

$OB = OD$ (given)

By Side-Angle-Side (SAS), the triangles are congruent.

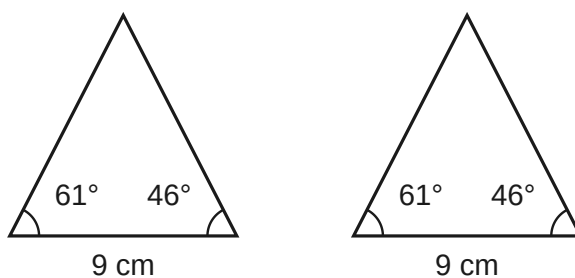
Since two sides and the included angle match, triangles AOB and COD are congruent.

Answer: Triangles are congruent (SAS).

Geometry › Congruence Proof › Congruent Triangle Proof (Foundation)

Topic 222 of 313 • id: congruent_triangle_proof_F

Q1. Shown below are two congruent triangles. Give a reason why they are congruent. A) SSS B) RHS C) ASA D) SAS



Q1

Look at which measurements are marked equal on both triangles - here they match the Angle-Side-Angle (ASA) pattern.

So the two triangles are congruent by Angle-Side-Angle (ASA).

Answer: C) ASA

Geometry › Vectors › Vector Arithmetic (Foundation)

Topic 223 of 313 • id: vectors_arithmetic_F

Q1. $\mathbf{a} = \begin{pmatrix} 8 \\ 8 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$. Find $3\mathbf{a} - 4\mathbf{b}$.

Q1

$$\mathbf{a} = \begin{pmatrix} 8 \\ 8 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$$

$$3\mathbf{a} = \begin{pmatrix} 24 \\ 24 \end{pmatrix} \text{ and } 4\mathbf{b} = \begin{pmatrix} 0 \\ -12 \end{pmatrix}$$

$$3\mathbf{a} - 4\mathbf{b} = \begin{pmatrix} 24 \\ 36 \end{pmatrix}$$

$$\text{Answer: } \begin{pmatrix} 24 \\ 36 \end{pmatrix}$$

Geometry › Vectors › Vector Arithmetic (Higher)

Topic 224 of 313 • id: vectors_arithmetic_H

Q1. $\mathbf{a} = \begin{pmatrix} 11 \\ -2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -9 \\ 2 \end{pmatrix}$. Find $-4\mathbf{a} + (-3\mathbf{b})$.

Q1

$$\mathbf{a} = \begin{pmatrix} 11 \\ -2 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} -9 \\ 2 \end{pmatrix}$$

$$-4\mathbf{a} = \begin{pmatrix} -44 \\ 8 \end{pmatrix} \text{ and } -3\mathbf{b} = \begin{pmatrix} 27 \\ -6 \end{pmatrix}$$

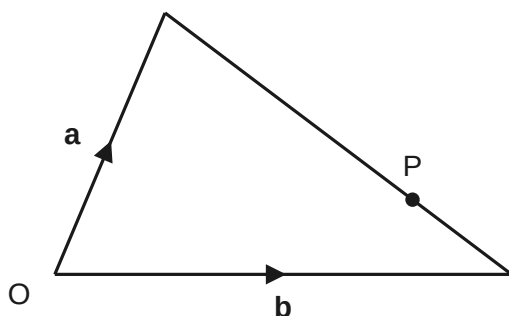
$$-4\mathbf{a} + (-3\mathbf{b}) = \begin{pmatrix} -17 \\ 2 \end{pmatrix}$$

$$\text{Answer: } \begin{pmatrix} -17 \\ 2 \end{pmatrix}$$

Geometry › Geometric Vectors › Vector Geometry (Higher)

Topic 225 of 313 • id: geometric_vectors_H

Q1. OAB is a triangle with $OA = \mathbf{a}$ and $OB = \mathbf{b}$. Point P lies on AB such that $AP:PB = 5:2$. Find the vector PB in terms of $\mathbf{a}$ and $\mathbf{b}$, giving your answer in its simplest form.



Q1

$$AB = OB - OA = \mathbf{b} - \mathbf{a}$$

$$AP = \left(\frac{5}{7}\right)AB = \left(\frac{5}{7}\right)(\mathbf{b} - \mathbf{a})$$

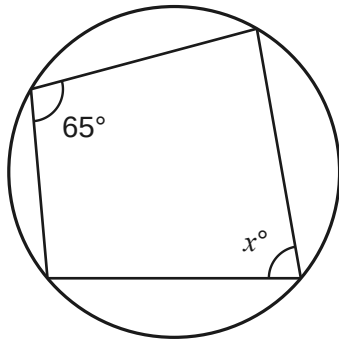
$$PB = OB - OP = -\left(\frac{2}{7}\right)\mathbf{a} + \left(\frac{2}{7}\right)\mathbf{b}$$

$$\text{Answer: } -\left(\frac{2}{7}\right)\mathbf{a} + \left(\frac{2}{7}\right)\mathbf{b}$$

Geometry › Circle Theorems › Circle Theorems (Higher)

Topic 226 of 313 • id: circle_theorems_H

Q1. Find the size of angle x .



Q1

Opposite angles in a cyclic quadrilateral sum to 180° .

$$x = 180 - 65 = 115$$

Answer: 115°

Geometry › 3D Shapes › Properties of 3D Shapes (Foundation)

Topic 227 of 313 • id: properties_3d_shapes_F

Q1. How many faces does a cone have?

Q1

A face is a flat or curved surface on the outside of a 3D shape.

By GCSE convention, a cone has 2 faces (1 flat circular base + 1 curved surface), 1 edge (where the curved surface meets the circular base), and 1 vertex (the apex point at the top).

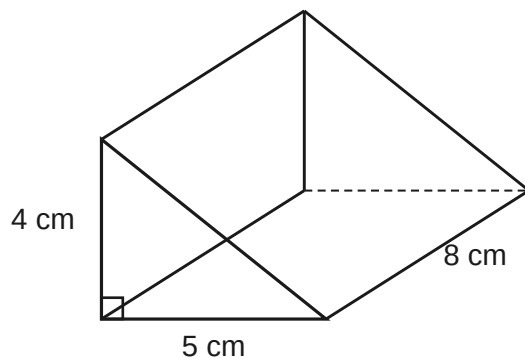
So a cone has 2 faces.

Answer: 2

Geometry › 3D Shapes › Properties of 3D Shapes (with Diagram) (Foundation)

Topic 228 of 313 • id: properties_3d_shapes_diagram_F

Q1. How many vertices does the triangular prism shown have?



Q1

A vertex is a corner point where edges meet.

A triangular prism has 5 faces (2 triangular ends + 3 rectangular sides), 9 edges, and 6 vertices (corners).

So a triangular prism has 6 vertices.

Answer: 6

Geometry › 3D Shapes › Nets of 3D Shapes (Foundation)

Topic 229 of 313 • id: nets_3d_shapes_F

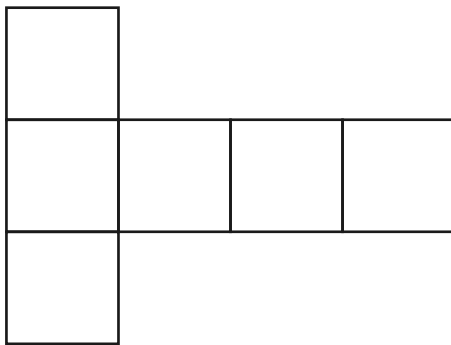
Q1. What 2D shapes make up the net of a cuboid?

Q1

Imagine unfolding a cuboid flat, so every one of its faces lies in the same plane with none overlapping - this flattened-out shape is called its net.

A cuboid's net is made up of: 6 rectangles (3 congruent pairs).

So the answer is: 6 rectangles (3 congruent pairs).

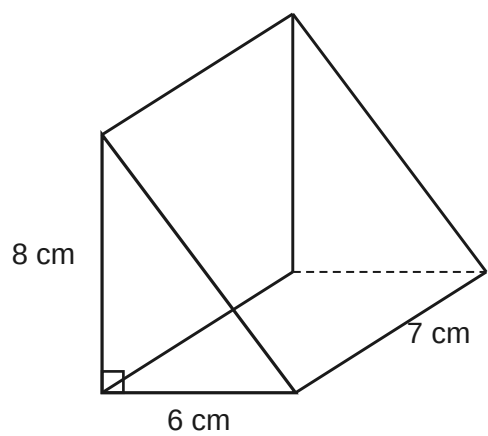


Answer: 6 rectangles (3 congruent pairs)

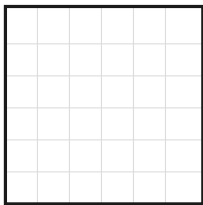
Geometry › 3D Shapes › Plans and Elevations (Foundation)

Topic 230 of 313 • id: plans_and_elevations_F

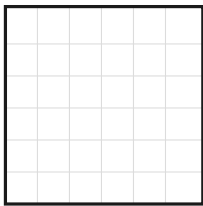
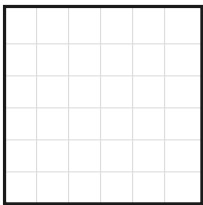
Q1. The diagram shows a solid with its dimensions labelled. Draw and label the front elevation, side elevation, and plan view of the solid.



Front elevation



Side elevation



Plan view

Q1

The front elevation is what you see looking at the solid straight on from the front.

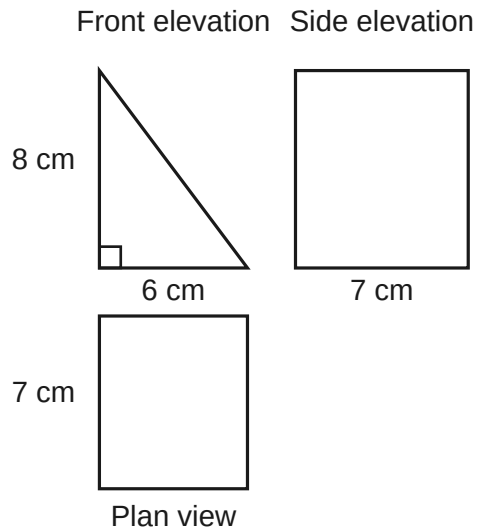
Front elevation: a right-angled triangle, base 6 cm and height 8 cm.

The side elevation is what you see looking at the solid straight on from the side.

Side elevation: a rectangle, 7 cm by 8 cm.

The plan view is what you see looking straight down at the solid from above.

Plan view: a rectangle, 6 cm by 7 cm.

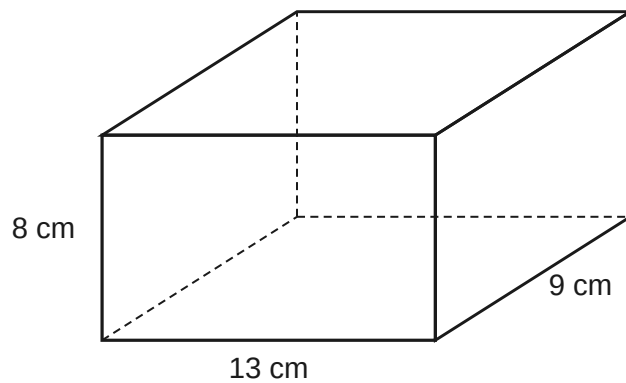


Answer: Front elevation: a right-angled triangle, base 6 cm and height 8 cm. Side elevation: a rectangle, 7 cm by 8 cm. Plan view: a rectangle, 6 cm by 7 cm.

Geometry › 3D Shapes › Volume & Surface Area of a Cuboid (Foundation)

Topic 231 of 313 • id: volume_surface_area_cuboid_F

Q1. A cuboid has length 9 cm, width 13 cm and height 8 cm. Find its volume.



Q1

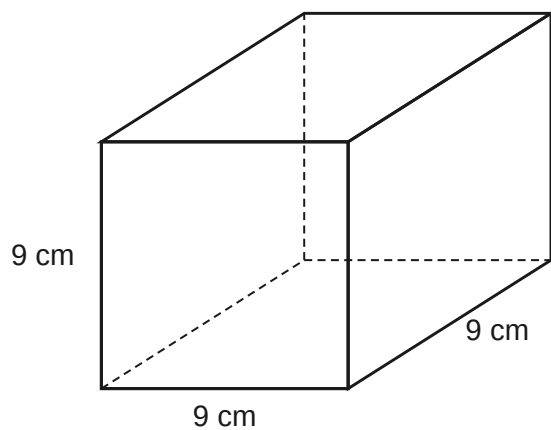
Volume = length $\times$ width $\times$ height = $9 \times 13 \times 8 = 936 \text{ cm}^3$

Answer: 936 cm^3

Geometry › 3D Shapes › Volume & Surface Area of a Cube (Foundation)

Topic 232 of 313 • id: volume_surface_area_cube_F

Q1. A cube has side length 9 cm. Find its surface area.



Q1

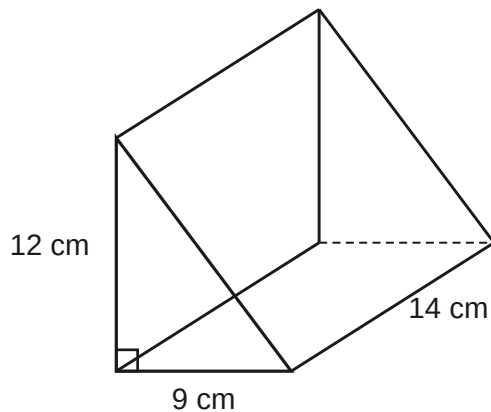
$$\text{Surface area} = 6 \times \text{side}^2 = 6 \times 9^2 = 6 \times 81 = 486 \text{ cm}^2$$

Answer: 486 cm²

Geometry › 3D Shapes › Volume & Surface Area of a Triangular Prism (Foundation)

Topic 233 of 313 • id: volume_surface_area_triangular_prism_F

Q1. A triangular prism has a right-angled triangular cross-section with sides 9 cm and 12 cm (hypotenuse 15 cm), and length 14 cm. Find its volume.



Q1

$$\text{Cross-section area} = \left(\frac{1}{2}\right) \times 9 \times 12 = 54 \text{ cm}^2$$

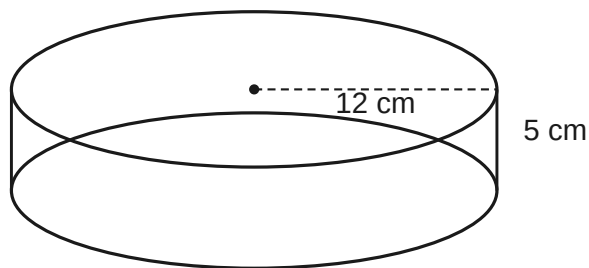
$$\text{Volume} = \text{cross-section area} \times \text{length} = 54 \times 14 = 756 \text{ cm}^3$$

Answer: 756 cm³

Geometry › 3D Shapes › Volume & Surface Area of a Cylinder (Calculator) (Foundation)

Topic 234 of 313 • id: volume_surface_area_cylinder_F

Q1. Find the total surface area of the cylinder, correct to 3 significant figures.



Q1

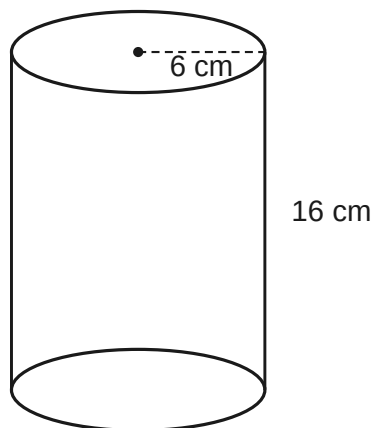
$$\begin{aligned}\text{Surface area} &= 2\pi r^2 + 2\pi rh = (2 \times \pi \times 12^2) + (2 \times \pi \times 12 \times 5) \\ &= 1280 \text{ cm}^2 \text{ (3 s.f., using a calculator value of } \pi)\end{aligned}$$

Answer: 1280 cm²

Geometry › 3D Shapes › Volume & Surface Area of a Cylinder (Higher)

Topic 235 of 313 • id: volume_surface_area_cylinder_H

Q1. Find the exact volume of the cylinder, in terms of π .



Q1

$$\text{Volume} = \pi \times r^2 \times h = \pi \times 6^2 \times 16$$

$$= 576\pi \text{ cm}^3$$

Answer: $576\pi \text{ cm}^3$

Geometry › 3D Shapes › Volume & Surface Area of a Cone (Higher)

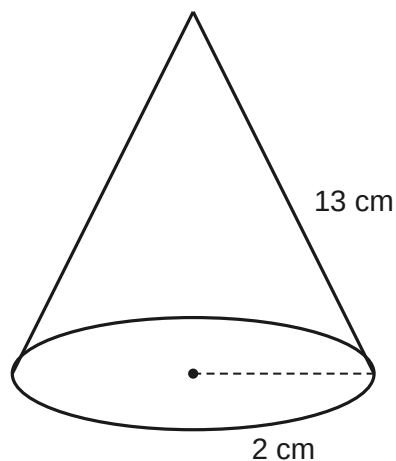
Topic 236 of 313 • id: volume_surface_area_cone_H

Formulae

Volume of a cone = $\left(\frac{1}{3}\right) \times \pi \times r^2 \times h$

Surface area of a cone = $\pi r(r + l)$, where l is the slant height

Q1. A cone has radius 2 cm and slant height 13 cm. Find its volume, correct to 2 decimal places.



Q1

$$\begin{aligned}\text{Volume} &= \left(\frac{1}{3}\right) \times \pi \times r^2 \times h = \left(\frac{1}{3}\right) \times \pi \times 2^2 \times 12.845 \\ &= 53.81 \text{ cm}^3 \text{ (2 d.p., using a calculator value of } \pi)\end{aligned}$$

Answer: 53.81 cm³

Geometry › 3D Shapes › Volume & Surface Area of a Sphere (Higher)

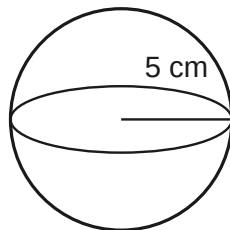
Topic 237 of 313 • id: volume_surface_area_sphere_H

Formulae

$$\text{Volume of a sphere} = \left(\frac{4}{3}\right) \times \pi \times r^3$$

$$\text{Surface area of a sphere} = 4 \times \pi \times r^2$$

Q1. A sphere has radius 5 cm. Find its volume, correct to 3 significant figures.



Q1

$$V = \left(\frac{4}{3}\right) \times \pi \times r^3 = \left(\frac{4}{3}\right) \times \pi \times 5^3$$

$$= 524 \text{ cm}^3 \text{ (3 s.f.)}$$

Answer: 524 cm³

Geometry › 3D Shapes › Volume & Surface Area of a Pyramid (Higher)

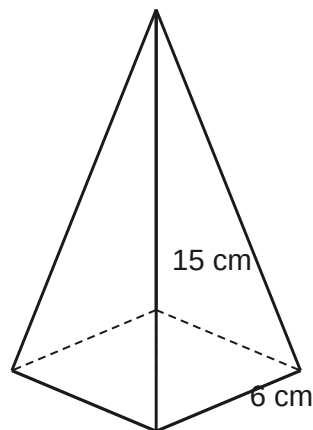
Topic 238 of 313 • id: volume_surface_area_pyramid_H

Formulae

Volume of a pyramid = $\left(\frac{1}{3}\right) \times \text{base}^2 \times \text{height}$

Surface area of a pyramid = $\text{base}^2 + 2 \times \text{base} \times l$, where l is the slant height of a face

Q1. A square-based pyramid has a base of side 6 cm and a perpendicular height of 15 cm. Find its total surface area, correct to 2 decimal places.



Q1

Slant height $l = \sqrt{(\text{height}^2 + (\text{base} \div 2)^2)} = \sqrt{(15^2 + (6 \div 2)^2)} \approx 15.30 \text{ cm}$

Surface area = $\text{base}^2 + 2 \times \text{base} \times l = 6^2 + 2 \times 6 \times 15.30$

= 219.56 cm^2 (2 d.p.)

Answer: 219.56 cm^2

Geometry › 3D Shapes › Volume & Surface Area of a Frustum (Higher)

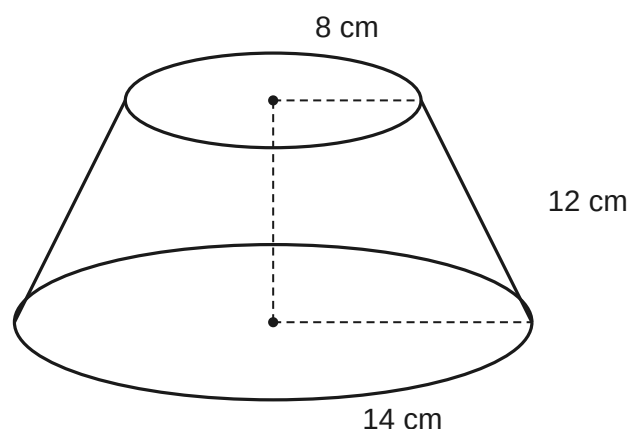
Topic 239 of 313 • id: frustum_volume_surface_area_H

Formulae

Volume of a frustum = $\left(\frac{1}{3}\right) \times \pi \times h \times (R^2 + Rr + r^2)$, where R and r are the two radii

Surface area of a frustum = $\pi(R + r)l + \pi R^2 + \pi r^2$, where l is the slant height

Q1. A frustum is formed from a cone with a bottom radius of 14 cm and a top radius of 8 cm, and a vertical height of 12 cm. Find its volume, correct to 2 decimal places.



Q1

Full cone height $H = R \times h \div (R - r) = 14 \times 12 \div (14 - 8) = 28.00$ cm (similar triangles)

Big cone volume = $\left(\frac{1}{3}\right) \times \pi \times R^2 \times H$, small cone volume = $\left(\frac{1}{3}\right) \times \pi \times r^2 \times (H - h)$

Frustum volume = $\left(\frac{1}{3}\right) \times \pi \times h \times (R^2 + Rr + r^2) = \left(\frac{1}{3}\right) \times \pi \times 12 \times (14^2 + 14 \times 8 + 8^2)$

= 4674.69 cm³ (2 d.p.)

Answer: 4674.69 cm³

Geometry › 3D Shapes › Compound 3D Shapes (Volume) (Higher)

Topic 240 of 313 • id: compound_3d_volume_H

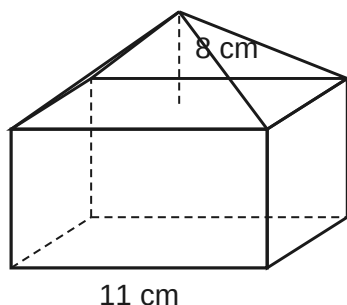
Formulae

Volume of a cone = $\left(\frac{1}{3}\right) \times \pi \times r^2 \times h$

Volume of a sphere = $\left(\frac{4}{3}\right) \times \pi \times r^3$ (a hemisphere is half this)

Volume of a pyramid = $\left(\frac{1}{3}\right) \times \text{base}^2 \times \text{height}$

Q1. A silo-shaped solid is made from a cuboid with a square base of side 11 cm and height 12 cm, with a square-based pyramid of height 8 cm on top (sharing the same square base). Find the total volume.



Q1

Cuboid volume = $\text{base}^2 \times \text{height} = 11^2 \times 12 = 1452 \text{ cm}^3$

Pyramid volume = $\left(\frac{1}{3}\right) \times \text{base}^2 \times \text{height} = \left(\frac{1}{3}\right) \times 11^2 \times 8 = \frac{968}{3} \text{ cm}^3$

Total volume = $1452 + \frac{968}{3} = \frac{5324}{3} \text{ cm}^3$

Answer: $\frac{5324}{3} \text{ cm}^3$

Geometry › 3D Shapes › Compound 3D Shapes (Surface Area) (Higher)

Topic 241 of 313 • id: compound_3d_surface_area_H

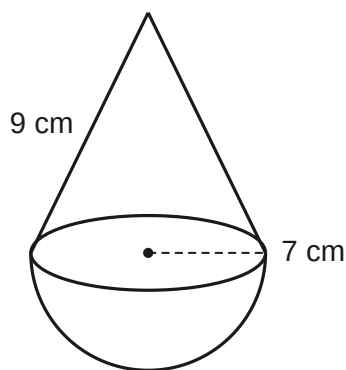
Formulae

Curved surface area of a cone = $\pi r l$, where l is the slant height

Curved surface area of a sphere = $4 \times \pi \times r^2$ (a hemisphere's curved part is half this)

Surface area of a pyramid's triangular faces = $2 \times \text{base} \times l$, where l is a face's slant height

Q1. An ice-cream-shaped solid is made from a cone of radius 7 cm and height 9 cm, with a hemisphere of the same radius attached to its open end. Find the total surface area, correct to 1 decimal place.



Q1

Slant height $l = \sqrt{(r^2 + h^2)} = \sqrt{(7^2 + 9^2)} \approx 11.4$ cm

Cone curved surface = $\pi \times r \times l = \pi \times 7 \times 11.4$

Hemisphere curved surface = $2 \times \pi \times r^2 = 2 \times \pi \times 7^2$ (the cone's base and the hemisphere's flat face are both glued together internally, so neither is exposed)

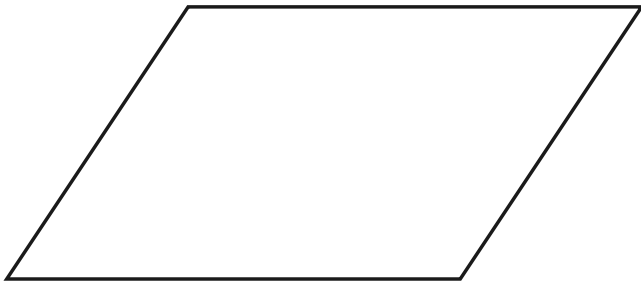
Total surface area ≈ 558.6 cm² (1 d.p.)

Answer: 558.6 cm²

Geometry › Symmetry › Line Symmetry (Foundation)

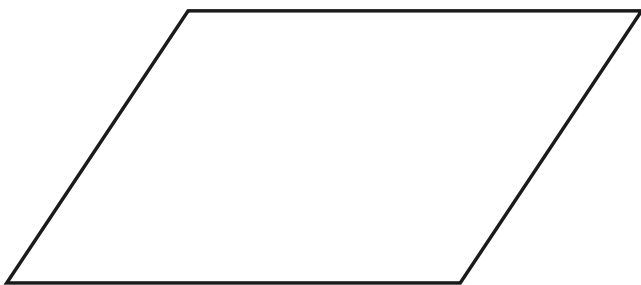
Topic 242 of 313 • id: symmetry_lines_F

Q1. The diagram shows a parallelogram. How many lines of symmetry does it have?



Q1

A parallelogram (of this kind) has no lines of symmetry - no mirror line maps it back onto itself.

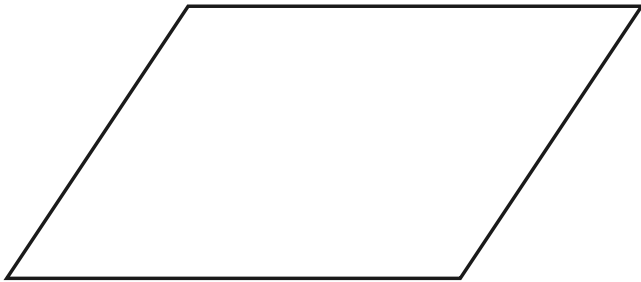


Answer: 0 lines of symmetry

Geometry › Symmetry › Rotational Symmetry (Foundation)

Topic 243 of 313 • id: symmetry_rotational_F

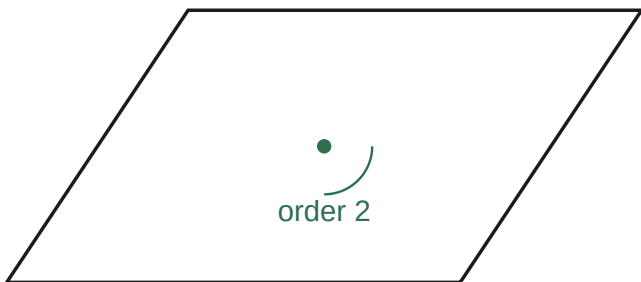
Q1. The diagram shows a parallelogram. State the order of rotational symmetry.



Q1

Rotating the parallelogram about its centre, it looks exactly the same 2 times during a full 360° turn.

So the order of rotational symmetry is 2.

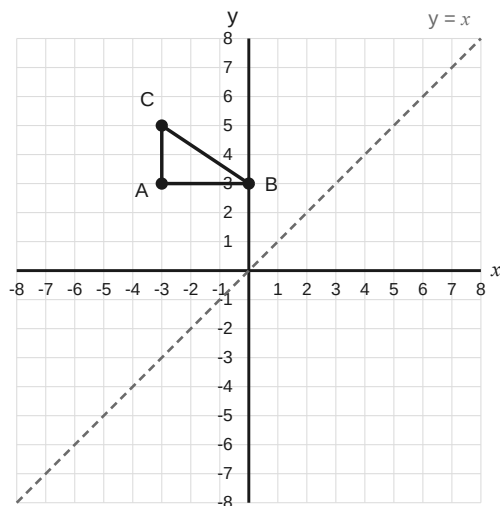


Answer: order 2

Geometry › Transformations › Reflection (Foundation)

Topic 244 of 313 • id: transform_reflect_complete_F

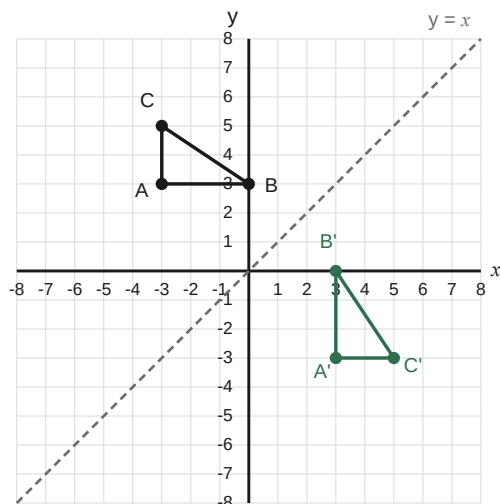
Q1. Reflect shape ABC in the line $y = x$.



Q1

Reflect each vertex in the line $y = x$: the mirror line is the perpendicular bisector of every point and its image, so each point stays the same distance from the line, on the other side.

$A(-3, 3) \rightarrow A'(3, -3)$, $B(0, 3) \rightarrow B'(3, 0)$, $C(-3, 5) \rightarrow C'(5, -3)$

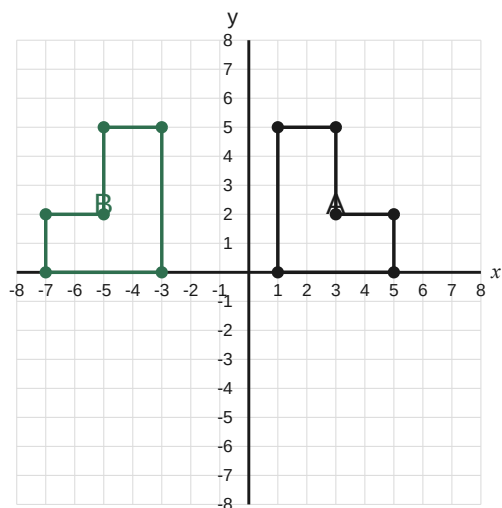


Answer: $A'(3, -3)$, $B'(3, 0)$, $C'(5, -3)$

Geometry › Transformations › Describe a Reflection (Foundation)

Topic 245 of 313 • id: transform_reflect_describe_F

Q1. Describe fully the single transformation that maps shape A onto shape B, shown on the grid.



Q1

Every vertex and its image are the same distance from a single line, on opposite sides of it - that line is the mirror line.

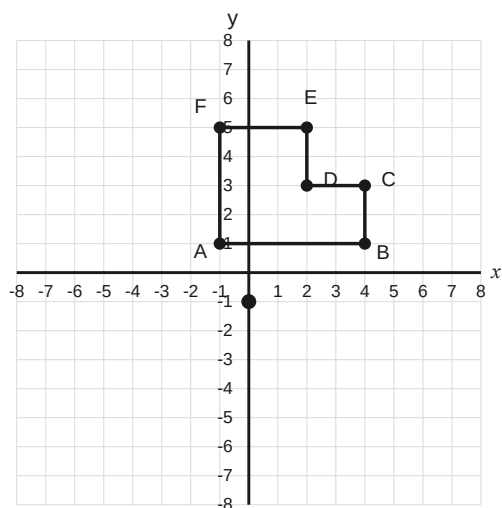
Comparing corresponding vertices shows the mirror line is $x = -1$.

Answer: Reflection in the line $x = -1$

Geometry › Transformations › Rotation (Foundation)

Topic 246 of 313 • id: transform_rotate_complete_F

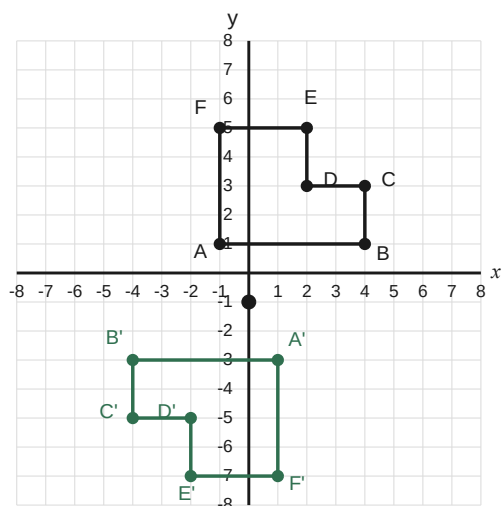
Q1. Rotate shape ABCDEF 180° about the centre (0, -1).



Q1

Rotate each vertex 180° about the centre (0, -1): every point stays the same distance from the centre and turns through the same angle.

A(-1, 1) → A'(1, -3), B(4, 1) → B'(-4, -3), C(4, 3) → C'(-4, -5), D(2, 3) → D'(-2, -5), E(2, 5) → E'(-2, -7), F(-1, 5) → F'(1, -7)

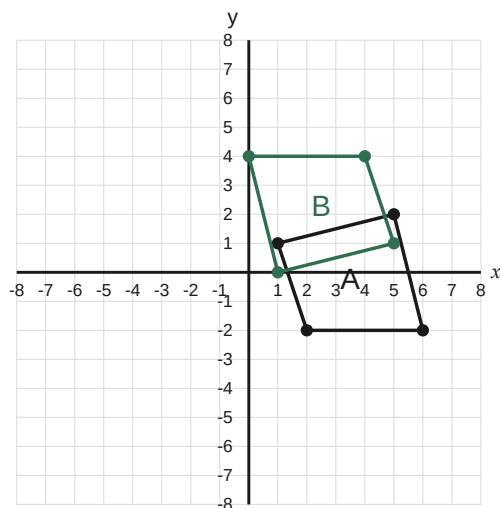


Answer: A'(1, -3), B'(-4, -3), C'(-4, -5), D'(-2, -5), E'(-2, -7), F'(1, -7)

Geometry › Transformations › Describe a Rotation (Higher)

Topic 247 of 313 • id: transform_rotate_describe_H

Q1. Describe fully the single transformation that maps shape A onto shape B, shown on the grid.



Q1

The centre of rotation is the one point that is the same distance from a vertex and its image, for every vertex - it can be found where the perpendicular bisectors of two such pairs cross.

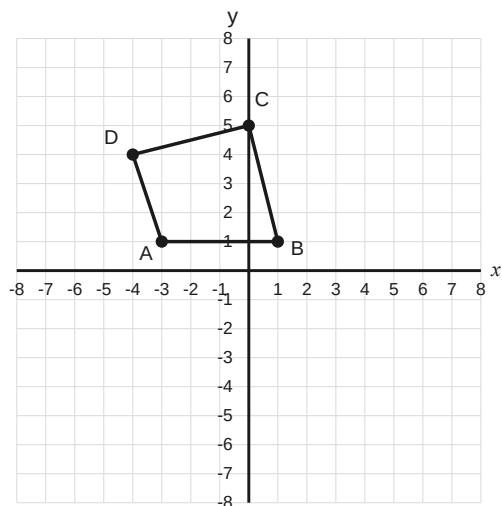
Measuring the turn from a vertex to its image about that point gives 180° about (3, 1).

Answer: Rotation 180° about (3, 1)

Geometry › Transformations › Translation (Foundation)

Topic 248 of 313 • id: transform_translate_complete_F

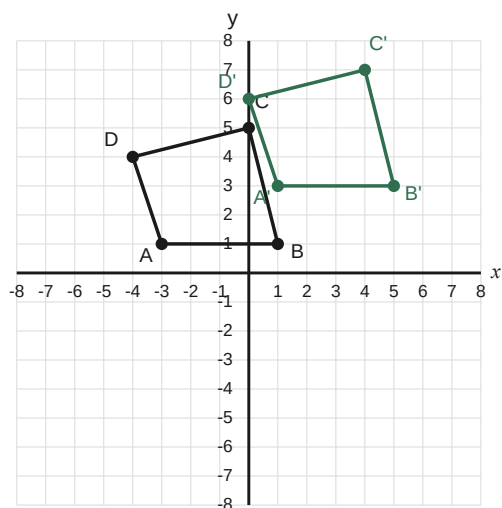
Q1. Translate shape ABCD by the vector $(4, 2)$.



Q1

Translate every vertex by the vector $(4, 2)$: move each point 4 squares right and 2 squares up.

$A(-3, 1) \rightarrow A'(1, 3)$, $B(1, 1) \rightarrow B'(5, 3)$, $C(0, 5) \rightarrow C'(4, 7)$, $D(-4, 4) \rightarrow D'(0, 6)$

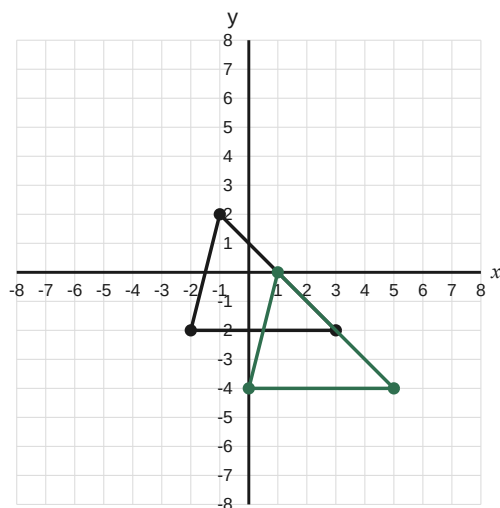


Answer: $A'(1, 3)$, $B'(5, 3)$, $C'(4, 7)$, $D'(0, 6)$

Geometry › Transformations › Describe a Translation (Foundation)

Topic 249 of 313 • id: transform_translate_describe_F

Q1. Describe fully the single transformation that maps shape A onto shape B, shown on the grid.



Q1

Every vertex moves by exactly the same amount: comparing a vertex of shape A, $(-2, -2)$, with the matching vertex of shape B, $(0, -4)$, gives the vector $(2, -2)$.

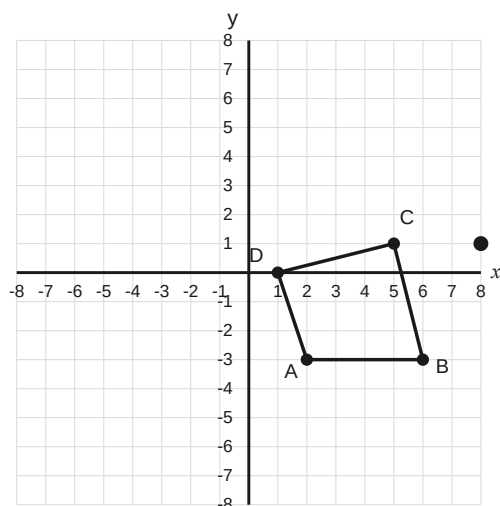
Checking the other vertices confirms the same vector applies throughout, so this is a translation.

Answer: Translation by the vector $(2, -2)$

Geometry › Transformations › Enlargement (Foundation)

Topic 250 of 313 • id: transform_enlarge_complete_F

Q1. Enlarge shape ABCD by scale factor 2, centre (8, 1).

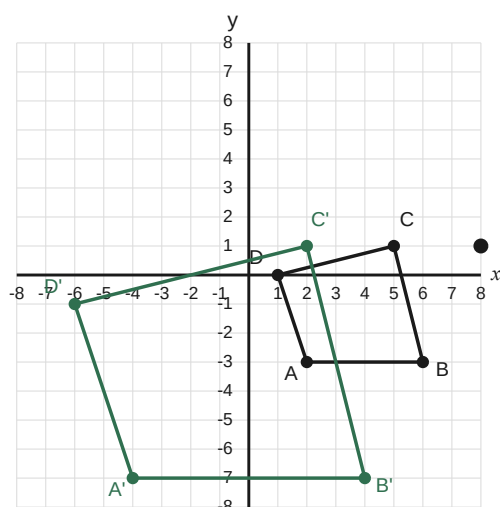


Q1

The centre of enlargement is (8, 1) and the scale factor is 2.

For each vertex, find its displacement from the centre and multiply by the scale factor: A is a displacement of (-6, -4) from the centre, so $A' = (8, 1) + 2 \times (-6, -4) = (-4, -7)$.

Repeat for every vertex: $A(2, -3) \rightarrow A'(-4, -7)$, $B(6, -3) \rightarrow B'(4, -7)$, $C(5, 1) \rightarrow C'(2, 1)$, $D(1, 0) \rightarrow D'(-6, -1)$

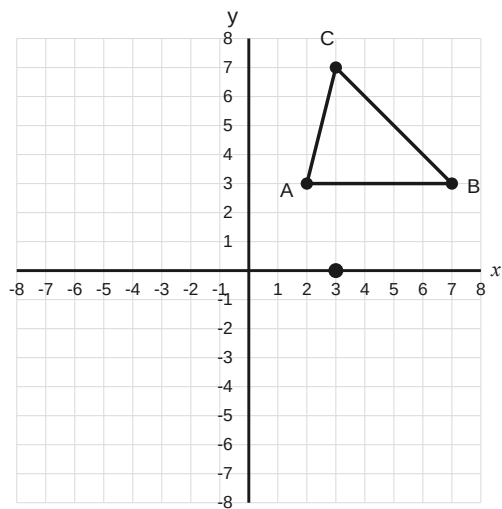


Answer: $A'(-4, -7)$, $B'(4, -7)$, $C'(2, 1)$, $D'(-6, -1)$

Geometry › Transformations › Enlargement (Higher)

Topic 251 of 313 • id: transform_enlarge_complete_H

Q1. Enlarge shape ABC by scale factor -1 , centre $(3, 0)$.



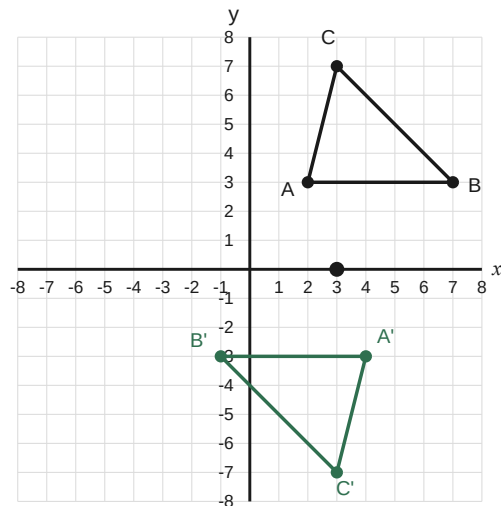
Q1

The centre of enlargement is (3, 0) and the scale factor is -1.

For each vertex, find its displacement from the centre and multiply by the scale factor: A is a displacement of (-1, 3) from the centre, so $A' = (3, 0) + -1 \times (-1, 3) = (4, -3)$.

Repeat for every vertex: $A(2, 3) \rightarrow A'(4, -3)$, $B(7, 3) \rightarrow B'(-1, -3)$, $C(3, 7) \rightarrow C'(3, -7)$

A negative scale factor puts the image on the opposite side of the centre from the original shape.

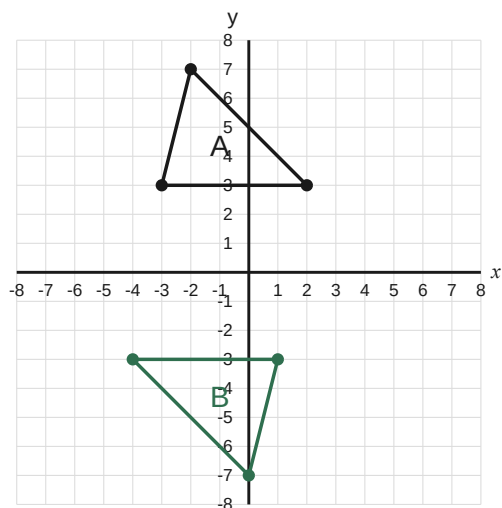


Answer: $A'(4, -3)$, $B'(-1, -3)$, $C'(3, -7)$

Geometry › Transformations › Describe an Enlargement (Higher)

Topic 252 of 313 • id: transform_enlarge_describe_H

Q1. Describe fully the single transformation that maps shape A onto shape B, shown on the grid.



Q1

The centre of enlargement is where lines joining each vertex to its image all meet (extended if needed) - the scale factor is the ratio of a side length in the image to the matching side in the original.

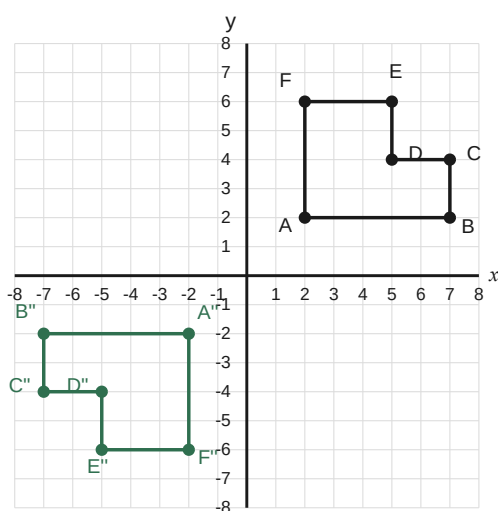
For this pair, the lines meet at $(-1, 0)$ and the ratio works out as -1 .

Answer: Enlargement, scale factor -1 , centre $(-1, 0)$

Geometry › Transformations › Combined Transformations (Higher)

Topic 253 of 313 • id: combined_transformations_H

Q1. Shape ABCDEF is reflected in the y -axis to give shape A'B'C'D'E'F'. Shape A'B'C'D'E'F' is then reflected in the x -axis to give shape A''B''C''D''E''F''. Describe fully the single transformation that maps ABCDEF directly onto A''B''C''D''E''F''.



Q1

Reflecting in the x -axis then the y -axis (or the other way round - the order doesn't matter here) sends every point (x, y) to $(-x, -y)$.

That is exactly the same effect as a single 180° rotation about the origin.

Answer: Rotation 180° about $(0, 0)$

Geometry › Bearings › Bearings (Foundation)

Topic 254 of 313 • id: bearings_F

Q1. The diagram shows the bearing of B from A. Write down the bearing of B from A as a three-figure bearing.



Q1

A bearing is always written as three figures, measured clockwise from north.

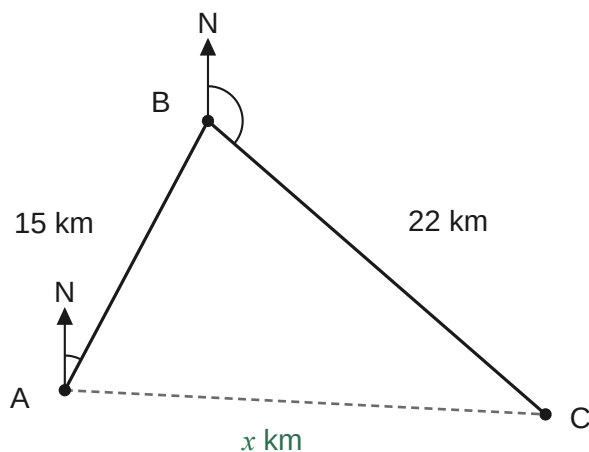
Reading the angle shown from north: 187° , written as a three-figure bearing: 187° .

Answer: 187°

Geometry › Bearings › Bearings (Higher)

Topic 255 of 313 • id: bearings_cosine_rule_H

Q1. A ship sails from port A on a bearing of 028° for 15 km to point B. It then changes course and sails on a bearing of 131° for 22 km to point C. Find the direct distance from A to C, correct to 3 significant figures.



Q1

The back bearing of A from B is $028^\circ + 180^\circ = 208^\circ$.

Angle ABC is the difference between this back bearing and the bearing of C from B: angle ABC = 77° .

Use the cosine rule: $AC^2 = AB^2 + BC^2 - 2 \times AB \times BC \times \cos(ABC)$

$$AC^2 = 15^2 + 22^2 - 2 \times 15 \times 22 \times \cos(77^\circ) = 560.5$$

$$AC = \sqrt{560.5} = 23.7 \text{ km (3 s.f.)}$$

Answer: 23.7 km

Geometry › Map Scales and Scale Drawings › Map Scales and Scale Drawings (Foundation)

Topic 256 of 313 • id: map_scale_drawings_F

Q1. A map has a scale of 1 cm represents 5 km. Two towns are 65 km apart in real life. Find the distance between the towns on the map, in cm.

Q1

The scale means 1 cm on the map represents 5 km in real life.

Map distance = $65 \div 5 = 13$ cm

Answer: 13 cm

Geometry › Constructions › Constructing an Angle Bisector (Foundation)

Topic 257 of 313 • id: construction_angle_bisector_F

Q1. Angle $DYV = 76^\circ$. Describe, step by step, how to construct the bisector of angle DYV using only a ruler and a pair of compasses.

Q1

Open the compasses to any radius and place the point on Y (the vertex of the angle).

Draw an arc that crosses both arms of the angle, at a point on YD and a point on YV.

Without changing the compass radius, place the point on each of these two crossing points in turn and draw two more arcs so that they intersect each other.

Draw a straight line from Y through the point where the two arcs intersect. This line is the bisector of angle DYV .

Answer: The straight line from Y through the intersection of the two equal-radius arcs bisects angle DYV into two equal angles.

Geometry › Constructions › Constructing a Perpendicular Bisector (Foundation)

Topic 258 of 313 • id: construction_perpendicular_bisector_F

Q1. Line segment AK has length 9 cm. Describe, step by step, how to construct the perpendicular bisector of AK using only a ruler and a pair of compasses.

Q1

Open the compasses to a radius more than half of 9 cm (more than half the length of AK).

With the compass point on A, draw an arc above and below the line.

Without changing the compass radius, place the point on K and draw two more arcs, crossing the first two.

Draw a straight line through the two points where the arcs intersect. This line is the perpendicular bisector of AK.

Answer: A straight line through the two arc intersections, crossing AK at its midpoint at a right angle.

Geometry › Constructions › Constructing a Perpendicular From/At a Point (Foundation)

Topic 259 of 313 • id: construction_perpendicular_from_point_F

Q1. Point L lies on line segment RQ (length 5 cm). Describe, step by step, how to construct the perpendicular to RQ at L using only a ruler and a pair of compasses.

Q1

With the compass point on L, draw two arcs of equal radius along the line RQ, one on each side of L, marking two new points that are the same distance from L.

Open the compasses to a larger radius than before.

With the point on each of the two new points in turn, draw two arcs above (or below) the line so that they intersect each other.

Draw a straight line from L through the point where these two arcs intersect. This line is perpendicular to RQ at L.

Answer: The straight line from L through the intersection of the two larger-radius arcs is perpendicular to RQ at L.

Geometry › Constructions › Constructing a Triangle (Foundation)

Topic 260 of 313 • id: construction_triangle_F

Q1. Triangle FPE has $FP = 10$ cm, $PE = 5$ cm, and $FE = 8$ cm. Describe, step by step, how to construct triangle FPE using only a ruler and a pair of compasses.

Q1

Draw the line segment $FP = 10$ cm using a ruler.

Open the compasses to 8 cm. With the point on F, draw an arc above the line.

Open the compasses to 5 cm. With the point on P, draw a second arc that crosses the first.

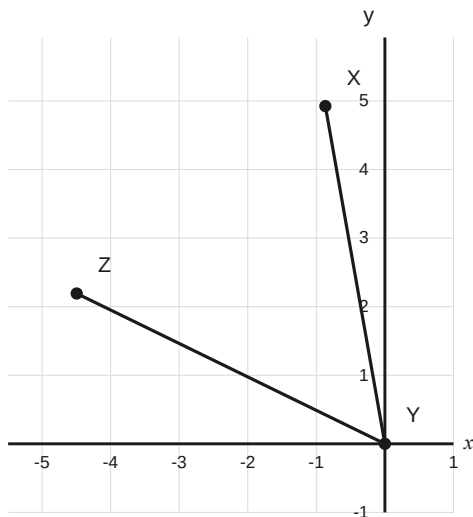
Label the point where the two arcs cross as E, then draw straight lines from F to E and from P to E to complete the triangle.

Answer: Triangle FPE, with E located at the intersection of the two compass arcs.

Geometry › Loci › Loci (Foundation)

Topic 261 of 313 • id: loci_constructions_F

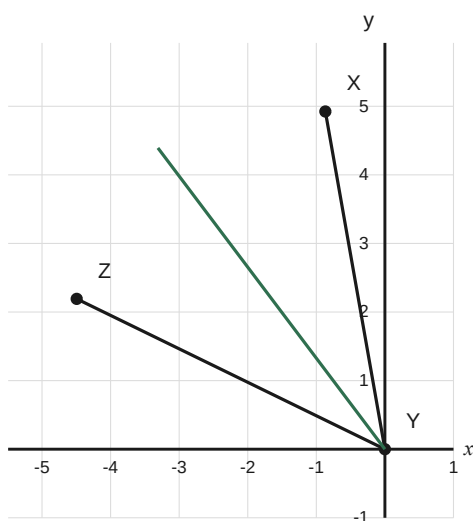
Q1. Two straight lines meet at Y, passing through X and Z as shown. Describe the locus of points that are equidistant from line XY and line ZY.



Q1

The locus of points equidistant from two intersecting lines is the bisector of the angle between them.

So the locus is the bisector of angle XYZ: the line from Y that splits the angle into two equal halves.

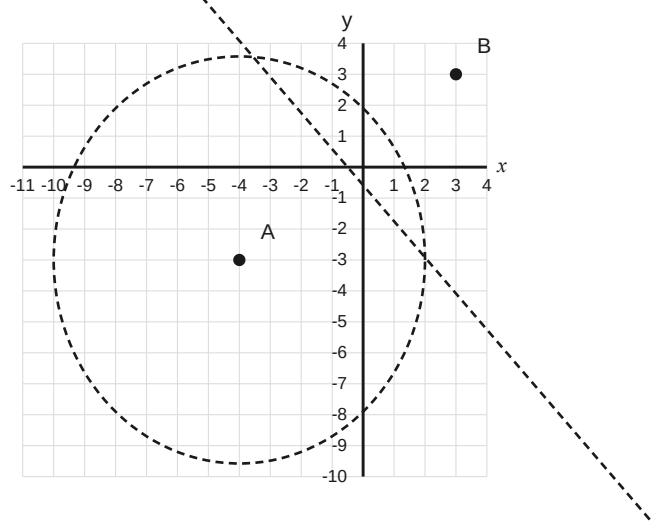


Answer: The angle bisector of angle XYZ.

Geometry › Loci › Loci Regions (Higher)

Topic 262 of 313 • id: loci_regions_H

Q1. A and B are two fixed points (shown). Shade the region containing points that are both within 6 cm of A AND closer to A than to B.

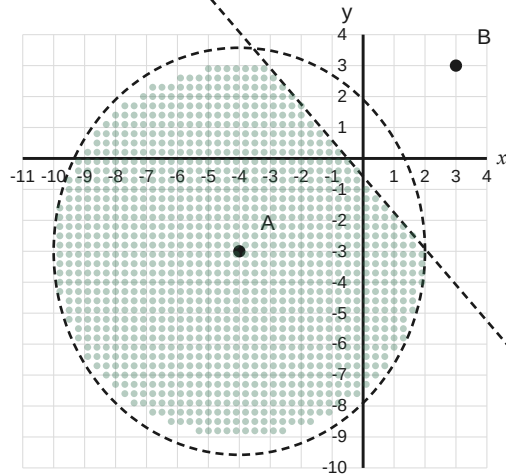


Q1

The boundary for 'within 6 cm of A' is a circle, centre A, radius 6 cm - the region is inside this circle.

The boundary for 'closer to A than to B' is the perpendicular bisector of AB - the region is on the A side of it.

The required region is where these two conditions overlap.

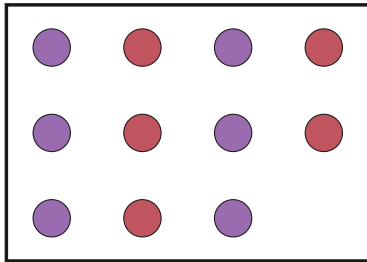


Answer: The region inside the circle of radius 6 cm centred on A, and on the A side of the perpendicular bisector of AB.

Probability › Probability › Single Event (Foundation)

Topic 263 of 313 • id: probability_single_event_F

Q1. A bag contains 6 purple, 5 red counters. A counter is picked at random. Find the probability that it is red.



Q1

Total counters = 11

Number of red counters = 5

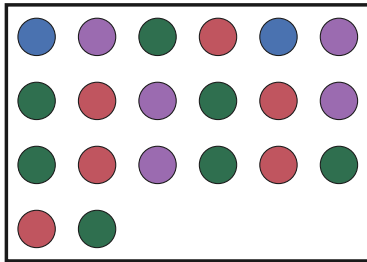
$$P(\text{red}) = \frac{5}{11} = \frac{5}{11}$$

Answer: $\frac{5}{11}$

Probability › Probability › Complement (Foundation)

Topic 264 of 313 • id: probability_complement_F

Q1. A bag contains 2 blue, 5 purple, 7 green, 6 red counters. A counter is picked at random. Find the probability that it is NOT green.



Q1

$$P(\text{green}) = \frac{7}{20} = \frac{7}{20}$$

$$P(\text{not green}) = 1 - \frac{7}{20} = \frac{13}{20}$$

Answer: $\frac{13}{20}$

Probability › Probability › Combined Dice (Higher)

Topic 265 of 313 • id: probability_combined_dice_H

Q1. Two fair six-sided dice are rolled. Find the probability that at least one die shows 4.

Q1

$$P(\text{neither die shows 4}) = \frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$$

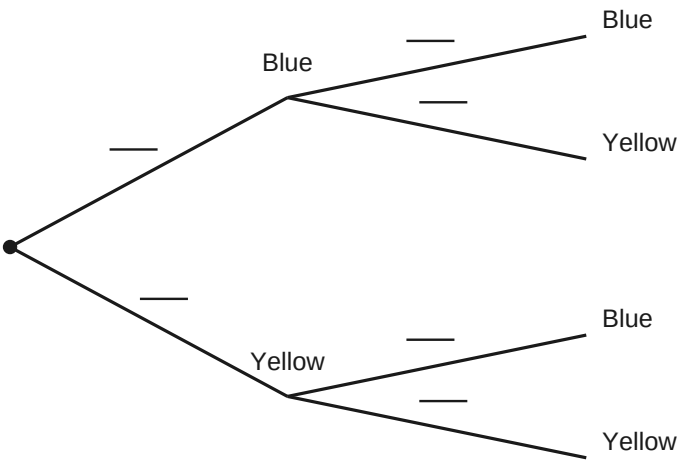
$$P(\text{at least one 4}) = 1 - \frac{25}{36} = \frac{11}{36}$$

Answer: $\frac{11}{36}$

Probability › Probability › Conditional (Without Replacement) (Higher)

Topic 266 of 313 • id: probability_conditional_H

Q1. A bag contains 5 yellow and 6 blue counters. Two counters are picked at random without replacement. Find the probability that both counters are the same colour. Use the tree diagram to help you.

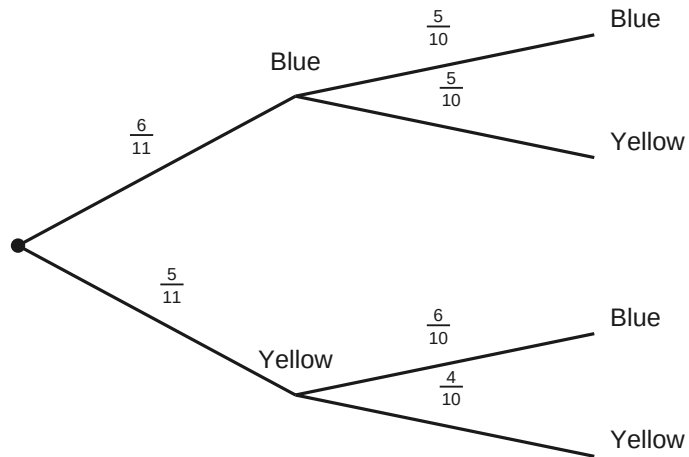


Q1

$$P(\text{both yellow}) = \frac{5}{11} \times \frac{4}{10} = \frac{2}{11}$$

$$P(\text{both blue}) = \frac{6}{11} \times \frac{5}{10} = \frac{3}{11}$$

$$P(\text{both same colour}) = \frac{2}{11} + \frac{3}{11} = \frac{5}{11}$$

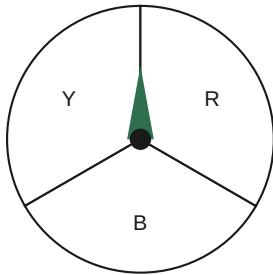


Answer: $\frac{5}{11}$

Probability › Probability › Listing Outcomes (Foundation)

Topic 267 of 313 • id: probability_listing_outcomes_F

Q1. A coin is flipped and a 3-sided spinner (labelled Y, B, R) is spun. List all the possible outcomes.



Q1

There are 2 possible outcomes for the first event and 3 for the second, so there are $2 \times 3 = 6$ possible combined outcomes.

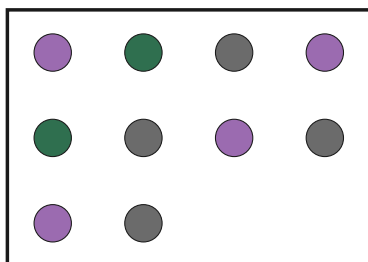
For each outcome of the first event, list every outcome of the second event: HY, HB, HR, TY, TB, TR

Answer: HY, HB, HR, TY, TB, TR

Probability › Probability › AND / OR Rule (Foundation)

Topic 268 of 313 • id: probability_and_or_rule_F

Q1. A bag contains coloured counters. $P(\text{purple}) = \frac{4}{10}$. $P(\text{green}) = \frac{2}{10}$. A counter cannot be both colours. Find $P(\text{purple or green})$.



Q1

A counter cannot be both purple and green, so these events are mutually exclusive.

For mutually exclusive events, $P(A \text{ or } B) = P(A) + P(B)$.

$$P(\text{purple or green}) = \frac{4}{10} + \frac{2}{10} = \frac{3}{5}$$

Answer: $\frac{3}{5}$

Probability › Probability › Expected Frequency (Foundation)

Topic 269 of 313 • id: probability_expectation_F

Q1. A biased coin has $P(\text{heads}) = \frac{5}{8}$. The coin is flipped 120 times. How many times would you expect to get heads?

Q1

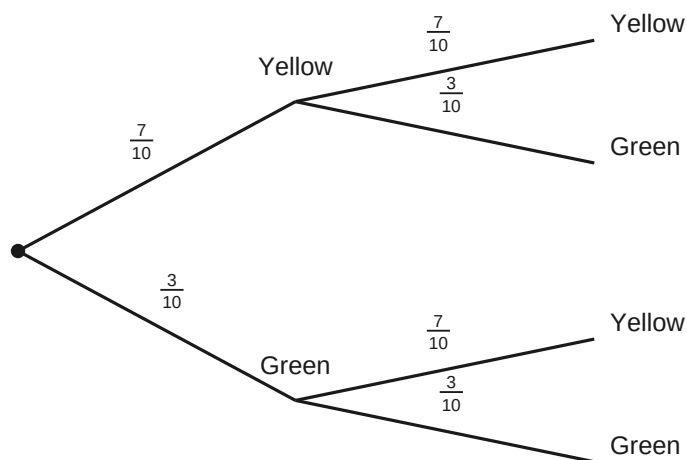
Expected number of times = probability $\times$ number of trials = $\frac{5}{8} \times 120$
= 75

Answer: 75

Probability › Tree Diagrams › Interpreting Tree Diagrams (Independent Events) (Foundation)

Topic 270 of 313 • id: tree_diagram_independent_F

Q1. A bag contains 3 green and 7 yellow counters. A counter is picked at random, replaced, and then a second counter is picked at random. The tree diagram shows this information. Find the probability that both counters are the same colour.



Q1

A counter is picked, replaced, then a second counter is picked. The probabilities stay the same each time: $P(\text{green}) = \frac{3}{10}$, $P(\text{yellow}) = \frac{7}{10}$.

$$P(\text{both green}) = \frac{3}{10} \times \frac{3}{10} = \frac{9}{100}$$

$$P(\text{both yellow}) = \frac{7}{10} \times \frac{7}{10} = \frac{49}{100}$$

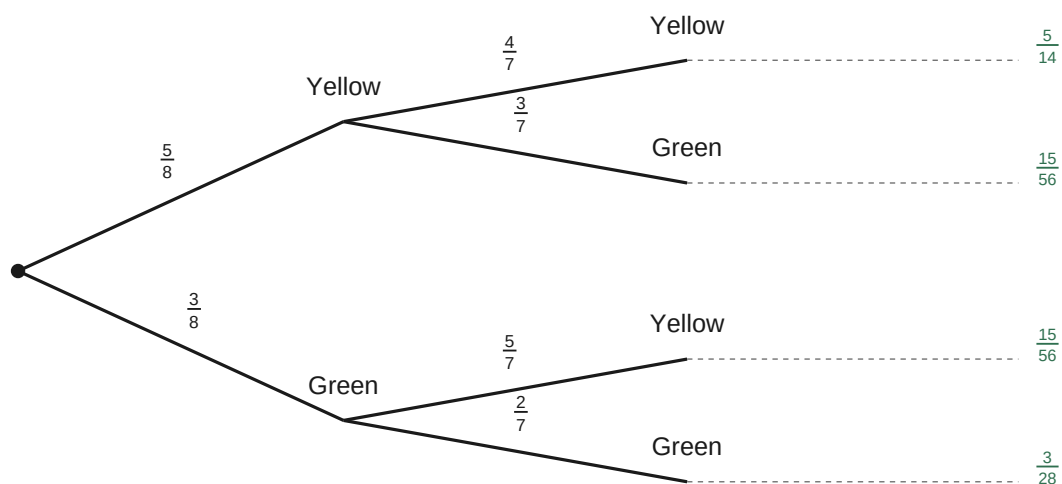
$$P(\text{same colour}) = \frac{9}{100} + \frac{49}{100} = \frac{29}{50}$$

Answer: $\frac{29}{50}$

Probability › Tree Diagrams › Interpreting Tree Diagrams (Dependent Events) (Higher)

Topic 271 of 313 • id: tree_diagram_dependent_H

Q1. A bag contains 3 green and 5 yellow counters. Two counters are picked at random, one after the other, without replacement. The tree diagram shows this information. Find the probability that the first counter is yellow and the second counter is yellow.



Q1

After the first counter is picked (without replacement), there are 7 counters left.

$$P(\text{green then green}) = \frac{3}{8} \times \frac{2}{7} = \frac{3}{28}$$

$$P(\text{green then yellow}) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$$

$$P(\text{yellow then green}) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56}$$

$$P(\text{yellow then yellow}) = \frac{5}{8} \times \frac{4}{7} = \frac{5}{14}$$

$$P(\text{the first counter is yellow and the second counter is yellow}) = \frac{5}{14}$$

Answer: $\frac{5}{14}$

Probability › Tree Diagrams › Drawing Tree Diagrams (Foundation)

Topic 272 of 313 • id: tree_diagram_drawing_F

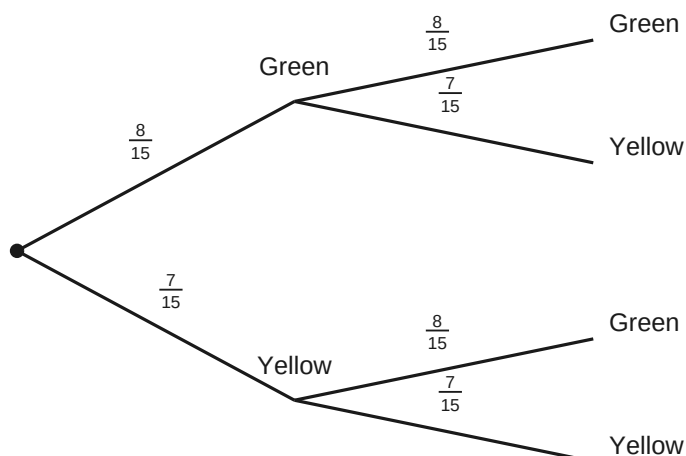
Q1. A bag contains 7 yellow and 8 green counters. A counter is picked at random, replaced, and then a second counter is picked at random. Draw a fully-labelled tree diagram to show this information, and use it to find the probability that the first counter is green and the second counter is yellow.

Q1

Draw two branches from the start: $P(\text{yellow}) = \frac{7}{15}$, $P(\text{green}) = \frac{8}{15}$.

From each of those branches, draw two more with the same probabilities (the counter is replaced).

$$P(\text{green then yellow}) = \frac{8}{15} \times \frac{7}{15} = \frac{56}{225}$$

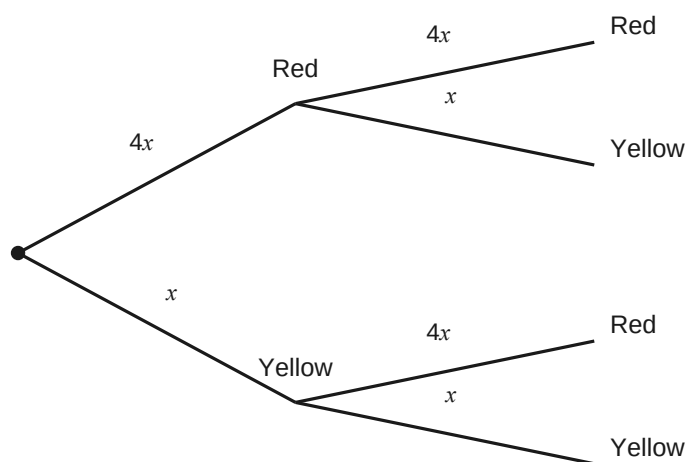


Answer: $\frac{56}{225}$

Probability › Tree Diagrams › Tree Diagrams with Algebraic Probabilities (Higher)

Topic 273 of 313 • id: tree_diagram_algebraic_H

Q1. A biased spinner can only land on yellow or red. The probability that it lands on yellow is x , and the probability that it lands on red is $4x$. The spinner is spun twice. Find the probability that both spins land on red.



Q1

A biased spinner can only land on yellow or red. The probability that it lands on yellow is x , and the probability that it lands on red is $4x$. Since the spinner can only land on yellow or red, the two probabilities must sum to 1: $x + 4x = 1$.

Solving this equation gives $x = \frac{1}{5}$.

Substituting back: $P(\text{yellow}) = \frac{1}{5}$, $P(\text{red}) = \frac{4}{5}$.

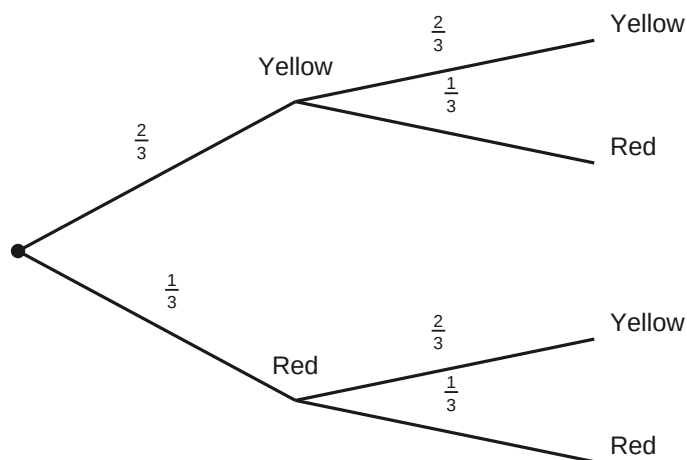
$P(\text{both spins land on red}) = \frac{4}{5} \times \frac{4}{5} = \frac{16}{25}$

Answer: $\frac{16}{25}$

Probability › Tree Diagrams › Mixed Tree Diagrams (Higher)

Topic 274 of 313 • id: tree_diagram_mixed_H

Q1. A bag contains 4 red and 8 yellow counters. A counter is picked at random, replaced, and then a second counter is picked at random. The tree diagram shows this information. Find the probability that both counters are the same colour.



Q1

A counter is picked, replaced, then a second counter is picked. The probabilities stay the same each time: $P(\text{red}) = \frac{1}{3}$, $P(\text{yellow}) = \frac{2}{3}$.

$$P(\text{both red}) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

$$P(\text{both yellow}) = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$$

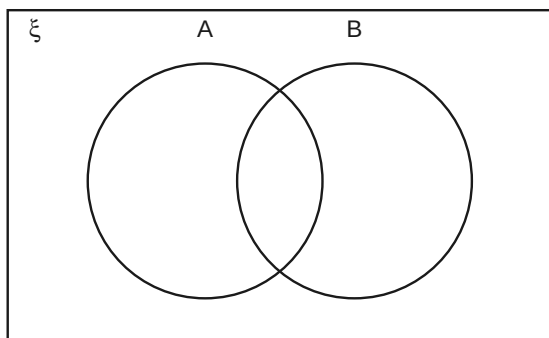
$$P(\text{same colour}) = \frac{1}{9} + \frac{4}{9} = \frac{5}{9}$$

Answer: $\frac{5}{9}$

Probability › Sets and Counting › Set Notation (Higher)

Topic 275 of 313 • id: set_notation_H

Q1. $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$, $A = \{1, 3, 4, 5, 7\}$, $B = \{2, 6, 8\}$. List the elements of $(A \cap B)'$.



Q1

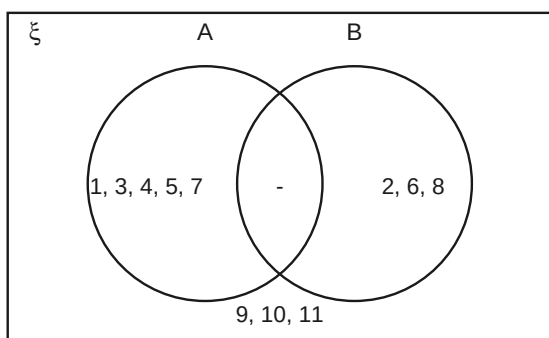
$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

$$A = \{1, 3, 4, 5, 7\}$$

$$B = \{2, 6, 8\}$$

$(A \cap B)'$ means the complement of A intersect B.

$$(A \cap B)' = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

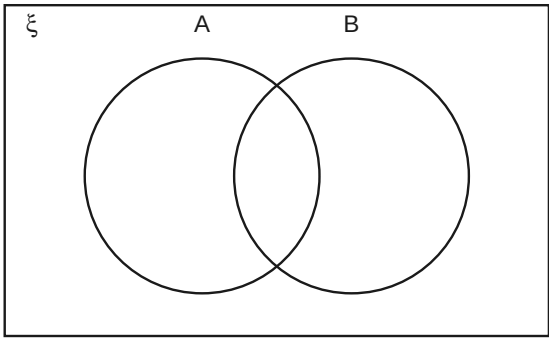


Answer: $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

Probability › Sets and Counting › Sets (Foundation)

Topic 276 of 313 • id: set_notation_F

Q1. $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 4, 5, 7, 9\}$, $B = \{1, 4, 5, 7, 8\}$. List the elements that are not in A.



Q1

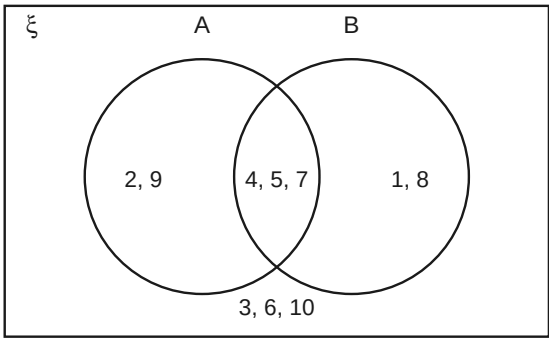
$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 5, 7, 9\}$$

$$B = \{1, 4, 5, 7, 8\}$$

Check each element of ξ in turn to see whether it are not in A.

Elements that are not in A: $\{1, 3, 6, 8, 10\}$



Answer: $\{1, 3, 6, 8, 10\}$

Probability › Sets and Counting › Product Rule for Counting (Higher)

Topic 277 of 313 • id: product_rule_counting_H

Q1. A shop sells 4 different shirts and 3 different pairs of trousers. Use the product rule for counting to find how many different combinations are possible, choosing exactly one of each.

Q1

Total combinations = $4 \times 3 = 12$

Answer: 12

Probability › Tables and Diagrams › Relative Frequency (Foundation)

Topic 278 of 313 • id: relative_frequency_F

Q1. A coin is tested 40 times, and it lands on heads 24 times. Use this relative frequency to estimate how many times this would happen in 75 trials.

Q1

$$\text{Relative frequency} = \text{frequency} \div \text{number of trials} = \frac{24}{40} = \frac{3}{5}$$

$$\text{Estimated number of times in 75 trials} = \frac{3}{5} \times 75 = 45$$

Answer: 45

Probability › Tables and Diagrams › Two-Way Tables (Foundation)

Topic 279 of 313 • id: two_way_tables_F

Q1. The two-way table shows the number of boys and girls who prefer bus or walk. Find the missing values.

	Bus	Walk	Total
Boys		13	23
Girls	18		27
Total	28	22	50

Q1

'Boys' / 'Bus': using the 'Boys' row total: $23 - 13 = 10$

'Girls' / 'Walk': using the 'Girls' row total: $27 - 18 = 9$

	Bus	Walk	Total
Boys	10	13	23
Girls	18	9	27
Total	28	22	50

Answer: 'Boys' / 'Bus' = 10; 'Girls' / 'Walk' = 9

Probability › Tables and Diagrams › Sample Space Diagrams (Foundation)

Topic 280 of 313 • id: sample_space_diagrams_F

Q1. Two fair spinners, one numbered 1 to 4 and the other numbered 1 to 4, are spun together. Complete the sample space diagram showing the product of the two spinners, and use it to find the probability that the product is 9.

	1	2	3	4
1				
2				
3				
4				

Q1

The sample space diagram shows every possible product of the two spinners.

Number of favourable outcomes (the product is 9) = 1

Total number of equally likely outcomes = 16

$P(\text{the product is 9}) = \frac{1}{16}$

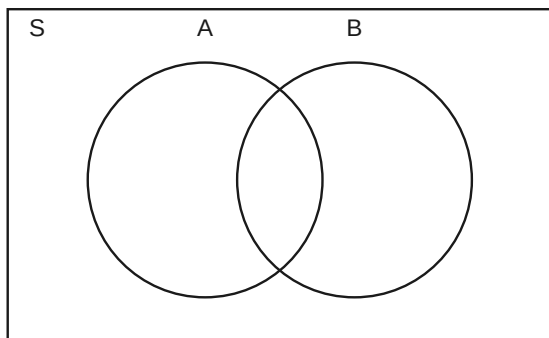
	1	2	3	4
1	1	2	3	4
2	2	4	6	8
3	3	6	9	12
4	4	8	12	16

Answer: $\frac{1}{16}$

Probability › Venn Diagrams › Venn Diagram Shading (Foundation)

Topic 281 of 313 • id: venn_diagram_shading_F

Q1. Shade the region that represents the elements in A but not in B.

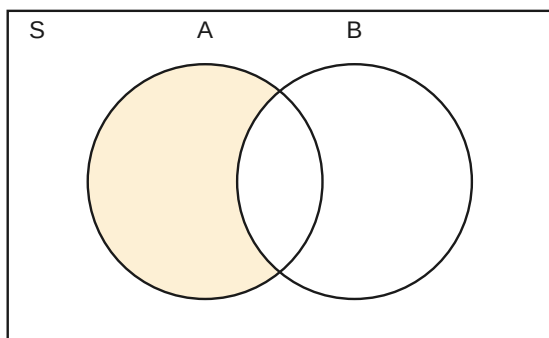


Q1

The Venn diagram has two circles, A and B, inside a rectangle representing everything being considered.

'Shade the region that represents the elements in A but not in B.' tells you which of the four regions (in A only, in B only, in both, or in neither) needs shading.

The region inside A only (not in B).

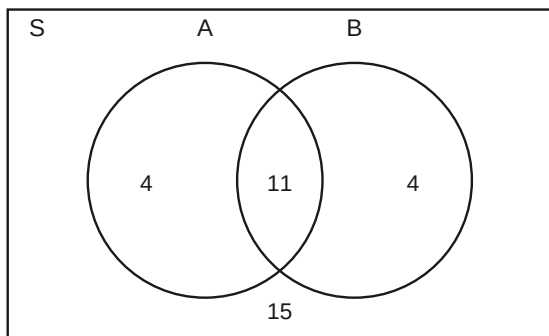


Answer: The region inside A only (not in B).

Probability › Venn Diagrams › Probability from a Venn Diagram (Foundation)

Topic 282 of 313 • id: venn_diagram_probability_F

Q1. The Venn diagram shows the number of students in each region for sets A and B. Find $P(\text{A only})$.



Q1

Total number of elements = $4 + 4 + 11 + 15 = 34$

Number of outcomes in A only = $4 = 4$

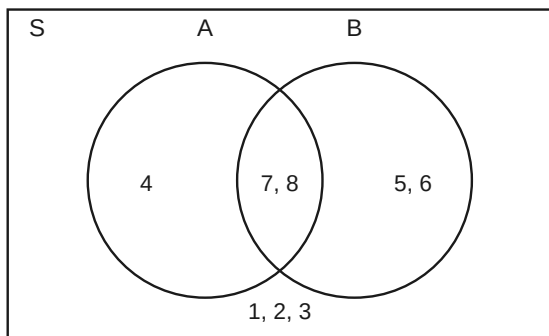
$$P(\text{A only}) = \frac{4}{34} = \frac{2}{17}$$

Answer: $\frac{2}{17}$

Probability › Venn Diagrams › Venn Diagram Set Notation (Higher)

Topic 283 of 313 • id: venn_diagram_notation_H

Q1. $\xi = \{1, 2, 3, 4, 5, 6, 7, 8\}$. The Venn diagram shows sets A and B. List the elements of $(A \cap B)'$.



Q1

$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{4, 7, 8\}, B = \{5, 6, 7, 8\}$$

$(A \cap B)'$ means the elements that are not in both A and B at once.

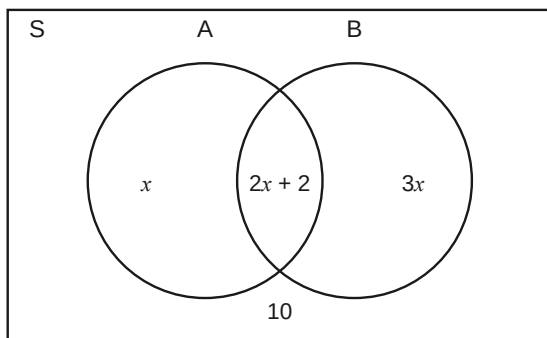
$$(A \cap B)' = \{1, 2, 3, 4, 5, 6\}$$

Answer: $\{1, 2, 3, 4, 5, 6\}$

Probability › Venn Diagrams › Algebraic Venn Diagrams (Higher)

Topic 284 of 313 • id: venn_diagram_algebra_H

Q1. The Venn diagram shows the number of elements in each region of sets A and B, given in terms of x . There are 54 elements in total. Find the value of x , and hence find the probability that a randomly chosen element is in A or B (or both).



Q1

$$\text{a_only} + \text{b_only} + \text{both} + \text{neither} = \text{total: } (x) + (3x) + (2x + 2) + 10 = 54$$

$$\text{Collecting like terms: } 6x + 12 = 54$$

$$x = 7$$

$$\text{Substituting } x = 7: \text{ A only} = 7, \text{ B only} = 21, \text{ both} = 16, \text{ neither} = 10$$

$$\text{The probability that a randomly chosen element is in A or B (or both)} = \frac{22}{27}$$

$$\text{Answer: } \frac{22}{27}$$

Statistics › Averages from a List › Mean (Foundation)

Topic 285 of 313 • id: stats_mean_F

Find the mean of this data set:

Q1. 26, 17, 6, 14, 18.

Q1. 26, 17, 6, 14, 18.

Mean = sum ÷ count = $(26 + 17 + 6 + 14 + 18) \div 5 = 81 \div 5 = 16.20$

Answer: 16.20

Statistics › Averages from a List › Mode (Foundation)

Topic 286 of 313 • id: stats_mode_F

Find the mode of this data set:

Q1. 11, 3, 11, 12, 8, 3, 3, 7.

Q1. 11, 3, 11, 12, 8, 3, 3, 7.

Count how often each value appears. 3 appears 3 times, more than any other value.

Mode = the most frequent value = 3

Answer: 3

Statistics › Averages from a List › Median (Foundation)

Topic 287 of 313 • id: stats_median_F

Find the median of this data set:

Q1. 22, 12, 15, 8, 29.

Q1. 22, 12, 15, 8, 29.

Sort the data: 8, 12, 15, 22, 29. The middle value is 15.

Answer: 15

Statistics › Averages from a List › Range (Foundation)

Topic 288 of 313 • id: stats_range_F

Find the range of this data set:

Q1. 2, 22, 29, 11, 13, 30.

Q1. 2, 22, 29, 11, 13, 30.

Range = largest - smallest = $30 - 2 = 28$

Answer: 28

Statistics › Averages from a List › Mean, Mode, Median & Range (Foundation)

Topic 289 of 313 • id: stats_averages_combined_F

Q1. For this data set, find (a) the mean (b) the mode (c) the median (d) the range: 9, 13, 18, 13, 13.

Q1

(a) Mean = $(9 + 13 + 18 + 13 + 13) \div 5 = 66 \div 5 = 13.20$

(b) Mode = the most frequent value = 13 (appears 3 times)

(c) Median: Sorted: 9, 13, 13, 13, 18. Middle value = 13.

(d) Range = $18 - 9 = 9$

Answer: Mean = 13.20, Mode = 13, Median = 13, Range = 9

Statistics › Averages from a List › Interquartile Range (Higher)

Topic 290 of 313 • id: stats_interquartile_range_H

Q1. Find the interquartile range (IQR) of this data set: 33, 29, 32, 3, 19, 13, 21, 33, 6, 7, 11.

Q1

Sort the data: 3, 6, 7, 11, 13, 19, 21, 29, 32, 33, 33.

Split into a lower half [3, 6, 7, 11, 13] and an upper half [21, 29, 32, 33, 33] (excluding the overall middle value when 11 is odd).

Q1 = median of the lower half = 7

Q3 = median of the upper half = 32

Interquartile range = $Q3 - Q1 = 32 - 7 = 25$

Answer: 25

Statistics › Frequency Tables › Mean from a Frequency Table (Foundation)

Topic 291 of 313 • id: stats_mean_frequency_table_F

Q1. Find the mean number of pets.

	Frequency
0	10
1	2
2	3
3	5
4	5
5	8

Q1

Multiply each value by its frequency and sum: $0 \times 10 + 1 \times 2 + 2 \times 3 + 3 \times 5 + 4 \times 5 + 5 \times 8 = 83$

Total frequency = $10 + 2 + 3 + 5 + 5 + 8 = 33$

Mean = $83 \div 33 = \frac{83}{33}$

Answer: $\frac{83}{33}$

Statistics › Frequency Tables › Mean from a Grouped Frequency Table (Higher)

Topic 292 of 313 • id: stats_mean_grouped_frequency_table_H

Q1. The table shows the times (in minutes) taken by a group of runners. Find an estimate of the mean time.

	Frequency
20-29	8
30-39	3
40-49	4
50-59	2

Q1

Midpoints of each class: 24.50, 34.50, 44.50, 54.50

Sum of (midpoint $\times$ frequency): $24.50 \times 8 + 34.50 \times 3 + 44.50 \times 4 + 54.50 \times 2 = 586.50$

Total frequency = $8 + 3 + 4 + 2 = 17$

Estimated mean = $586.50 \div 17 = 34.50$

Answer: ≈ 34.50

Statistics › Frequency Tables › Mean from a Grouped Frequency Table (Foundation)

Topic 293 of 313 • id: stats_mean_grouped_frequency_table_F

Q1. The table shows the times (in minutes) taken by a group of runners. Find an estimate of the mean time.

	Frequency
0-9	5
10-19	7
20-29	4

Q1

Midpoints of each class: 4.50, 14.50, 24.50

Sum of (midpoint $\times$ frequency): $4.50 \times 5 + 14.50 \times 7 + 24.50 \times 4 = 222$

Total frequency = $5 + 7 + 4 = 16$

Estimated mean = $222 \div 16 = 13.88$

Answer: ≈ 13.88

Statistics › Frequency Tables › Mode from a Frequency Table (Foundation)

Topic 294 of 313 • id: stats_mode_frequency_table_F

Q1. Find the modal number of pets.

	Frequency
0	3
1	2
2	4
3	6

Q1

The frequency shows how often each value occurs: 0 (3 times), 1 (2 times), 2 (4 times), 3 (6 times).

The highest frequency is 6, for the value 3.

Mode = 3

Answer: 3

Statistics › Frequency Tables › Median from a Frequency Table (Foundation)

Topic 295 of 313 • id: stats_median_frequency_table_F

Q1. Find the median number of pets.

	Frequency
0	5
1	6
2	8
3	8
4	9
5	8

Q1

Total frequency = $5 + 6 + 8 + 8 + 9 + 8 = 44$ data items.

Imagine the data written out in full, in order (each value repeated as many times as its frequency), then find the middle of that list.

Median = 3

Answer: 3

Statistics › Frequency Tables › Range from a Frequency Table (Foundation)

Topic 296 of 313 • id: stats_range_frequency_table_F

Q1. Find the range of the number of pets.

	Frequency
2	2
3	8
4	3
5	10
6	2

Q1

The listed values run from 2 to 6 (every value in between has a frequency of at least 1, so these are the true extremes).

Range = largest - smallest = $6 - 2 = 4$

Answer: 4

Statistics › Working Backwards › Reverse Mean (Higher)

Topic 297 of 313 • id: stats_reverse_mean_H

Q1. The mean of four numbers is 16. Three of the numbers are 30, 17, 4. Find the missing number.

Q1

Total of all 4 numbers = mean $\times$ count = $16 \times 4 = 64$

Sum of the known numbers = $30 + 17 + 4 = 51$

Missing number = $64 - 51 = 13$

Answer: 13

Statistics › Working Backwards › Reverse Mean (Foundation)

Topic 298 of 313 • id: stats_reverse_mean_F

Q1. The mean of four numbers is 7. Three of the numbers are 10, 2, 5. Find the missing number.

Q1

Total of all 4 numbers = mean $\times$ count = $7 \times 4 = 28$

Sum of the known numbers = $10 + 2 + 5 = 17$

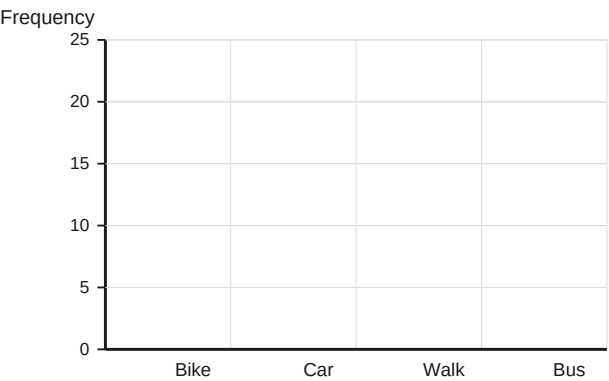
Missing number = $28 - 17 = 11$

Answer: 11

Statistics › Charts and Graphs › Constructing Bar Charts (Foundation)

Topic 299 of 313 • id: bar_chart_construct_F

Q1. The table shows the mode of transport to school of a group of students: Bike: 7, Car: 19, Walk: 9, Bus: 6. Draw a bar chart to show this information.

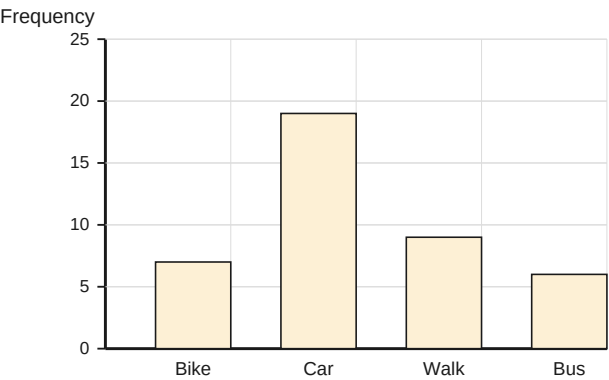


Q1

Draw one bar per category (Bike, Car, Walk, Bus), each as tall as its frequency.

Heights: Bike: 7, Car: 19, Walk: 9, Bus: 6.

Bars of height 7 (Bike), 19 (Car), 9 (Walk), 6 (Bus)

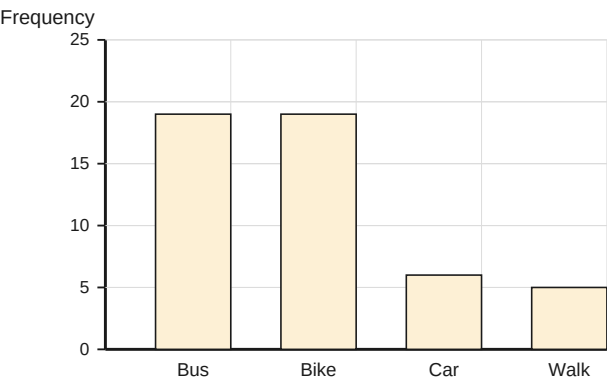


Answer: Bars of height 7 (Bike), 19 (Car), 9 (Walk), 6 (Bus)

Statistics › Charts and Graphs › Interpreting Bar Charts (Foundation)

Topic 300 of 313 • id: bar_chart_interpret_F

Q1. The bar chart shows the mode of transport to school of a group of students. How many chose Bus?



Q1

Find the Bus bar on the chart.

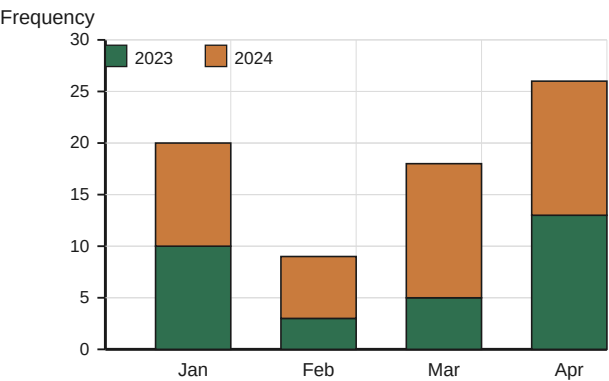
Read its height: 19.

Answer: 19

Statistics › Charts and Graphs › Composite Bar Charts (Foundation)

Topic 301 of 313 • id: composite_bar_chart_F

Q1. The composite bar chart shows the number of sales split by 2023 and 2024. Find the difference between 2023 and 2024 for Feb.



Q1

2023: 3, 2024: 6.

Difference = 3.

Answer: 3

Statistics › Charts and Graphs › Constructing Pie Charts (Foundation)

Topic 302 of 313 • id: pie_chart_construct_F

Q1. A survey asked 18 people about their favourite sport. Calculate the angle for each category, and draw a pie chart.

	Frequency	Angle
Football	6	
Tennis	2	
Swimming	10	

Q1

Total = 18.

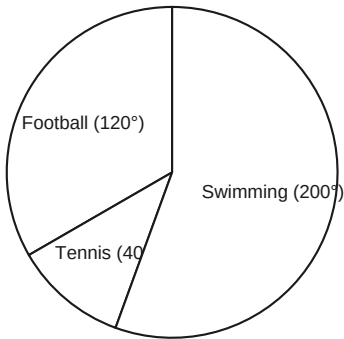
Football: $\frac{6}{18} \times 360^\circ = 120^\circ$

Tennis: $\frac{2}{18} \times 360^\circ = 40^\circ$

Swimming: $\frac{10}{18} \times 360^\circ = 200^\circ$

Football: 120°, Tennis: 40°, Swimming: 200°

	Frequency	Angle
Football	6	120°
Tennis	2	40°
Swimming	10	200°

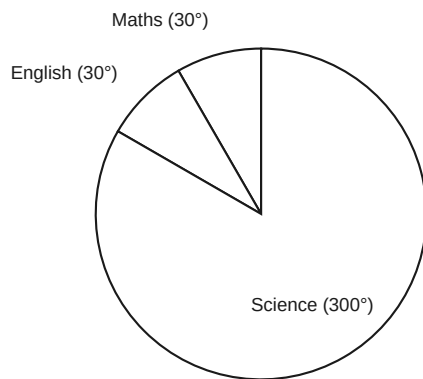


Answer: Football: 120°, Tennis: 40°, Swimming: 200°

Statistics › Charts and Graphs › Interpreting Pie Charts (Foundation)

Topic 303 of 313 • id: pie_chart_interpret_F

Q1. The pie chart shows the favourite subject of 12 people. What fraction of the total chose Science? Give your answer in its simplest form.



Q1

Science represents 10 out of 12: $\frac{10}{12}$.

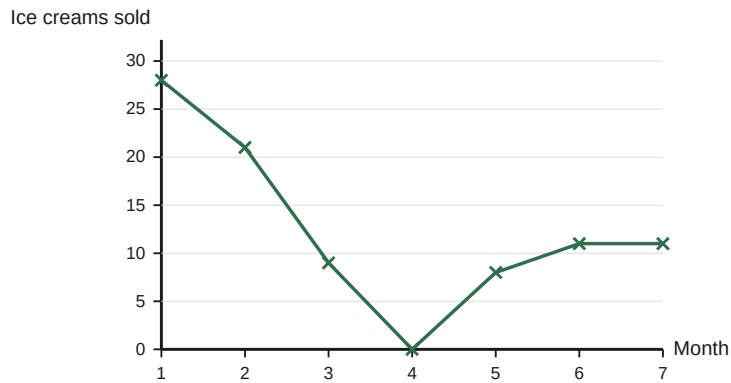
Simplify: $\frac{5}{6}$.

Answer: $\frac{5}{6}$

Statistics › Charts and Graphs › Time Series Graphs (Foundation)

Topic 304 of 313 • id: time_series_graph_F

Q1. The time series graph shows the ice creams sold over time. At which period was the value highest?



Q1

Compare every plotted value: 28, 21, 9, 0, 8, 11, 11.

The highest value, 28, occurs at period 1.

Answer: 1

Statistics › Charts and Graphs › Constructing Scatter Graphs (Foundation)

Topic 305 of 313 • id: scatter_graph_construct_F

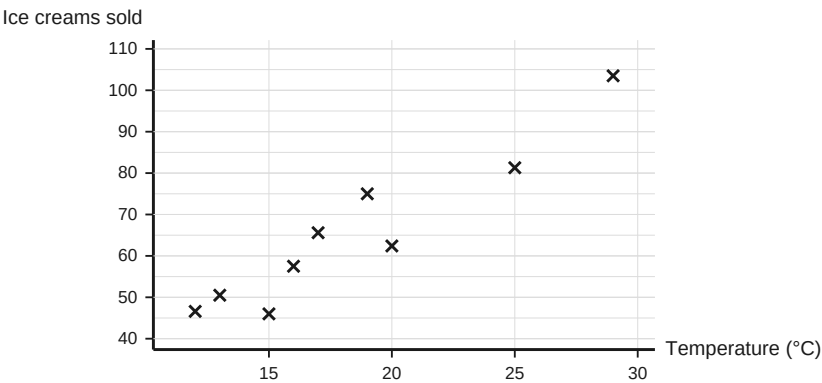
Q1. The table shows the midday temperature on a number of days and the number of ice creams a shop sold. Draw a scatter graph of this data, and describe the correlation.

	1	2	3	4	5	6	7	8	9
Temperature (°C)	12	13	15	16	17	19	20	25	29
Ice creams sold	46.6	50.5	46.0	57.5	65.6	75.0	62.4	81.3	103.5

Q1

Plot each (Temperature (°C), Ice creams sold) pair as a single point - do not join the points up with a line or curve.

As temperature (°c) increases, ice creams sold tends to increase too, so the points show positive correlation.

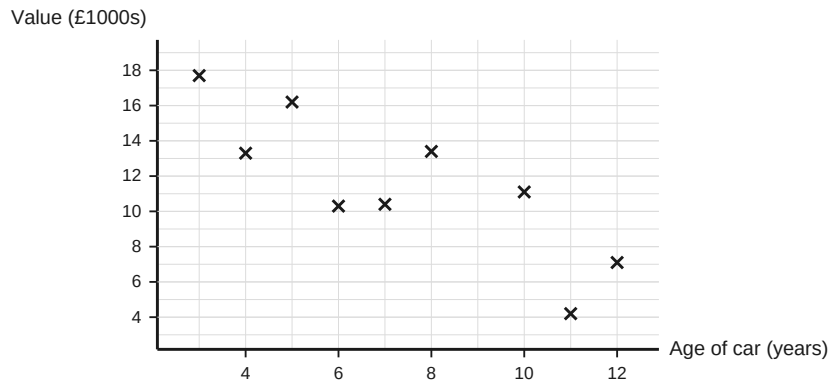


Answer: Positive correlation

Statistics › Charts and Graphs › Interpreting Scatter Graphs (Foundation)

Topic 306 of 313 • id: scatter_graph_interpret_F

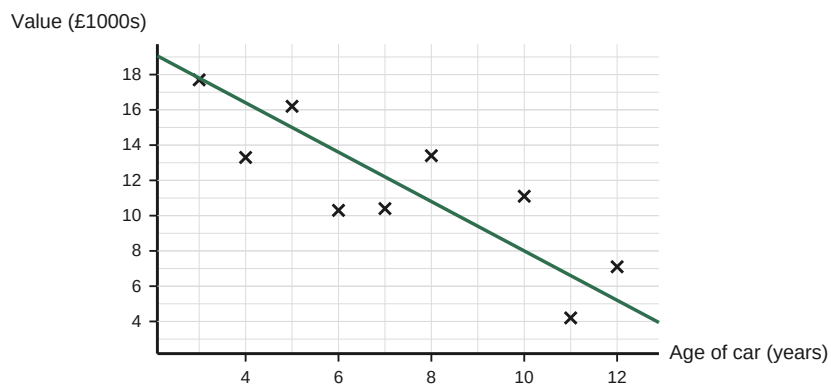
Q1. The scatter graph shows the age of a number of second-hand cars and their value. Draw a line of best fit, and use it to estimate value (£1000s) when age of car (years) is 8.



Q1

Find 8 on the age of car (years) axis and read up to your line of best fit.

Read across to the value (£1000s) axis: approximately 11.



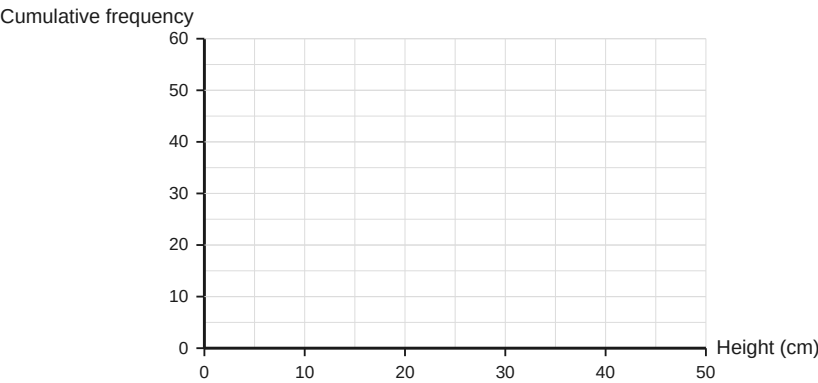
Answer: Approximately 11

Statistics › Cumulative Frequency & Box Plots › Plotting Cumulative Frequency Graphs (Higher)

Topic 307 of 313 • id: cumulative_frequency_plot_H

Q1. The table shows the heights (in cm) of a group of plants. Draw a cumulative frequency graph to show this information.

	Frequency
0-10	15
10-20	10
20-30	12
30-40	2
40-50	15

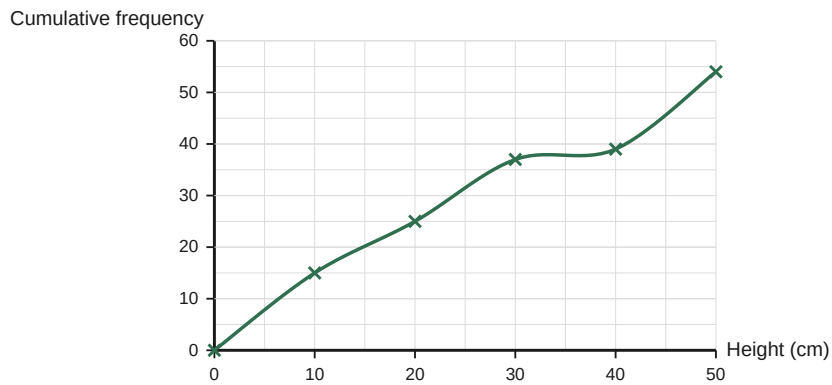


Q1

Running total of the frequencies, class by class: 0-10: 15, 10-20: 10, 20-30: 12, 30-40: 2, 40-50: 15.

Cumulative frequencies: 15, 25, 37, 39, 54

Plot each (upper class boundary, cumulative frequency) point and join with a smooth curve.

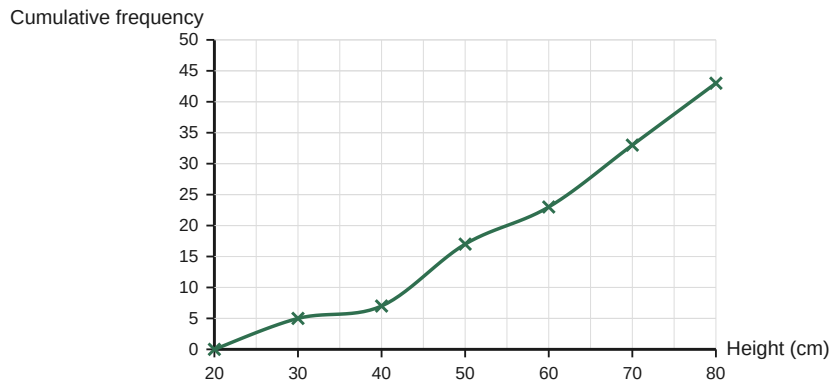


Answer: Cumulative frequencies: 15, 25, 37, 39, 54

Statistics › Cumulative Frequency & Box Plots › Interpreting Cumulative Frequency Graphs (Higher)

Topic 308 of 313 • id: cumulative_frequency_interpret_H

Q1. The cumulative frequency graph shows the heights (in cm) of a group of plants. Use the graph to estimate the interquartile range.



Q1

Lower quartile (at cumulative frequency 10.8) ≈ 43.8

Upper quartile (at cumulative frequency 32.2) ≈ 69.2

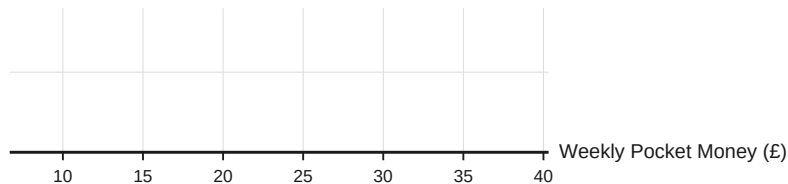
$IQR = 69.2 - 43.8 = 25.5$

Answer: 25.5

Statistics › Cumulative Frequency & Box Plots › Constructing Box Plots (Higher)

Topic 309 of 313 • id: box_plot_construct_H

Q1. Here are seven weekly pocket money (£): [9, 15, 19, 24, 26, 30, 38]. Draw a box plot for this data.



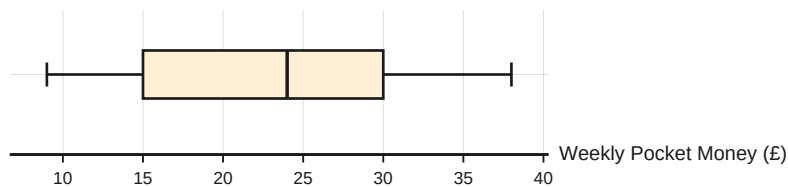
Q1

Sorted data: [9, 15, 19, 24, 26, 30, 38].

Min = 9, Max = 38.

Median splits the data in half; Q1 is the median of the lower half, Q3 the median of the upper half.

Min = 9, Q1 = 15, Median = 24, Q3 = 30, Max = 38

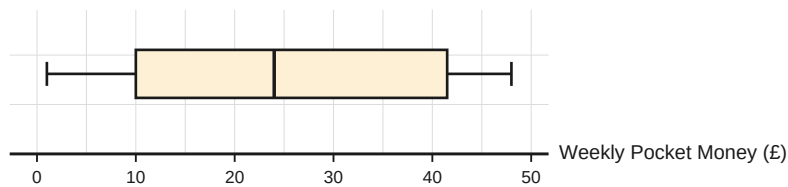


Answer: Min = 9, Q1 = 15, Median = 24, Q3 = 30, Max = 38

Statistics › Cumulative Frequency & Box Plots › Interpreting Box Plots (Higher)

Topic 310 of 313 • id: box_plot_interpret_H

Q1. The box plot shows weekly pocket money (£). Find the interquartile range.



Q1

$$Q1 = 10.0, Q3 = 41.5$$

$$IQR = 41.5 - 10.0 = 31.5$$

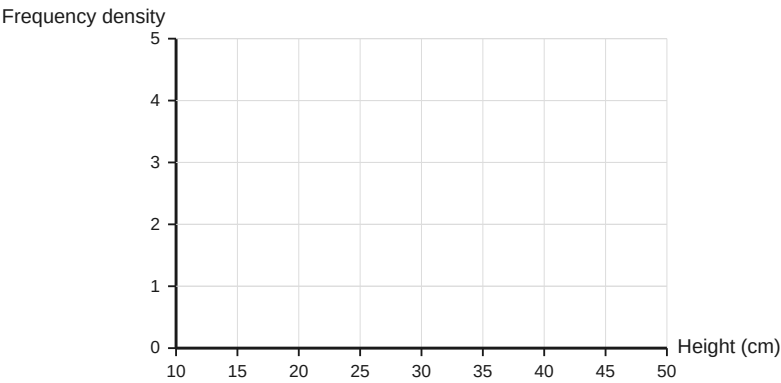
Answer: 31.5

Statistics › Histograms › Plotting Histograms (Higher)

Topic 311 of 313 • id: histogram_plot_H

Q1. The table shows the heights (in cm) of a group of plants. Complete a frequency density column and draw a histogram.

	Frequency
10-15	6
15-35	18
35-45	35
45-50	18



Q1

Frequency density = frequency $\div$ class width, for each class:

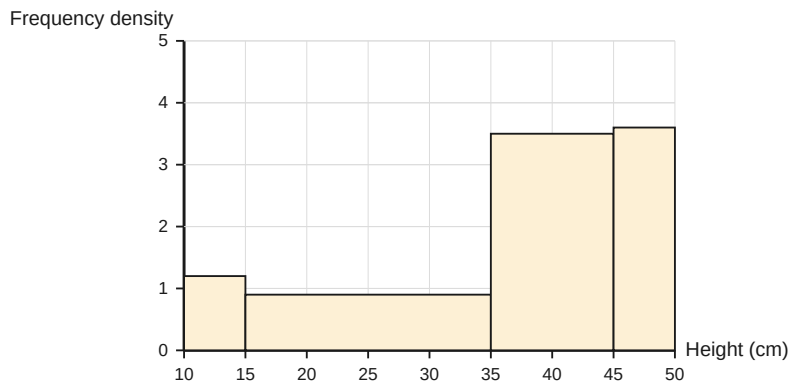
$$10-15: 6 \div 5 = 1.2$$

$$15-35: 18 \div 20 = 0.9$$

$$35-45: 35 \div 10 = 3.5$$

$$45-50: 18 \div 5 = 3.6$$

Frequency densities: 1.2, 0.9, 3.5, 3.6

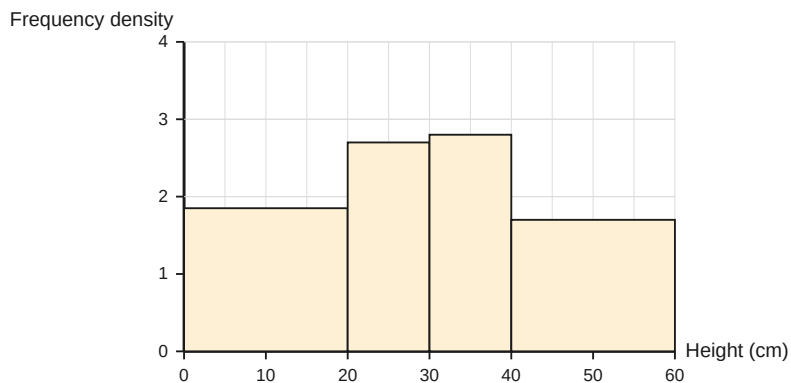


Answer: Frequency densities: 1.2, 0.9, 3.5, 3.6

Statistics › Histograms › Interpreting Histograms (Higher)

Topic 312 of 313 • id: histogram_interpret_H

Q1. The histogram shows the heights (in cm) of a group of plants. Which class has the highest frequency?



Q1

The tallest bar is not necessarily the highest frequency, since classes have different widths.

Frequency = density $\times$ width for each class: 0-20: $1.85 \times 20 = 37$, 20-30: $2.70 \times 10 = 27$, 30-40: $2.80 \times 10 = 28$, 40-60: $1.70 \times 20 = 34$.

The highest frequency is in class 0-20.

Answer: 0-20

Statistics › Sampling and Populations › Sampling and Populations (Foundation)

Topic 313 of 313 • id: sampling_methods_F

Q1. A school has 393 students: Year 9: 220, Year 7: 173. A stratified sample of 55 students is taken. How many students from Year 7 should be in the sample?

Q1

Year 7 makes up 173 out of 393 students.

Sample size needed = $(173 \div 393) \times 55$

= 24 students (rounded to the nearest whole number)

Answer: 24